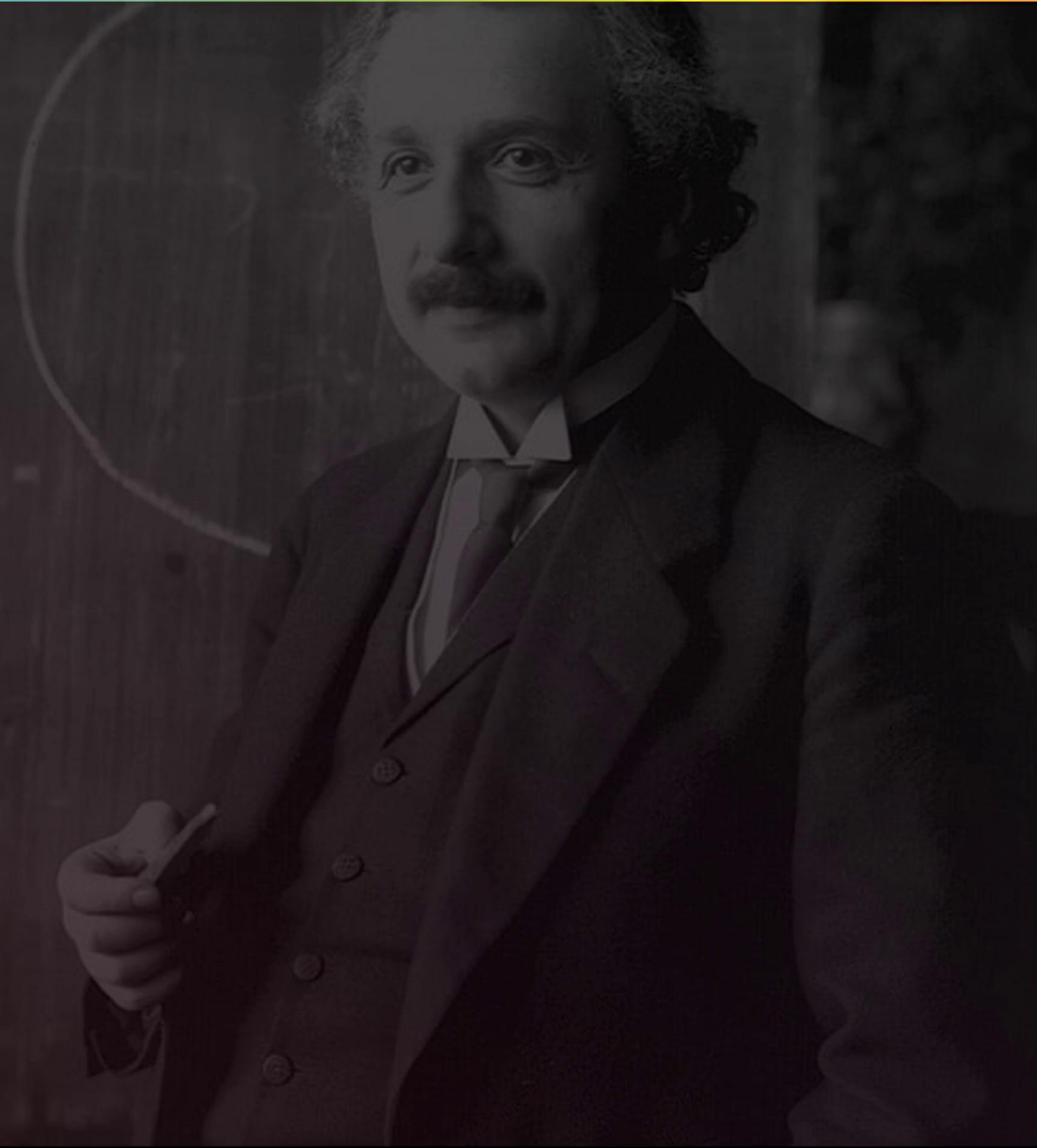
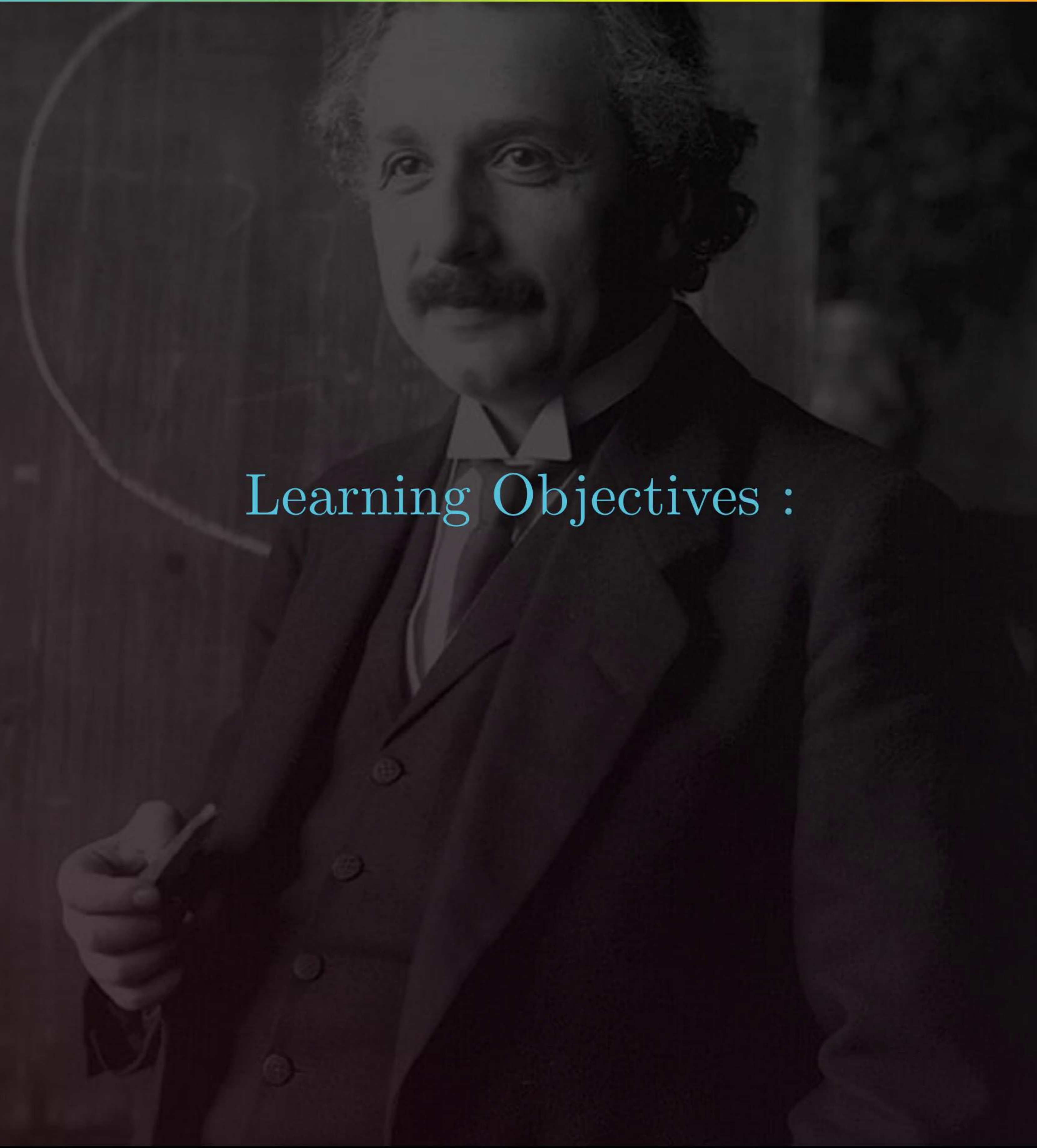


CHAPTER 11 : DUAL NATURE OF RADIATION AND MATTER



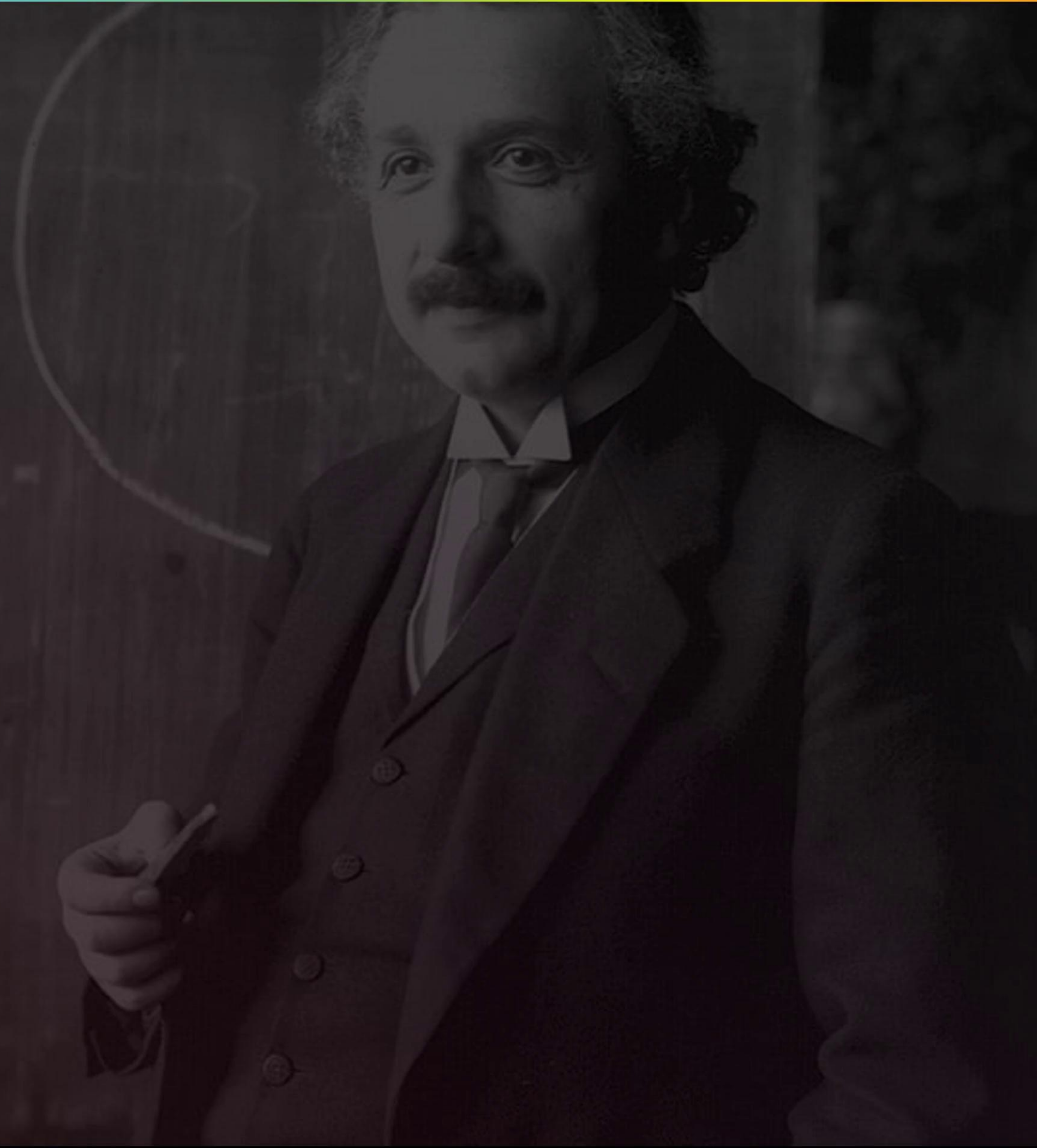
CHAPTER 11 : DUAL NATURE OF RADIATION AND MATTER

Learning Objectives :



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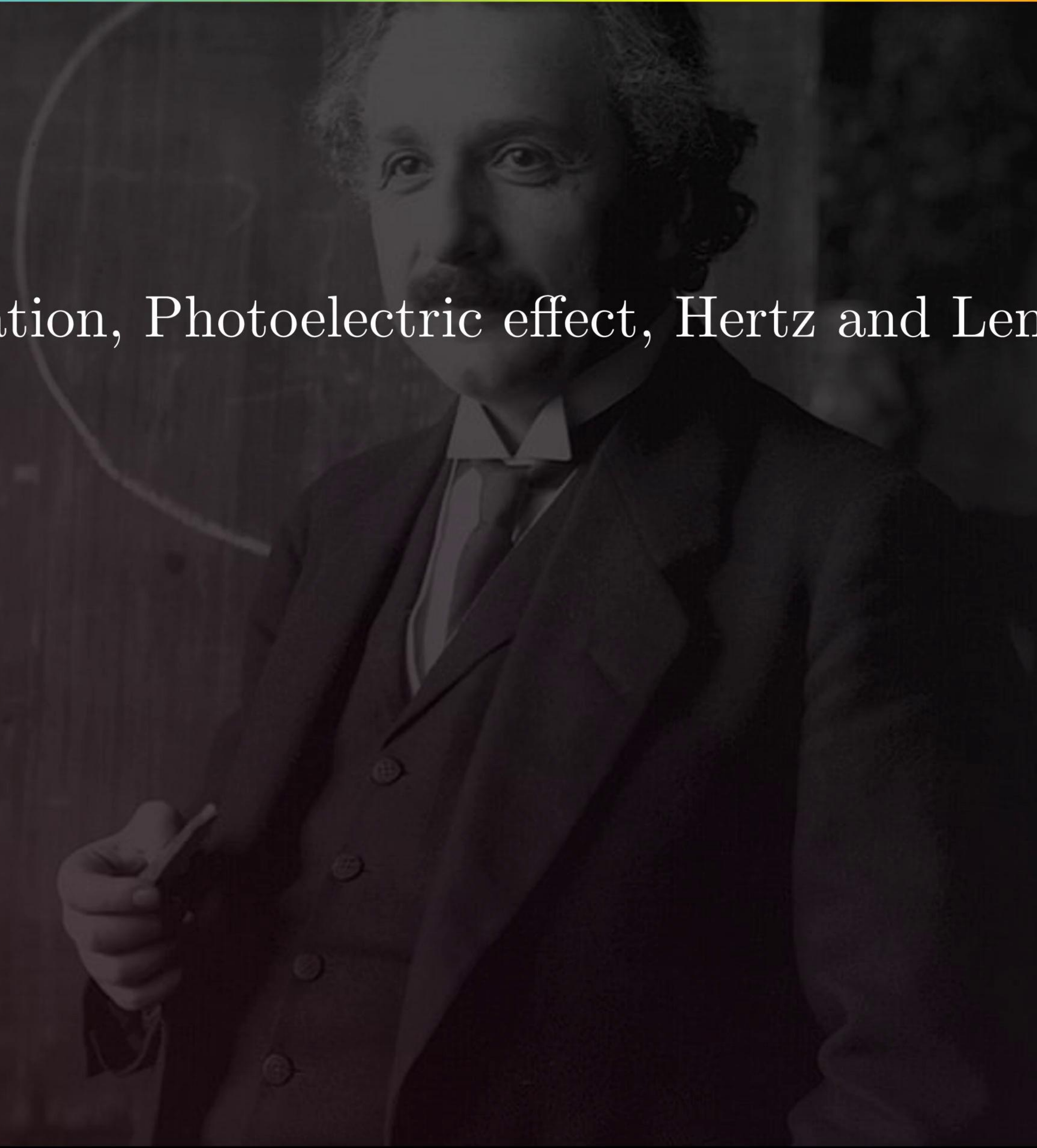
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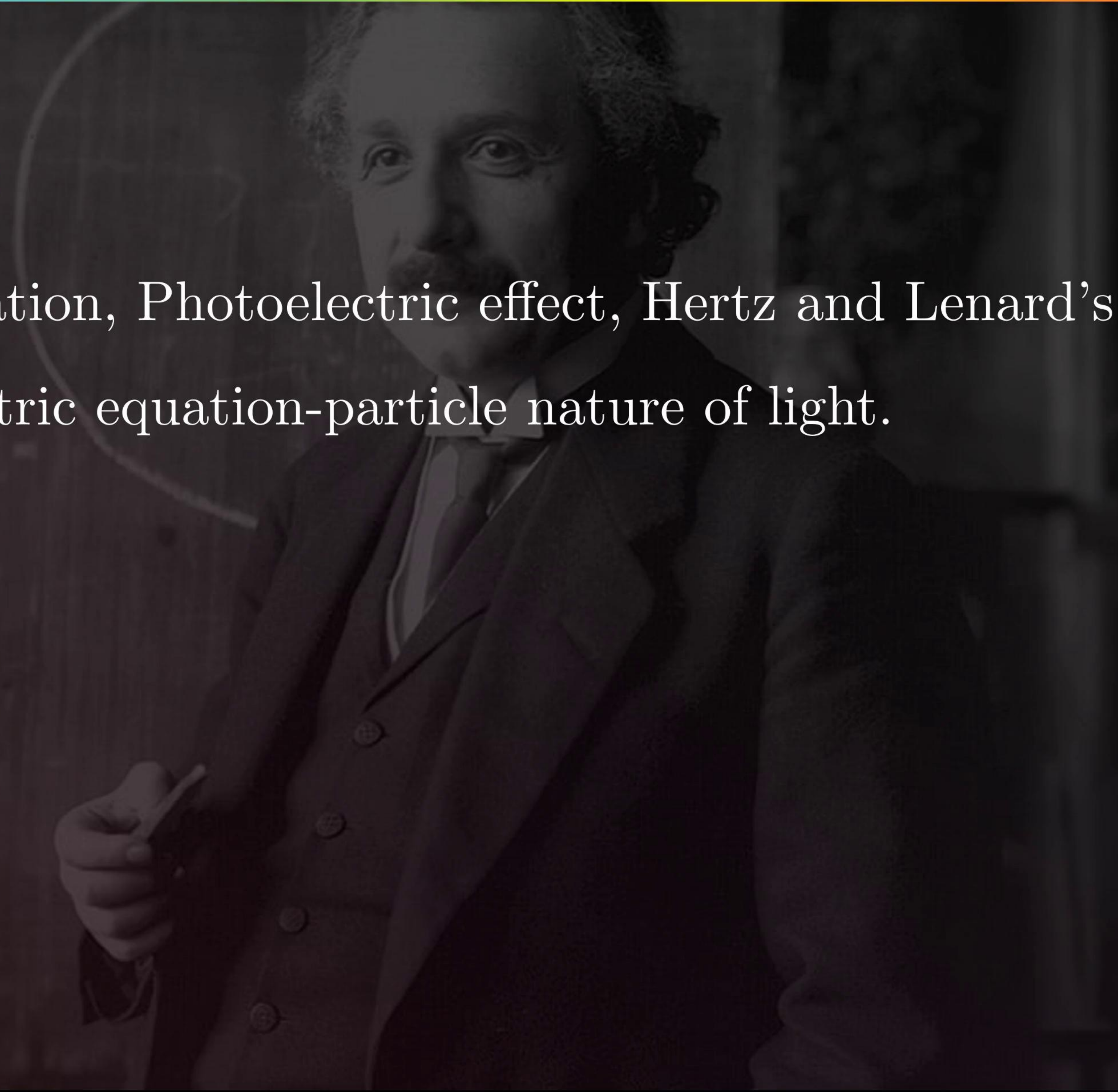
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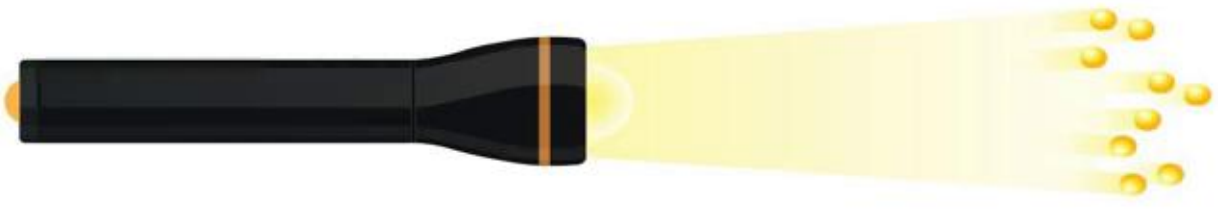
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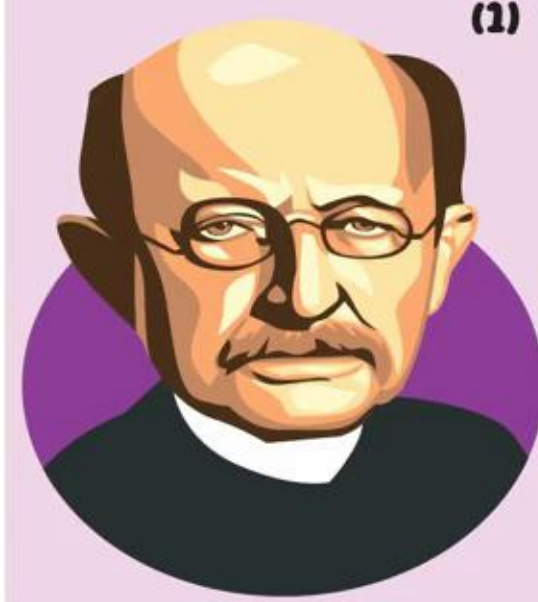
- Dual nature of radiation, Photoelectric effect, Hertz and Lenard's observations;
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- Matter waves-wave nature of particles, de-Broglie relation.

DUAL NATURE OF RADIATION AND MATTER



Particles

PLANCK'S QUANTUM THEORY OF LIGHT



- (1) The energy of one photon is proportional to its frequency
- (2) $E \propto \nu$, $E = h\nu$
 $h = \text{Planck's constant} = 6.62 \times 10^{-34} \text{ JS}$
- (3) Energy of any light or radiation is one integral multiple of $h\nu$.
 $E = nh\nu$
- (4) Energy of one photon.
 $E = h\nu = \frac{1240}{\lambda(\text{nm})} \text{ eV}$

PROPERTIES OF PHOTONS

- (1) Photon is just a packet of energy.
- (2) Energy of photon does not change with medium.
- (3) Photon can not be deflected by electric field and magnetic field.
- (4) Momentum of photon $|p| = m \times c = \frac{h}{\lambda} = \frac{E}{c}$
- (5) Intensity of light beam = $\frac{\text{Energy}}{\text{area} \times \text{time}}$



PHOTON EMITTED PER SECOND

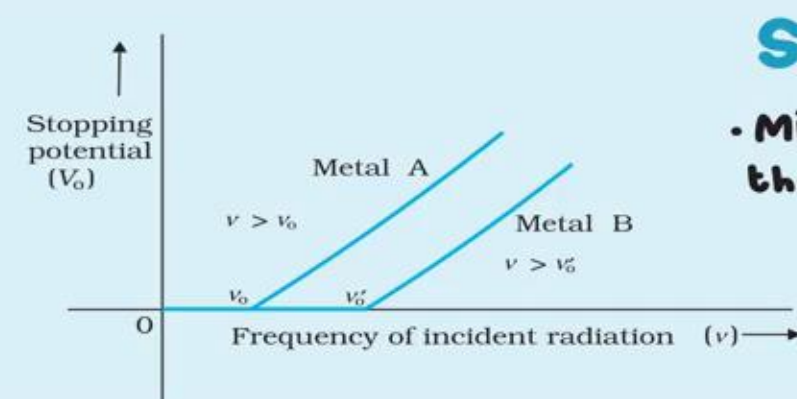
(1) $E = nh\nu$
 (2) Power, $P = nh\nu$
 $\Rightarrow = \frac{p}{h\nu} = \frac{p\lambda}{hc}$
 Number of photon per second = $\frac{\text{Power}}{\text{energy of one photon}}$

PHOTON FLUX

• Photon flux is no. of photon incident normally to a surface per seconds
 $\phi = \frac{n}{A} = \frac{P}{h\nu}$

FORCE AND RADIATION PRESSURE EXERTED BY A LIGHT BEAM

(1) $n = \frac{p\lambda}{hc}$
 (2) momentum of one photon = $\frac{h}{\lambda}$ momentum imparted per second = $n \times \frac{h}{\lambda} = \frac{P}{c}$
 $\Rightarrow \text{Force exerted} = \frac{P}{c}$
 (3) Radiation Pressure = $\frac{F}{A} = \frac{P}{CA} = \frac{I}{c}$



STOPPING POTENTIAL

• Minimum negative potential required to stop the electron of maximum K.E.

$$V_0 = \frac{K.E_{\text{max}}}{e} = \frac{h}{e} (\nu - \nu_0) \text{ Volts}$$

PHOTOELECTRIC EFFECT

- (1) It is a phenomenon of ejecting electrons by falling light of suitable frequency on a metal
- (2) Ejected electrons are called photoelectrons.
- (3) Current flowing due to the photoelectrons is called photocentric current

LAWS

- (1) No emission takes place below the threshold frequency.
- (2) Above threshold frequency, no. of photoelectrons emitted per seconds is directly proportional to intensity of radiation
- (3) The emission of photoelectrons is an instantaneous process.
- (4) Above threshold frequency, K.E (max) depends on frequency

WORK FUNCTION

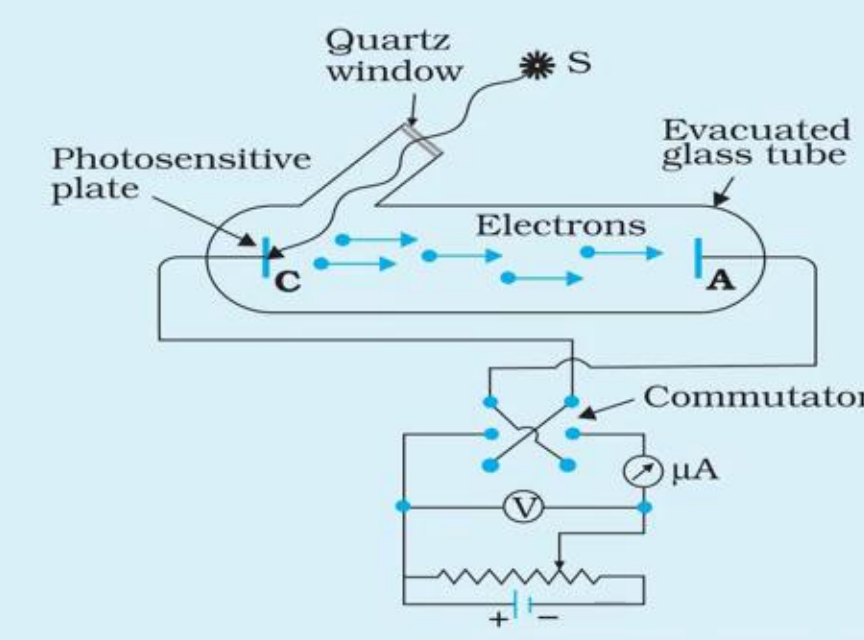
- Minimum energy required for getting a free electron away from the metal surface.
- Work function (ϕ_0) = $h\nu_0$
 $\nu_0 = \frac{\phi}{h} = \text{threshold frequency}$

EINSTEIN'S PHOTOELECTRIC EQUATION

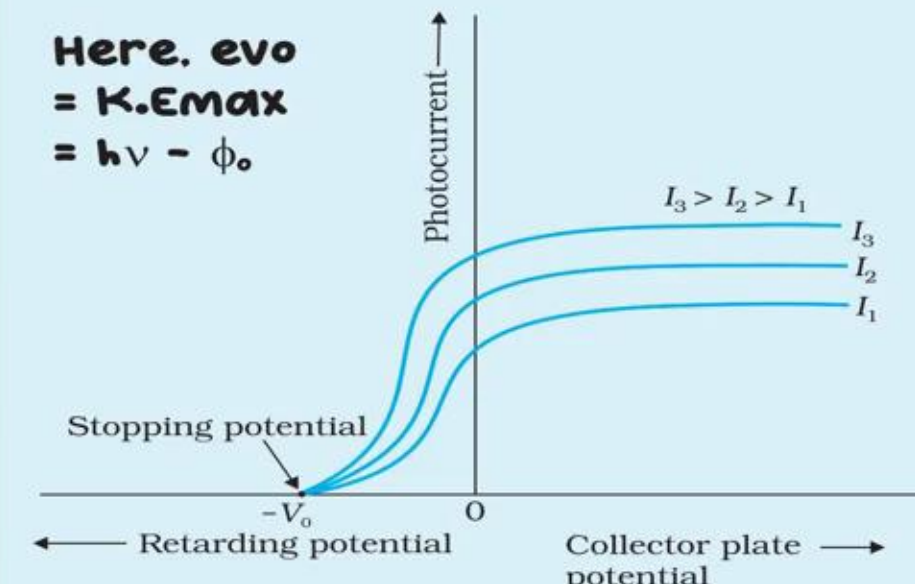
- The electron is emitted with maximum K.E
 $K.E_{\text{max}} = h\nu - \phi_0$
 $h\nu = K.E_{\text{max}} + \phi_0$
- Range of K.E.
 $0 \leq K.E_{\text{photoelectrons}} \leq h\nu - \phi_0$

EXPERIMENTAL STUDY

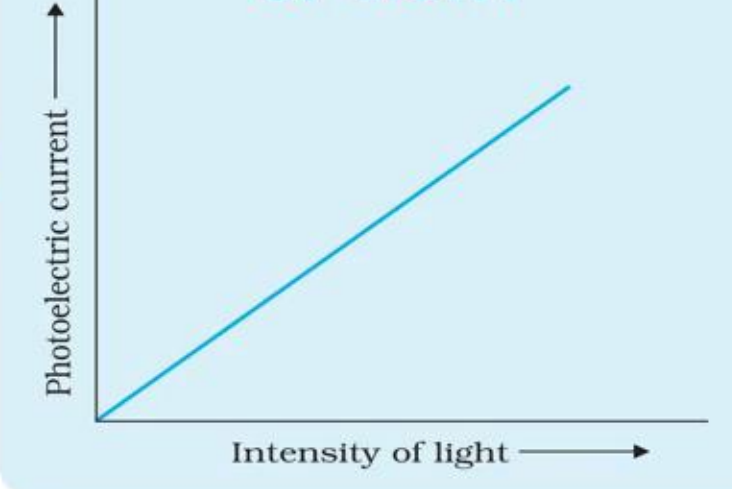
• The emission of electrons causes flow of electric current in the circuit.



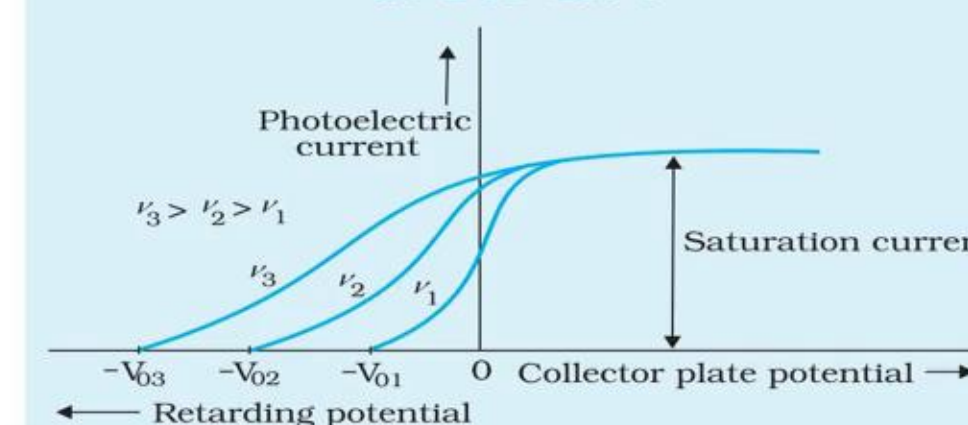
EFFECT OF POTENTIAL



EFFECT OF INTENSITY OF LIGHT



EFFECT OF FREQUENCY OF INCIDENT



DUAL NATURE OF LIGHT



de - Broglie wavelength, $\lambda = \frac{h}{mv}$ & $2\pi r = n\lambda$
 $\bullet mvr = \frac{nh}{2\pi}$
 This is Bohr quantisation Condition

PARTICLE NATURE OF LIGHT

- (1) In interaction of radiation with matter, radiation behaves as if it is made of particles called photons
- (2) $E = h\nu$ and $p = \frac{h\nu}{c}$
- (3) In a photon - particle collision, total energy and total momentum are conserved.

MATTER WAVE THEORY

- de - Broglie wavelength associated with moving particles, $\lambda = \frac{h}{p}$
- K.E of particle = $\frac{1}{2}mv^2 = \frac{p^2}{2m}$
- momentum, $p = mv = \sqrt{2m \times K.E}$

FOR UNCHARGED PARTICLES

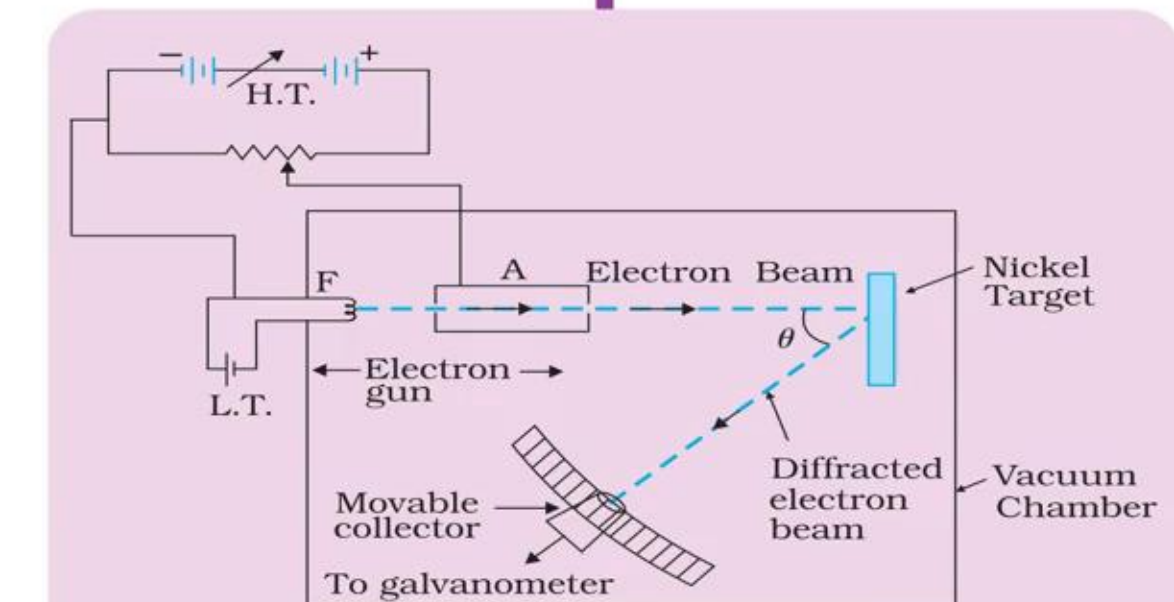
$$\lambda = \frac{h}{mv} = \frac{h}{\sqrt{2m \times K.E}}$$

FOR ACCELERATED CHARGED PARTICLES

$$\lambda = \frac{h}{\sqrt{2m \times K.E}}$$

V = potential difference

DAVISON-GERMER EXPERIMENT



- at $\phi = 50^\circ$ and accelerating potential = 54 V, maxima is obtained
- This experiment confirmed the wave nature of electron.

SPECIAL CASE FOR ELECTRON

$$\lambda = \frac{1.227}{\sqrt{V}} \text{ nm}$$

AND
 $V = \frac{150.1}{[\lambda(\text{\AA})]^2} \text{ Volt}$

FOR GASEOUS MOLECULES

$$K.E = \frac{3}{2}KT$$

$$\Rightarrow \lambda = \frac{h}{\sqrt{2m \times \frac{3}{2}KT}}$$

$$\Rightarrow \lambda = \frac{h}{\sqrt{3mKT}}$$

INTRODUCTION

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- * By the end of the 19th century, light was widely accepted as a wave.
- * Maxwell's equations predicted electromagnetic waves.
- * Hertz's experiments (1887) confirmed the existence of electromagnetic waves.
- * Thus, the wave theory of light was strongly supported.

INTRODUCTION

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(i) X-rays — Roentgen (1895)

- * X-rays were discovered during experiments on gas discharge.
- * This became an important tool for understanding atomic structure.

(iii) Electron — J.J. Thomson (1897)

- * Thomson confirmed cathode rays are made of negatively charged particles (electrons).
 - * He used crossed electric and magnetic fields to determine:
 - * Speed of electrons: about 0.1–0.2 times the speed of light
 - * e/m ratio: 1.76×10^{11} C/kg
 - * The e/m ratio was the same regardless of the gas or metal used.
 - ✓ This proved electrons are universal constituents of matter.
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★ More Evidence Supporting the Existence of Electrons

- * Electrons were found to be emitted when metals were:
 - * Heated (thermionic emission)
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★ Millikan's Oil Drop Experiment (1913)

R.A. Millikan measured the charge of the electron accurately.

Found that: Charge (e) = 1.602×10^{-19} C

Charge appears only in integral multiples of this value → charge is quantised.

Using e and e/m , the mass of the electron was calculated.

ELECTRON EMISSION

★ Meaning of Electron Emission

- * Metals contain free electrons that are responsible for electrical conductivity.
- * These free electrons cannot normally escape from the metal surface.
- * When an electron tries to leave, the metal surface becomes positively charged and pulls the electron back.
- * An electron can come out of the metal only when it gains sufficient energy to overcome this attractive force.
- * The minimum energy required for an electron to escape from the metal surface is called the work function.

★ 2. Work Function (ϕ_0)

- * Definition: The minimum energy required by an electron to escape from the metal surface.
- * Unit: electron volt (eV) $1\text{eV} = 1.602 \times 10^{-19} \text{ J}$
- * The value of ϕ_0 depends on the metal and the nature of its surface.
- * Different metals have different work functions.
- * Example :
 - Cesium (Cs): 2.14 eV
 - Platinum (Pt): 5.65 eV — highest among the listed metals.

ELECTRON EMISSION

TABLE 11.1 WORK FUNCTIONS OF SOME METALS

Metal	Work function ϕ_0 (eV)	Metal	Work function ϕ_0 (eV)
Cs	2.14	Al	4.28
K	2.30	Hg	4.49
Na	2.75	Cu	4.65
Ca	3.20	Ag	4.70
Mo	4.17	Ni	5.15
Pb	4.25	Pt	5.65

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(iii) Photoelectric Emission

- * Electrons are emitted when light of suitable frequency falls on the metal surface.
- * The emitted electrons are called photoelectrons.
- * The minimum frequency required is called threshold frequency (ν_0).

PHOTOELECTRIC EFFECT

- * When light of suitable frequency falls on a metal surface, electrons are emitted.
- * The emitted electrons are called photoelectrons.
- * The phenomenon is called the photoelectric effect.

PHOTOELECTRIC EFFECT

(1) Hertz's observations

PHOTOELECTRIC EFFECT

(1) Hertz's observations

- * Hertz (1887) discovered the photoelectric effect accidentally during experiments on electromagnetic waves.
- * He observed that sparks across a detector loop became stronger when the emitter plate was illuminated with ultraviolet light.
- * UV light helped electrons escape from the metal surface, allowing the spark discharge to occur more easily.
- * When light shines on a metal, electrons near the surface absorb energy from incident light.
- * If the absorbed energy is greater than the work function, electrons escape from the metal surface.
- * Conclusion: Light can cause the emission of charged particles (electrons) from a metal surface.

PHOTOELECTRIC EFFECT

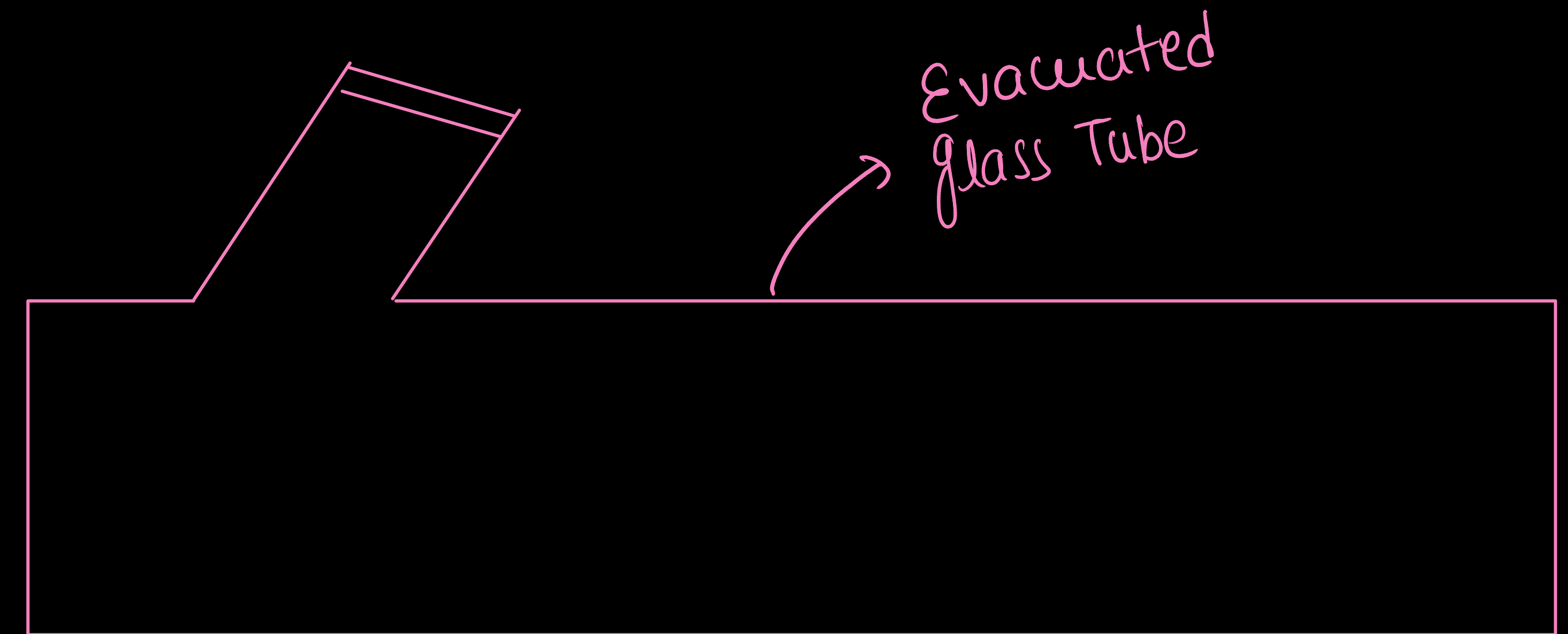
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Lenard's Observations :

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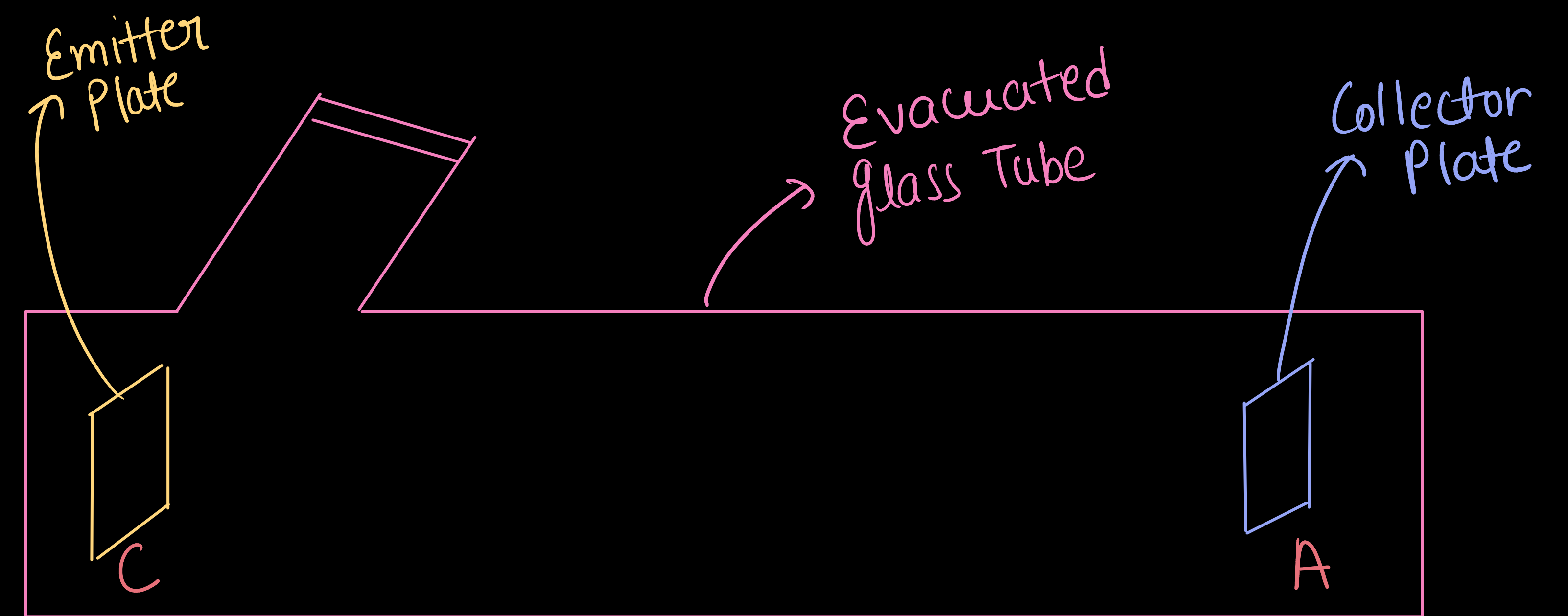
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Lenard's Observations :

* Lenard used an evacuated glass tube with:

Emitter plate (C)
Collector plate (A)



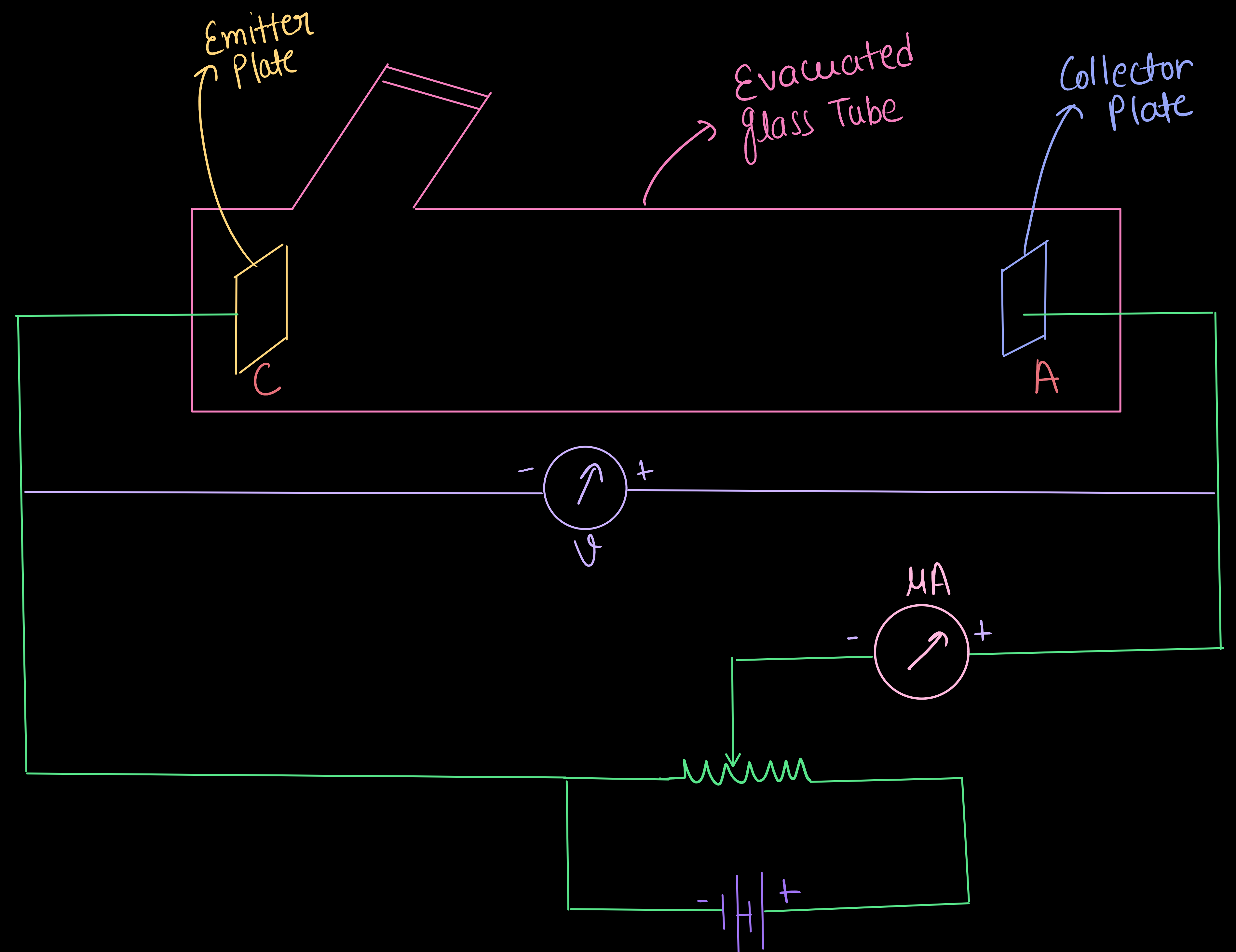
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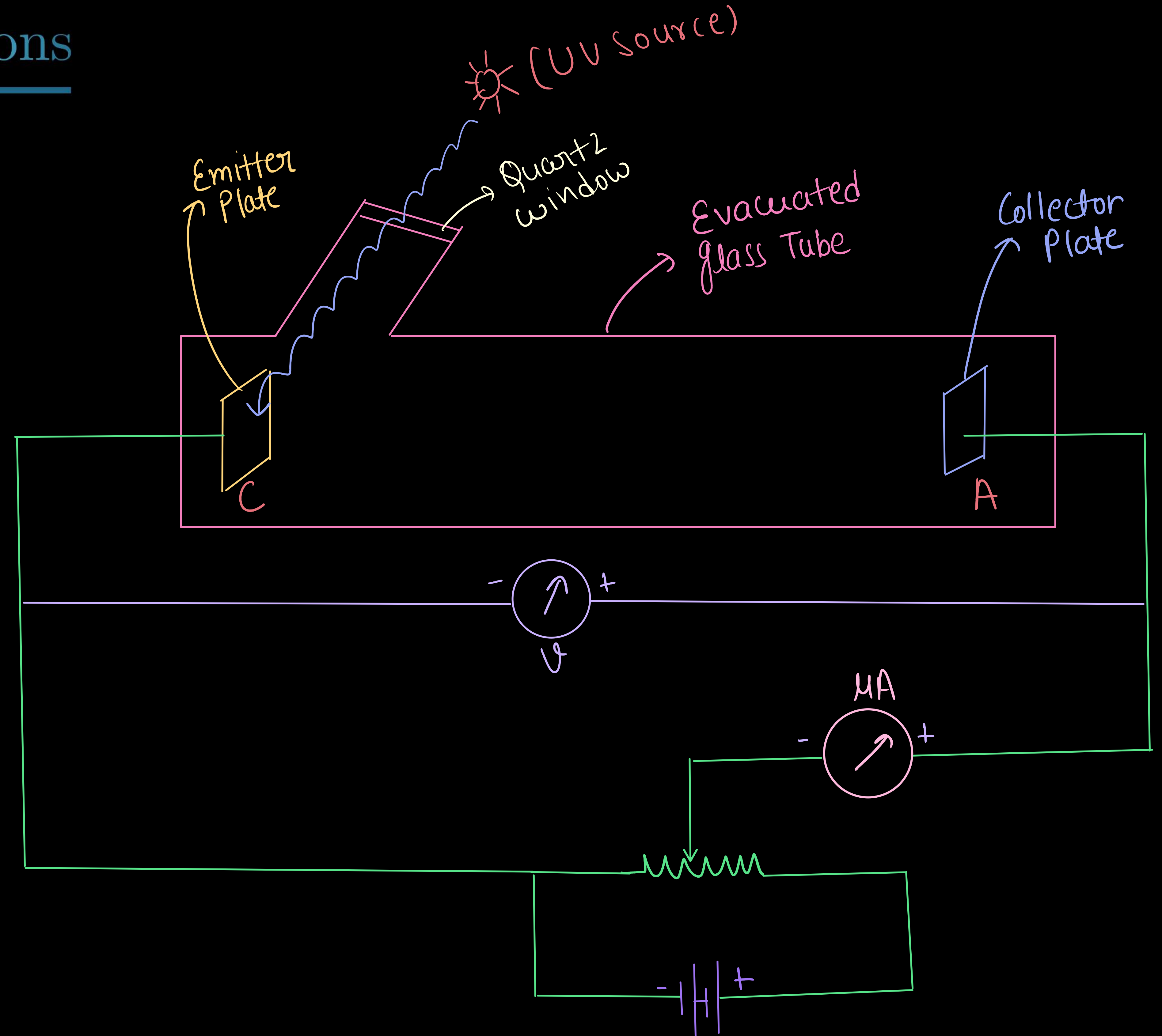
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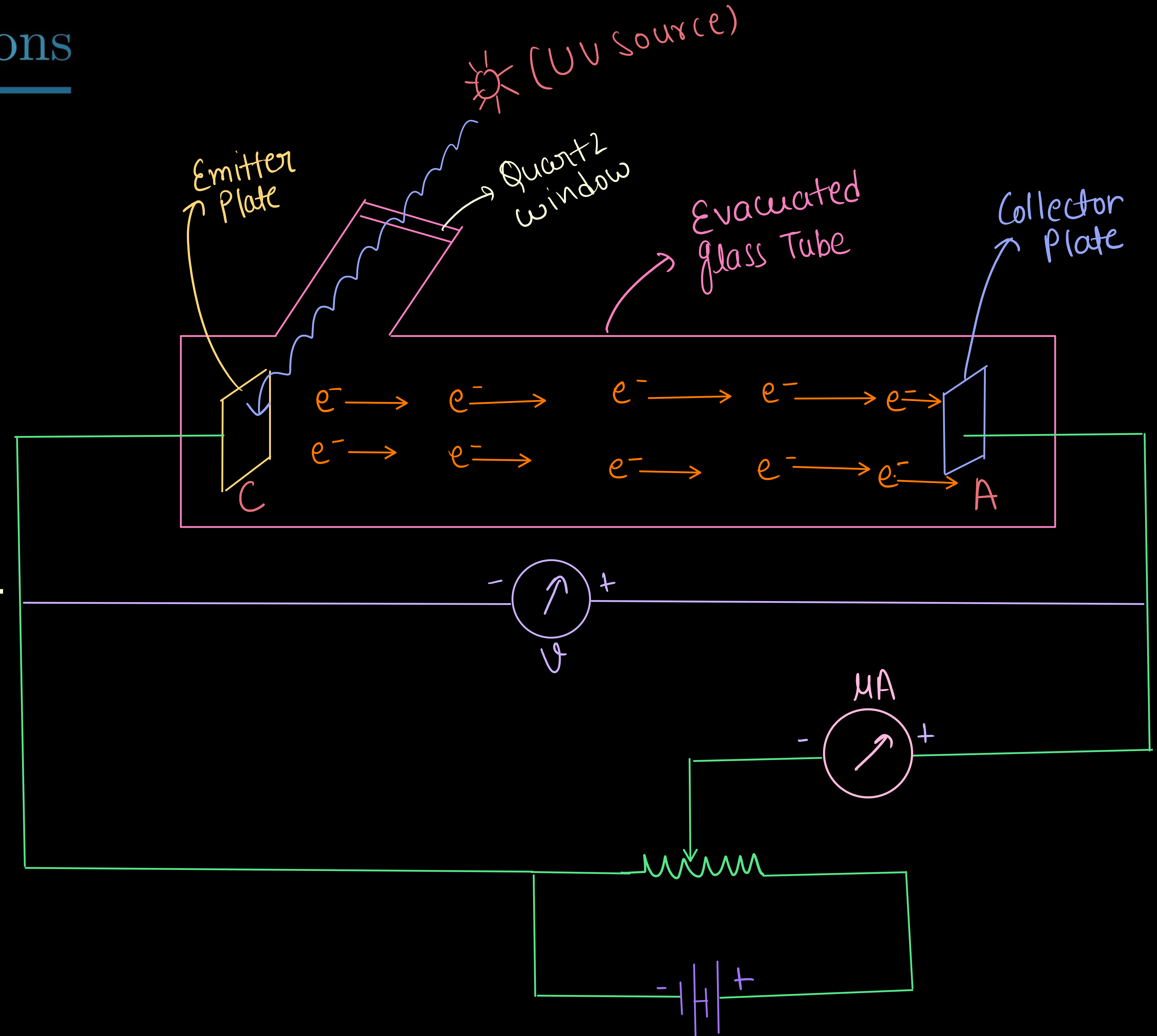
* Lenard used an evacuated glass tube with:

Emitter plate (C)
Collector plate (A)

* When UV light fell on plate C:

* Electrons were emitted and attracted to plate A.

* A current flowed in the external circuit (photoelectric current).



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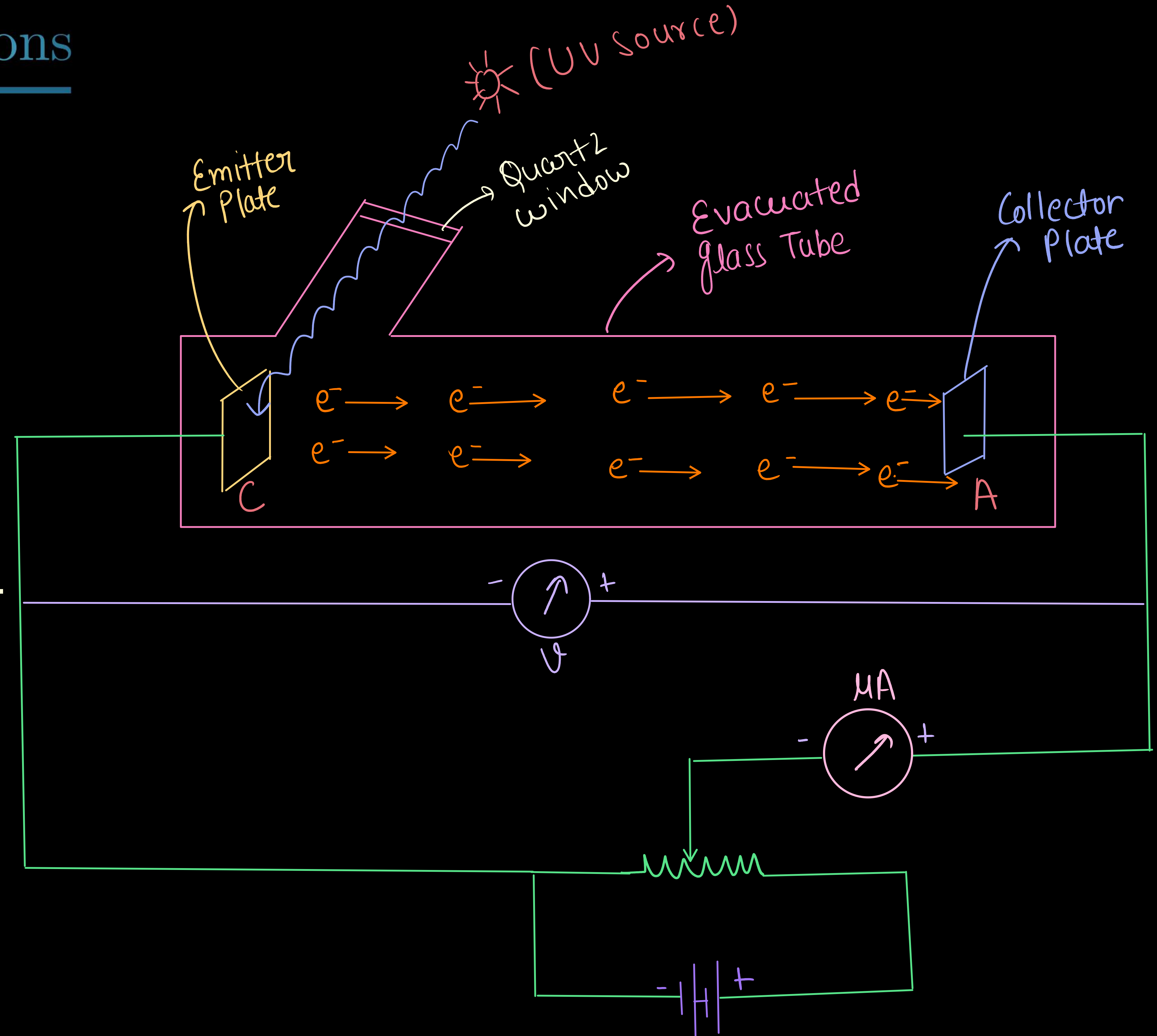
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Lenard's Observations :

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Emitter plate (C)
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- * When UV light fell on plate C:
- * Electrons were emitted and attracted to plate A.
- * A current flowed in the external circuit (photoelectric current).
- * Current stopped immediately when UV light was switched off.



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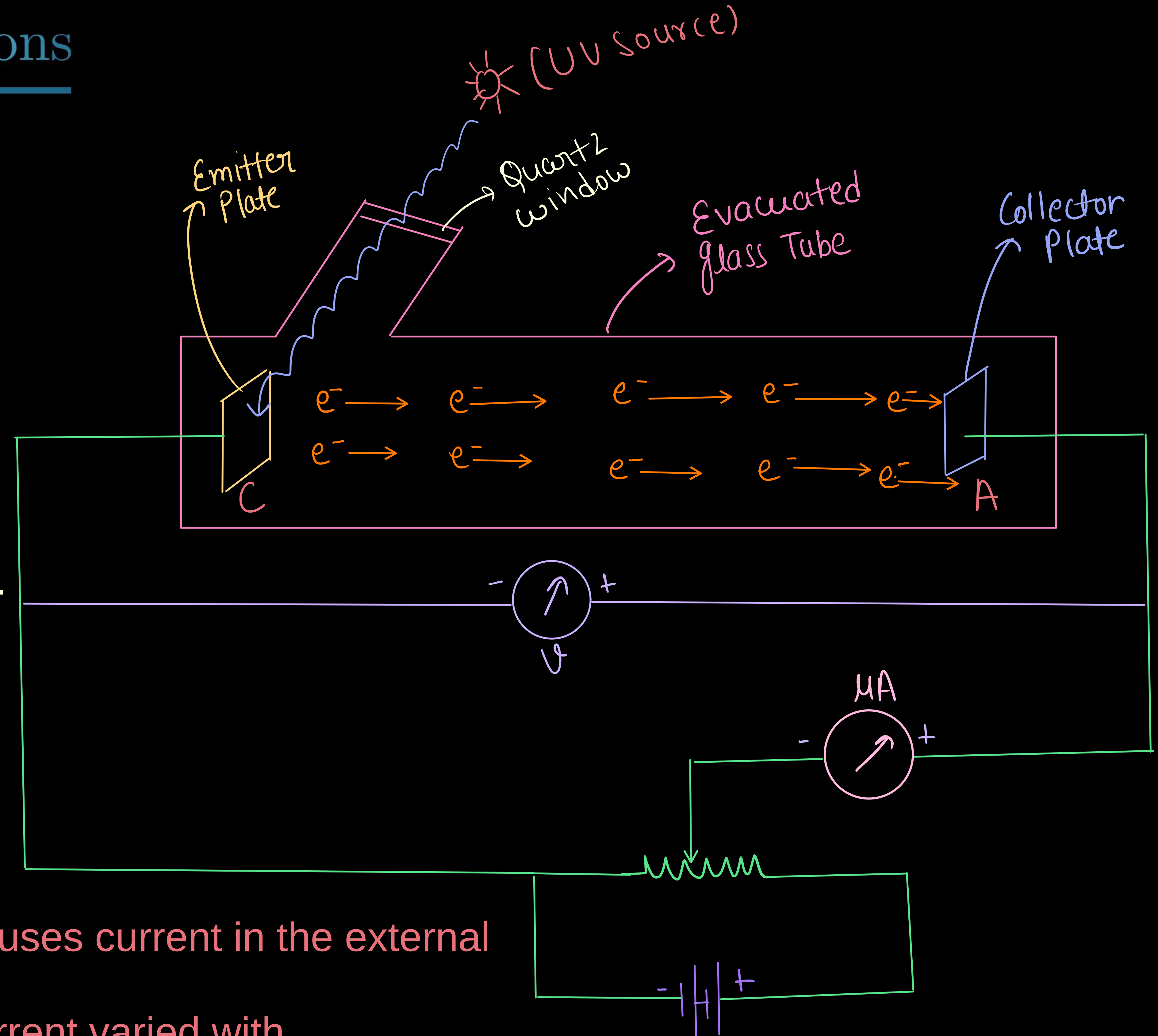
- * Electrons were emitted and attracted to plate A.

- * A current flowed in the external circuit (photoelectric current).

- * Current stopped immediately when UV light was switched off.

- * Thus, light falling on the surface of the emitter causes current in the external circuit.

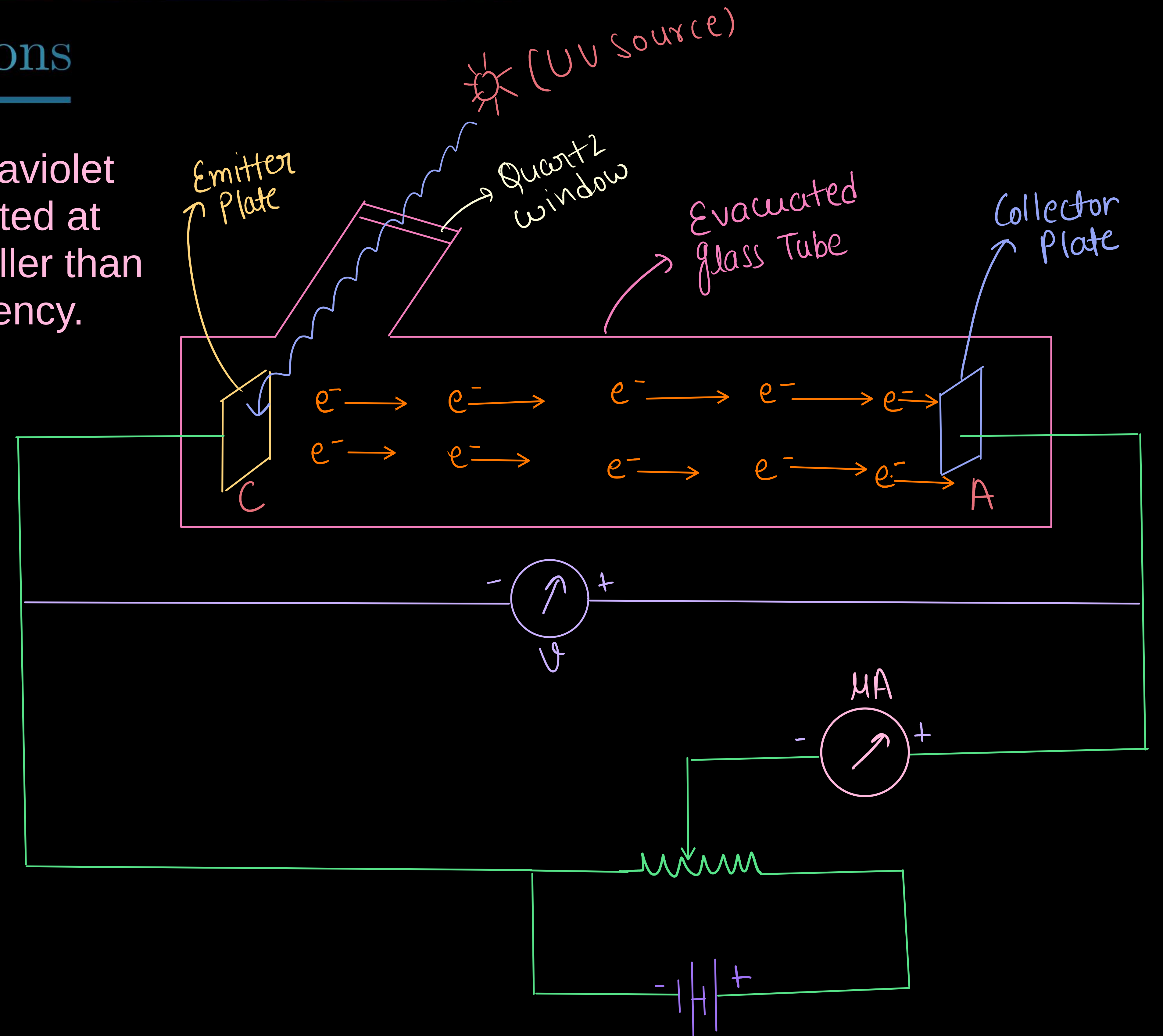
- * Hallwachs and Lenard studied how this photo current varied with collector plate potential, and with frequency and intensity of incident light.



PHOTOELECTRIC EFFECT

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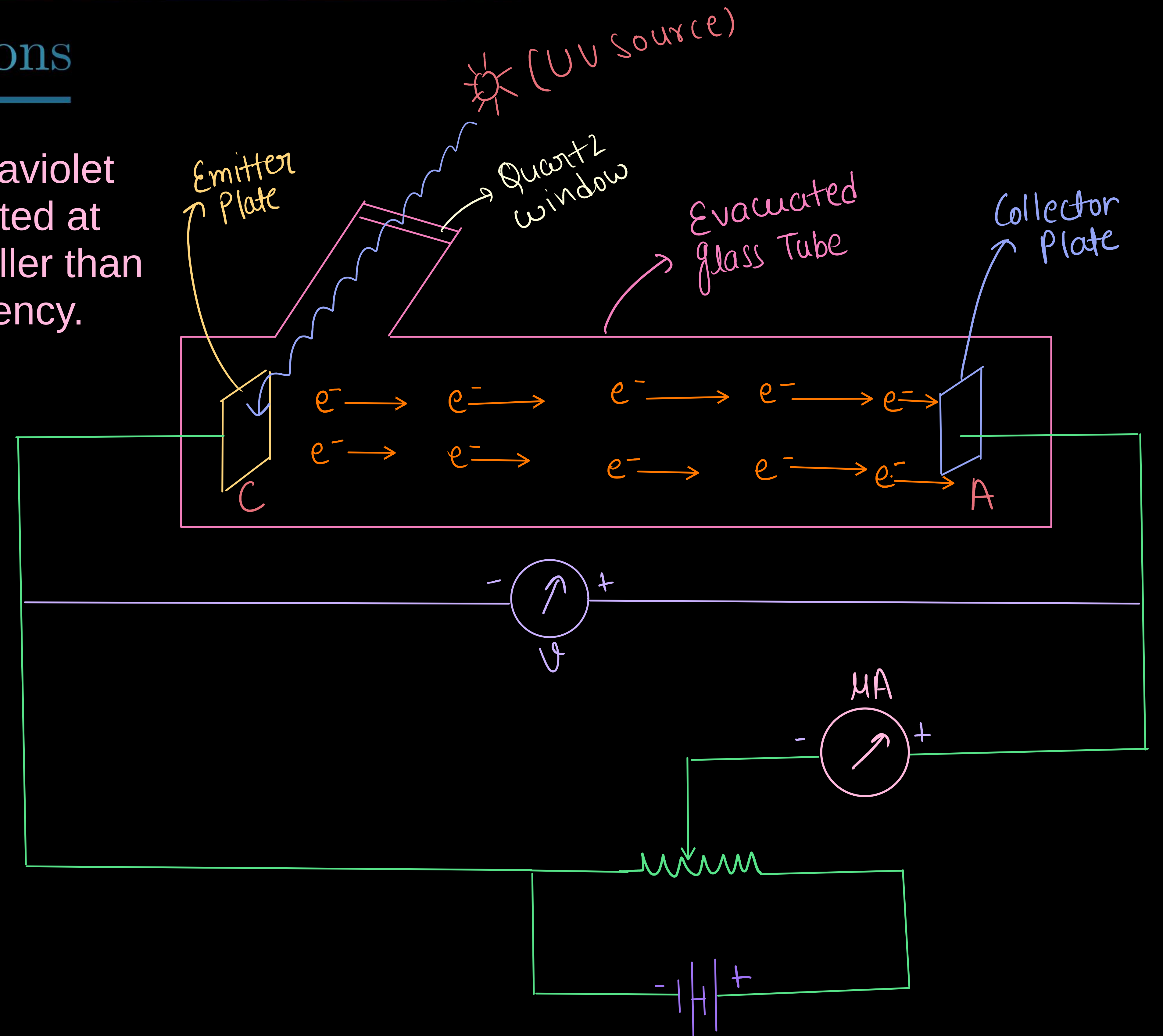
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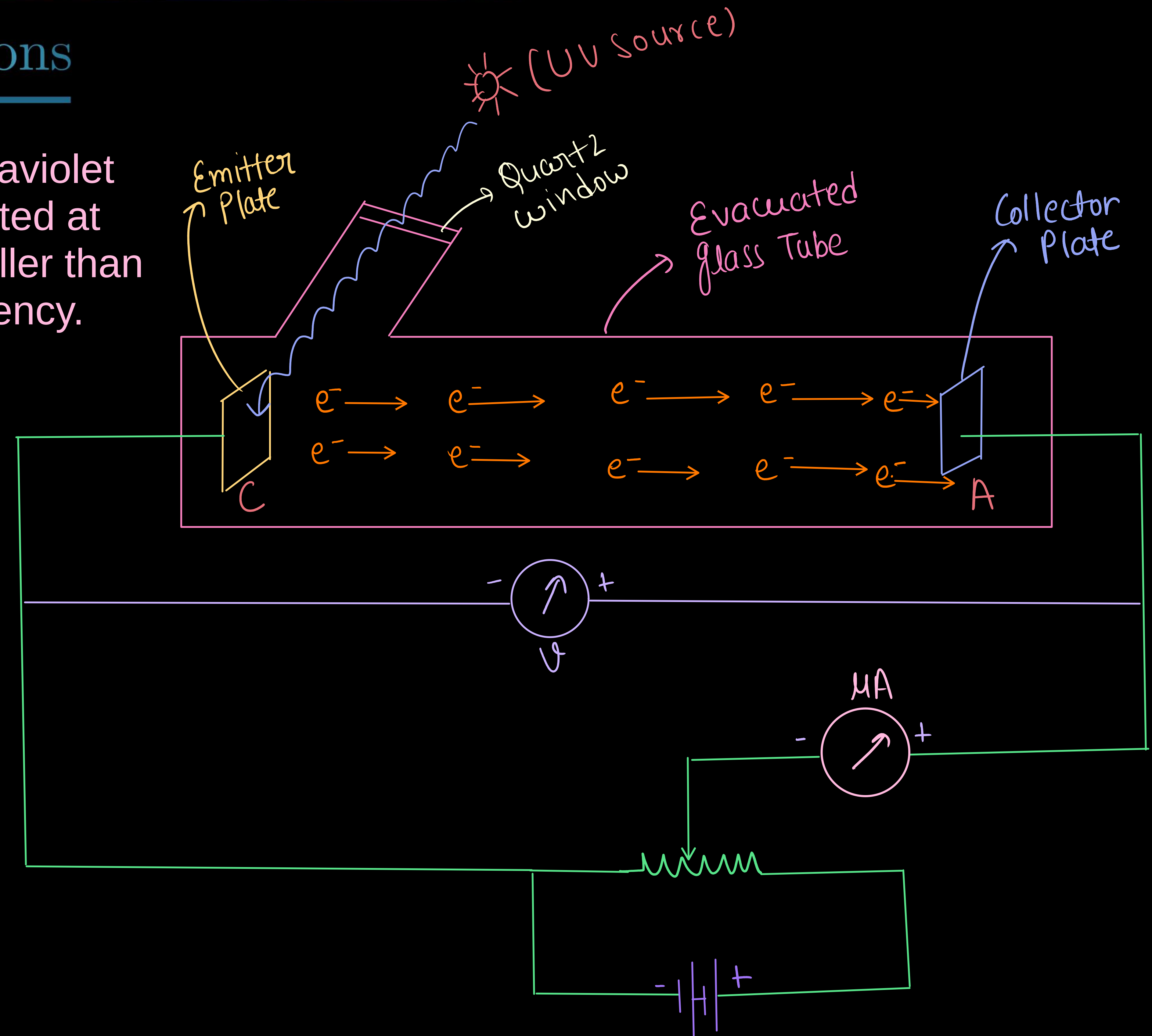
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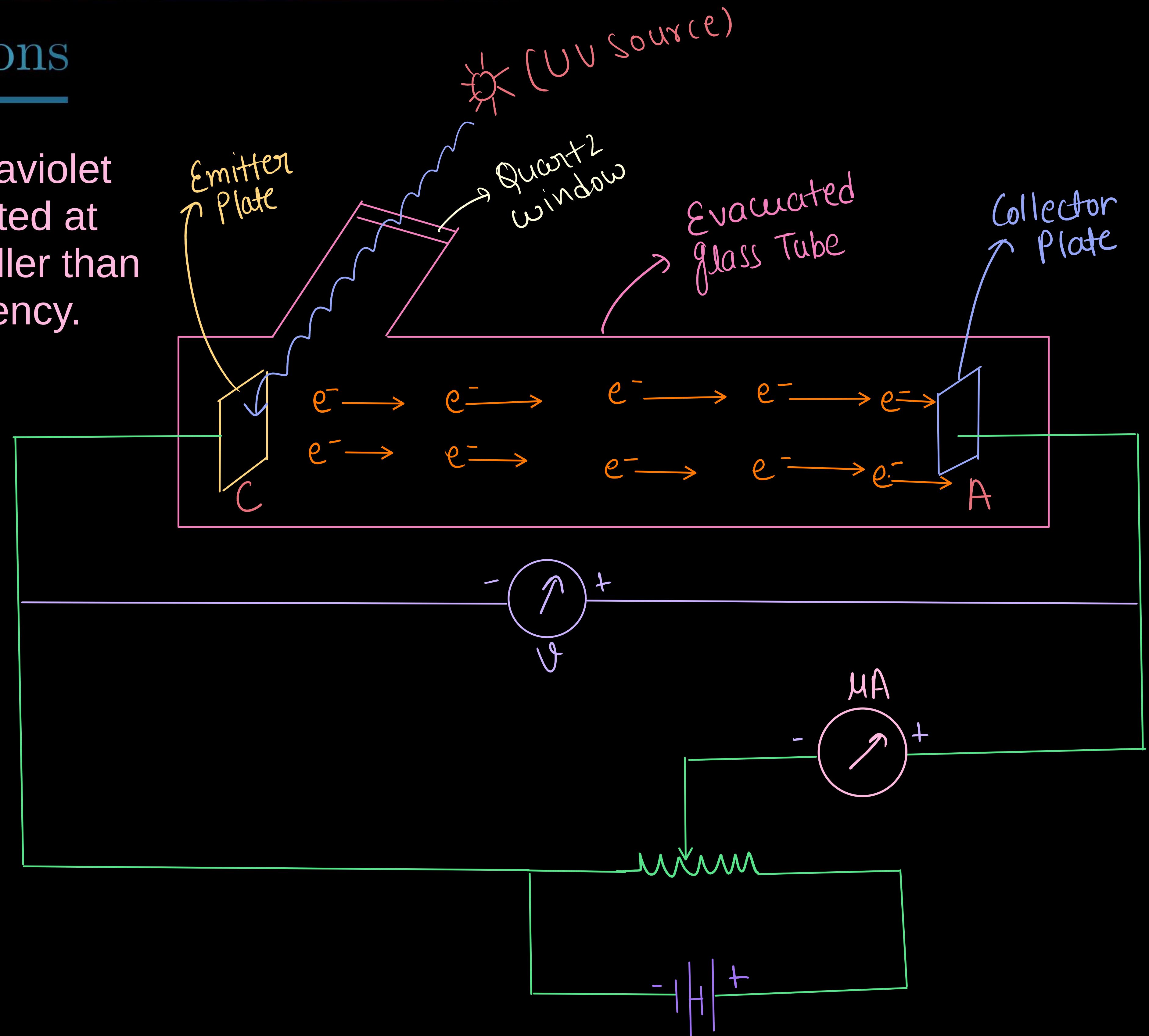
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- * It was found that certain metals like zinc, cadmium, magnesium, etc., responded only to ultraviolet light, having short wavelength, to cause electron emission from the surface.



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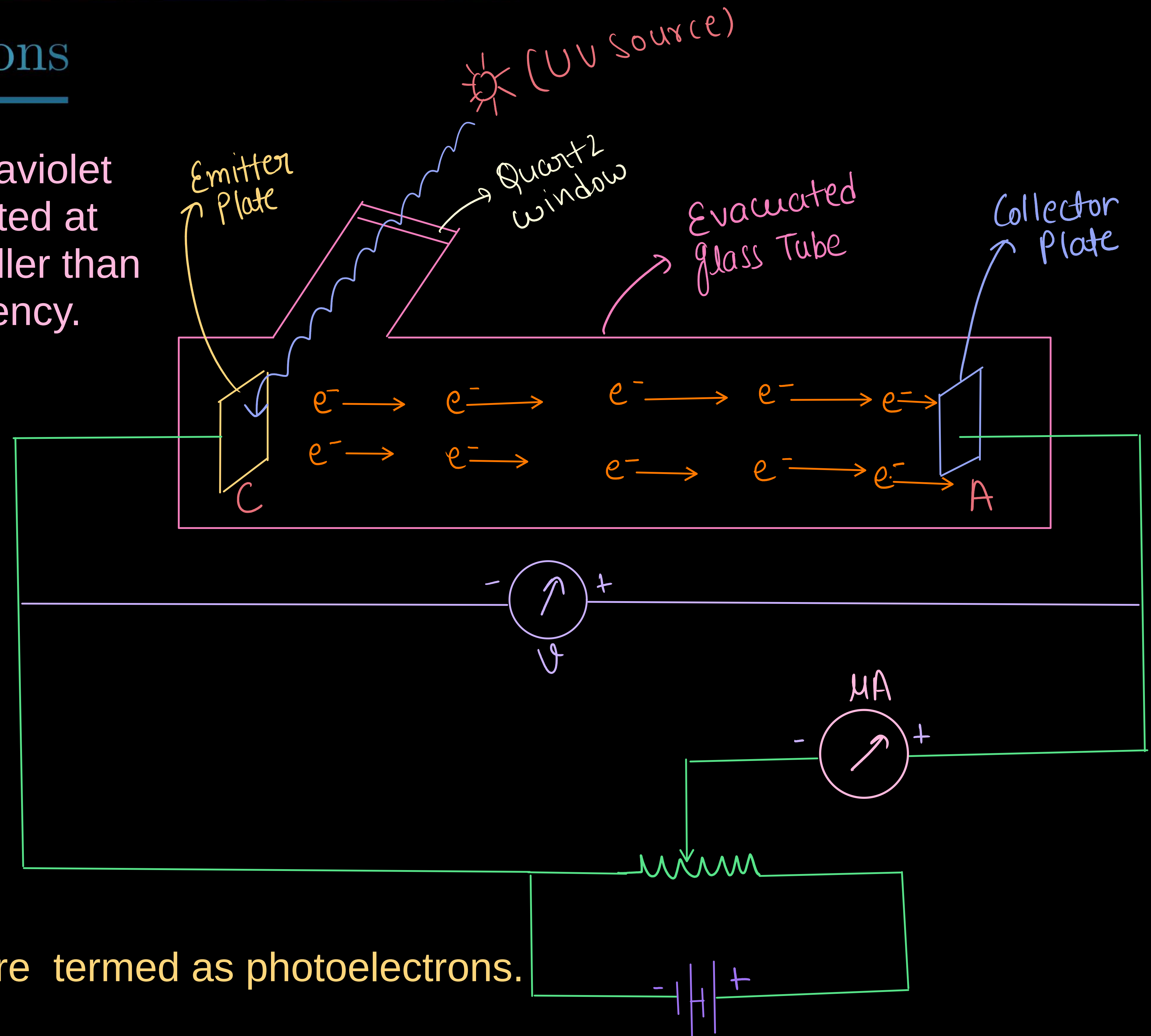
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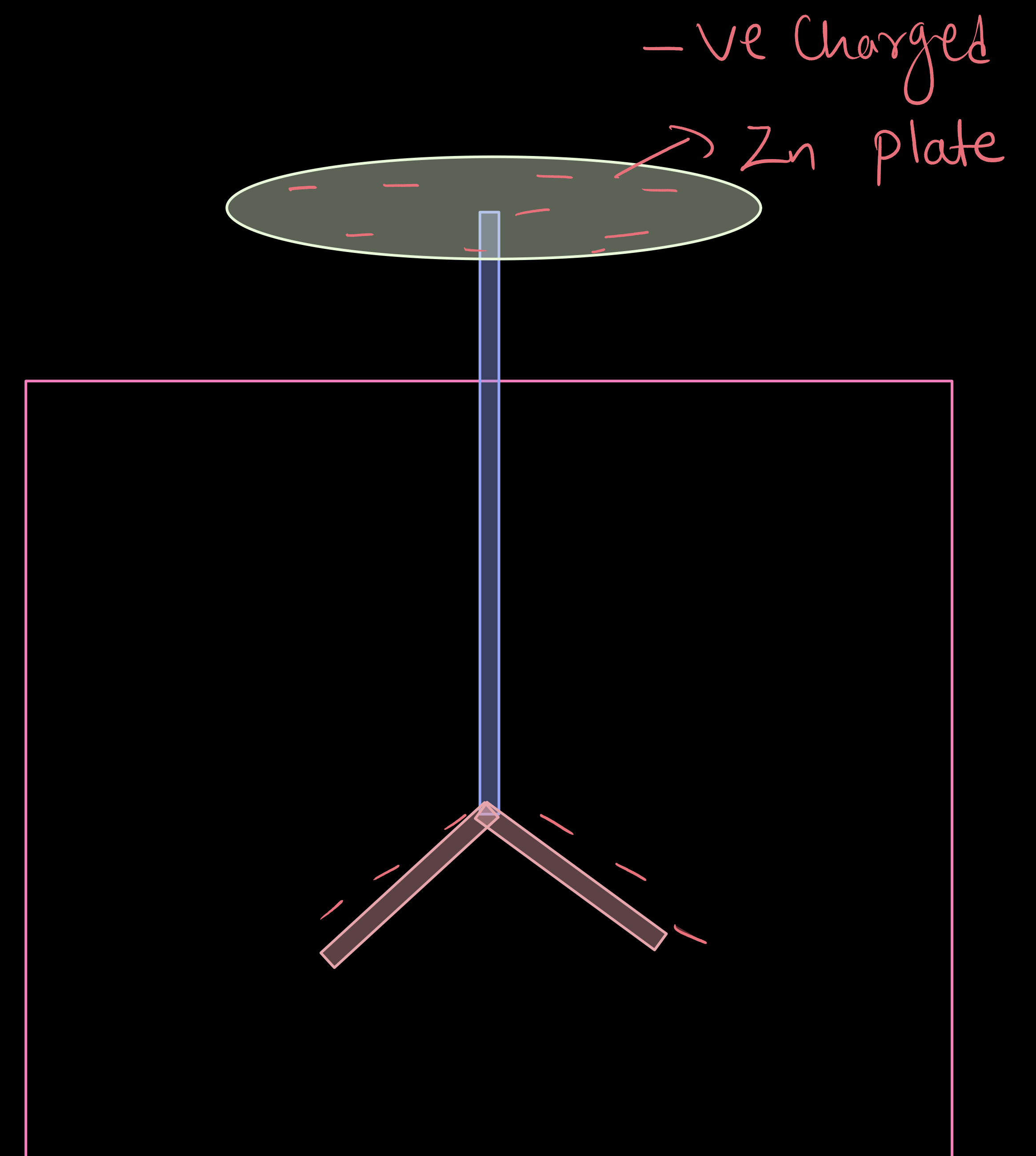
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- * It was found that certain metals like zinc, cadmium, magnesium, etc., responded only to ultraviolet light, having short wavelength, to cause electron emission from the surface.
- * However, some alkali metals such as lithium, sodium, potassium, caesium and rubidium were sensitive even to visible light.
- * After the discovery of electrons, these electrons were termed as photoelectrons. The phenomenon is called photoelectric effect.



PHOTOELECTRIC EFFECT

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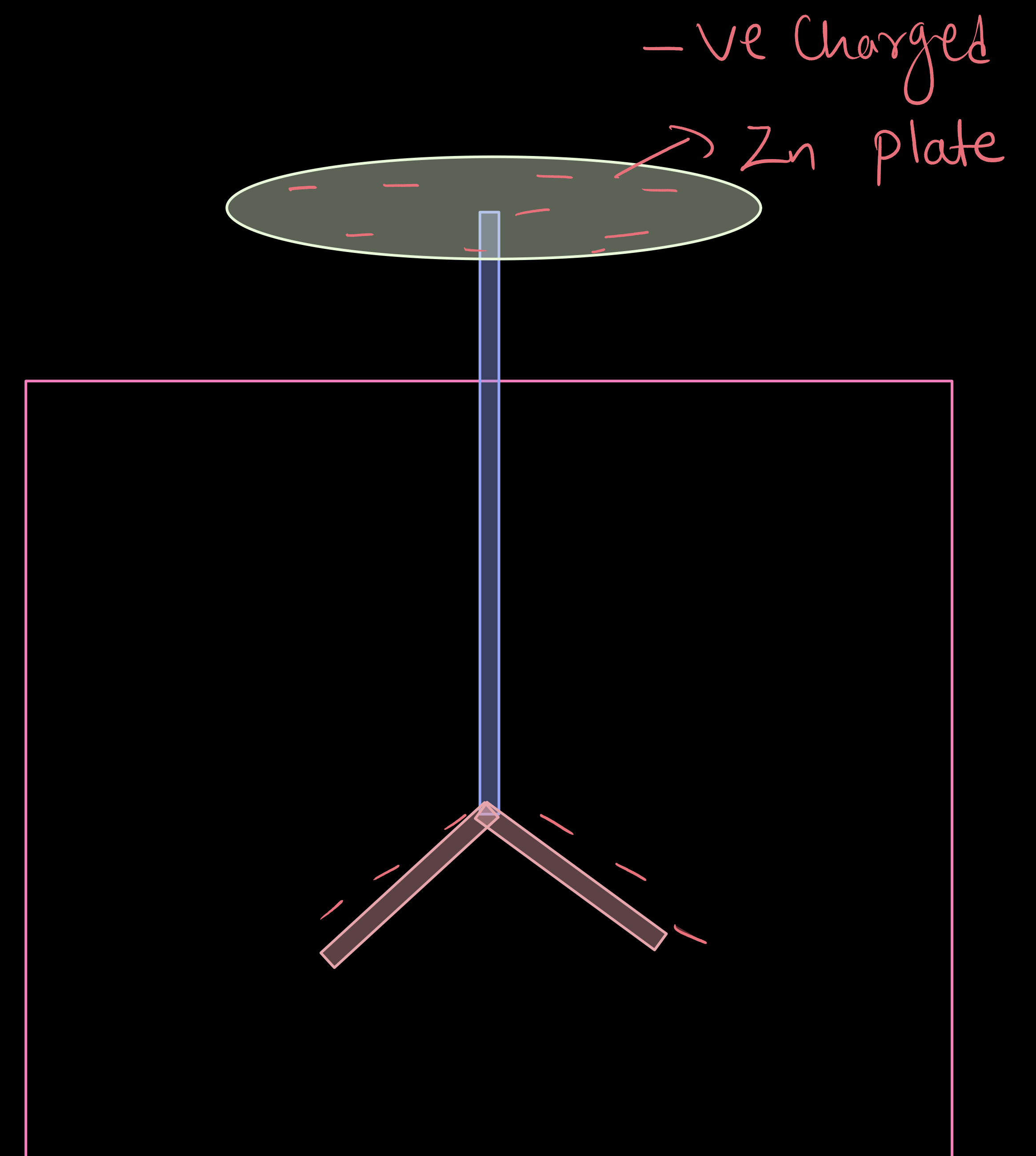
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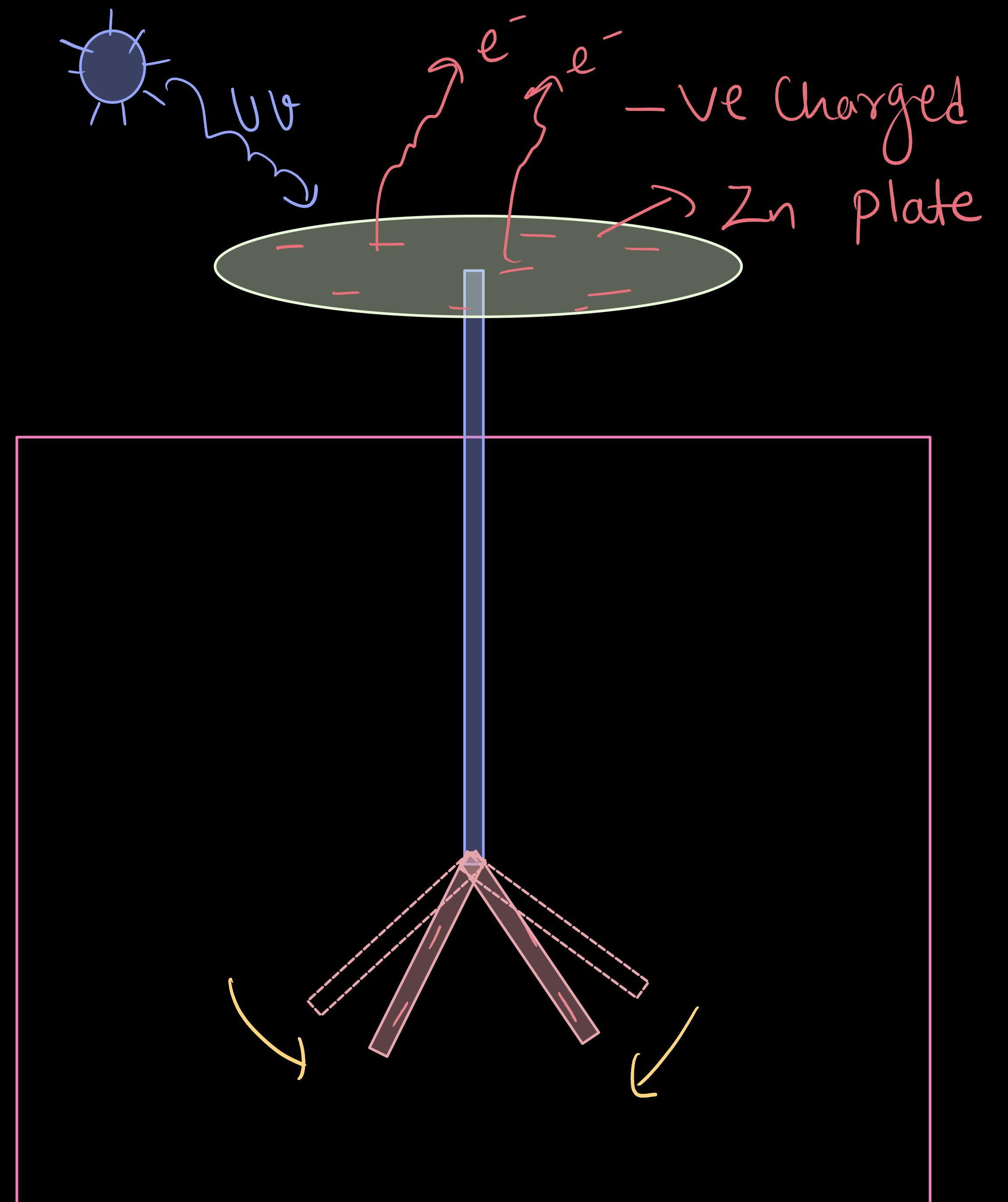
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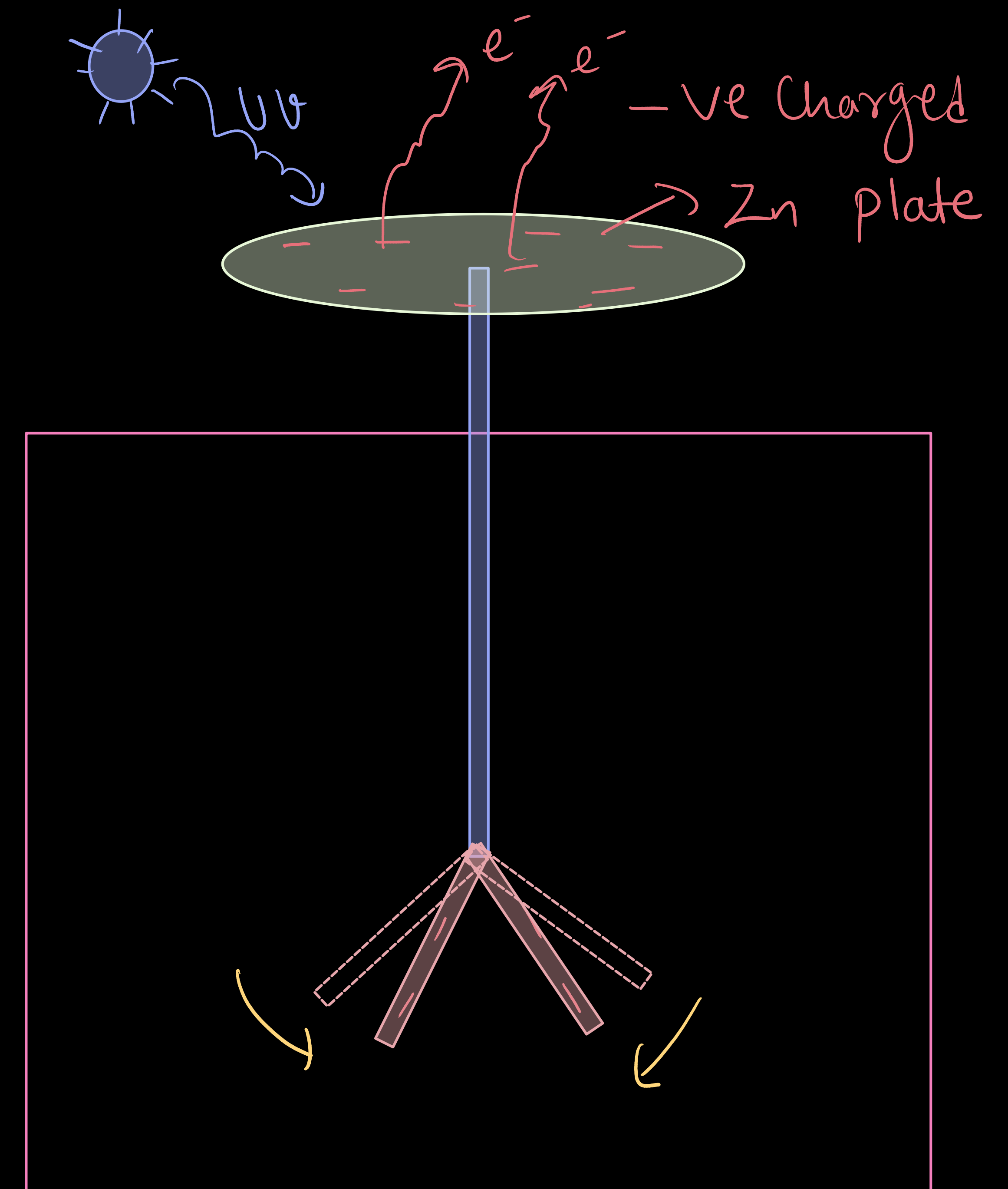
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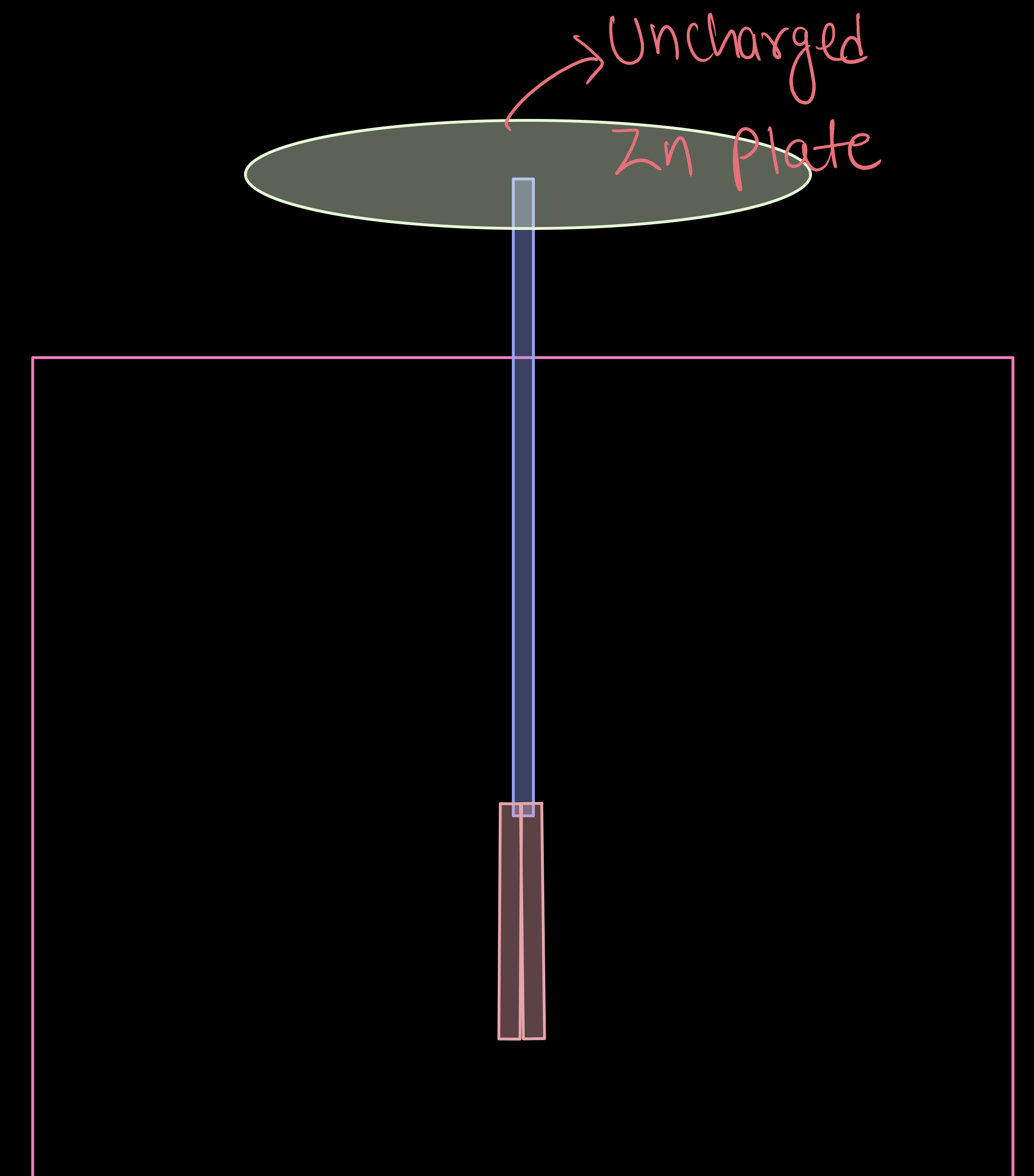


PHOTOELECTRIC EFFECT

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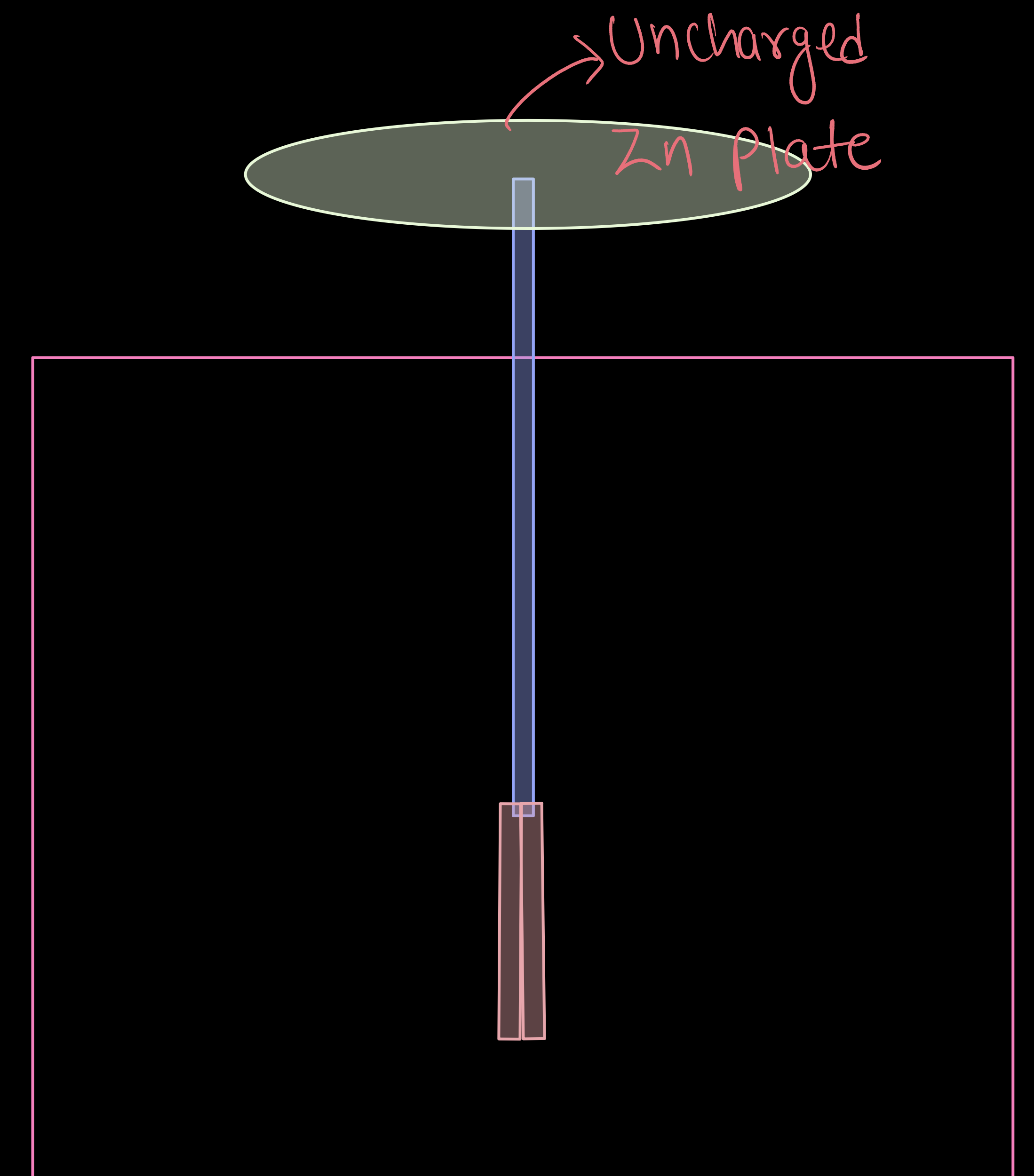
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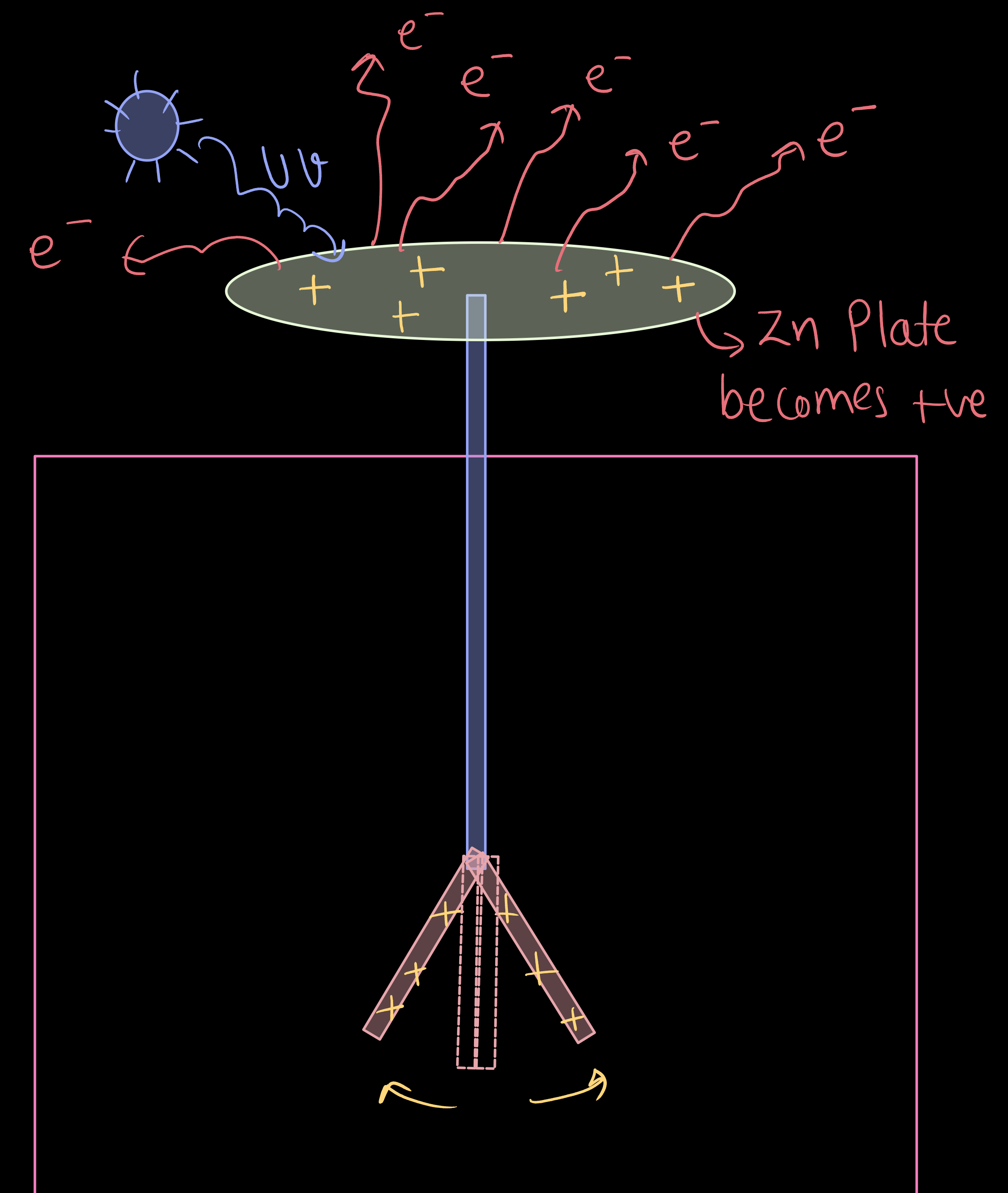


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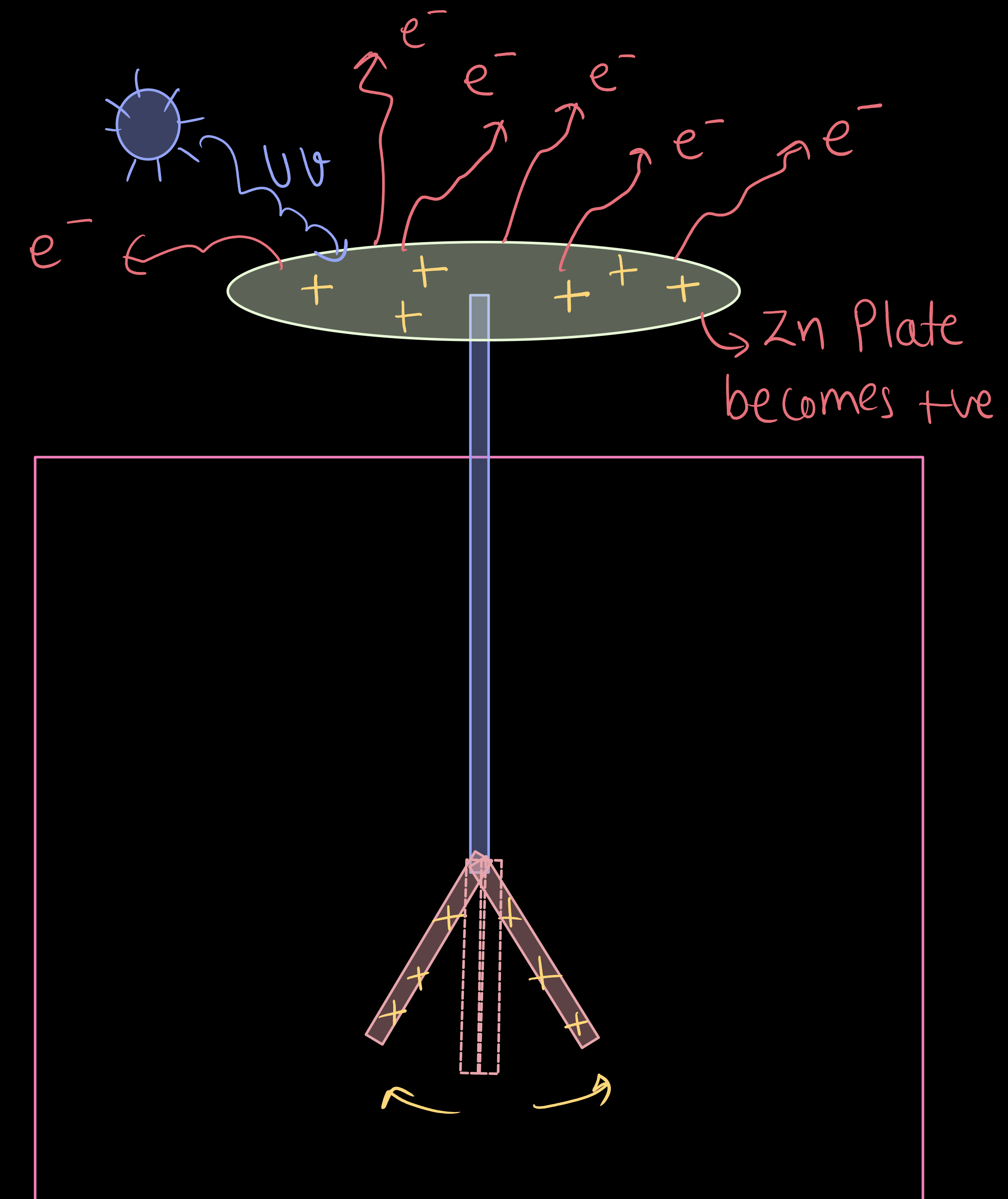
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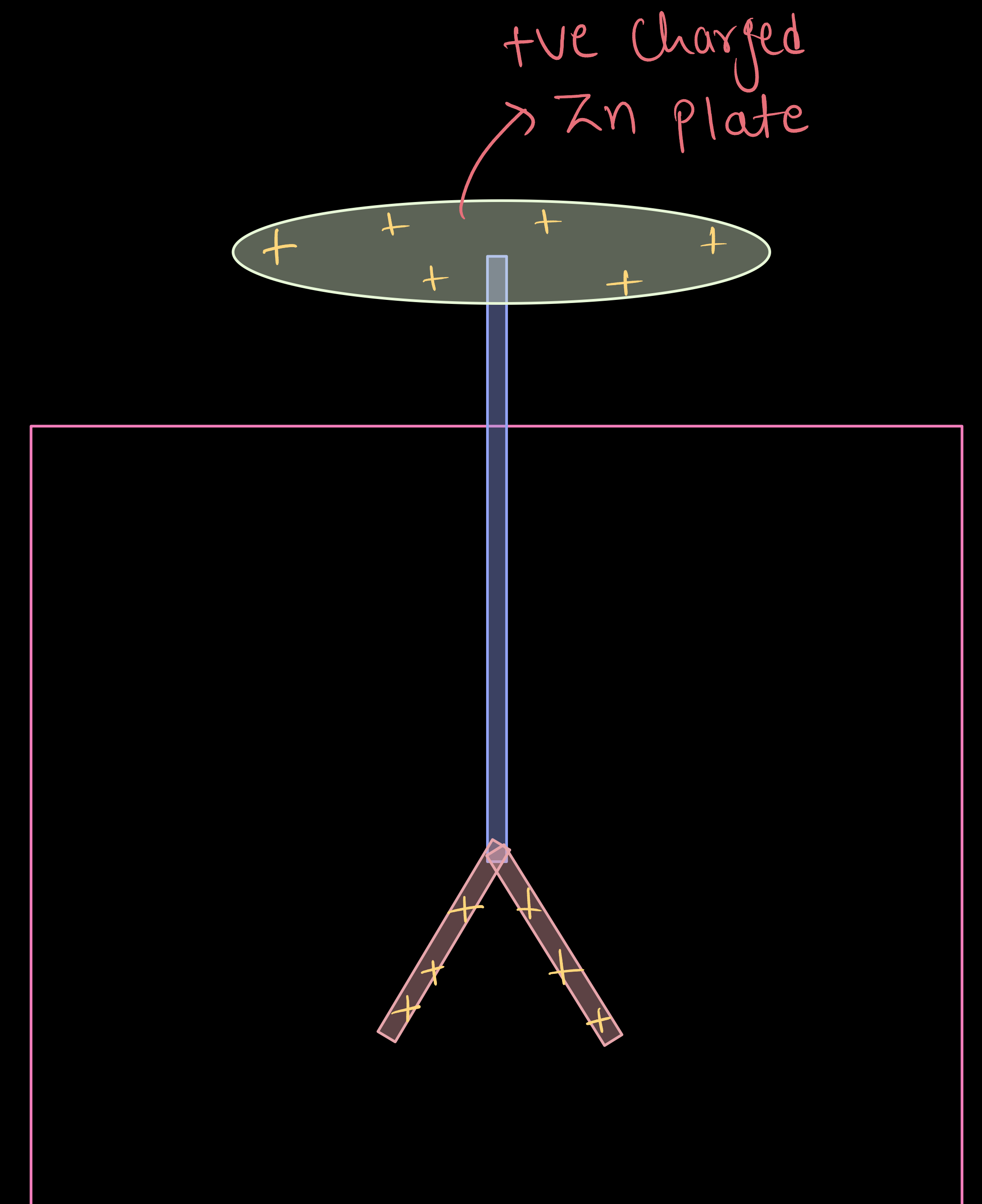
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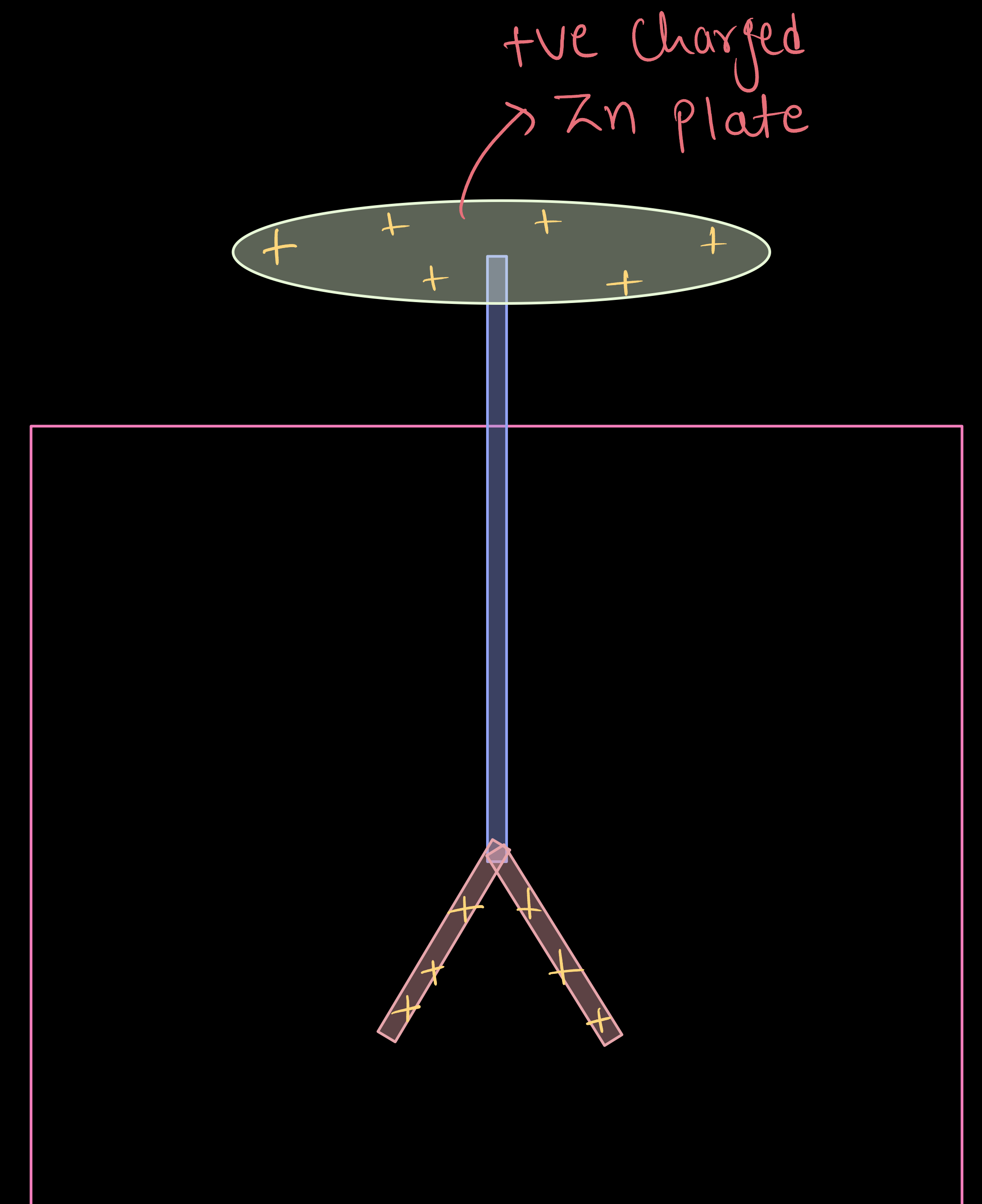
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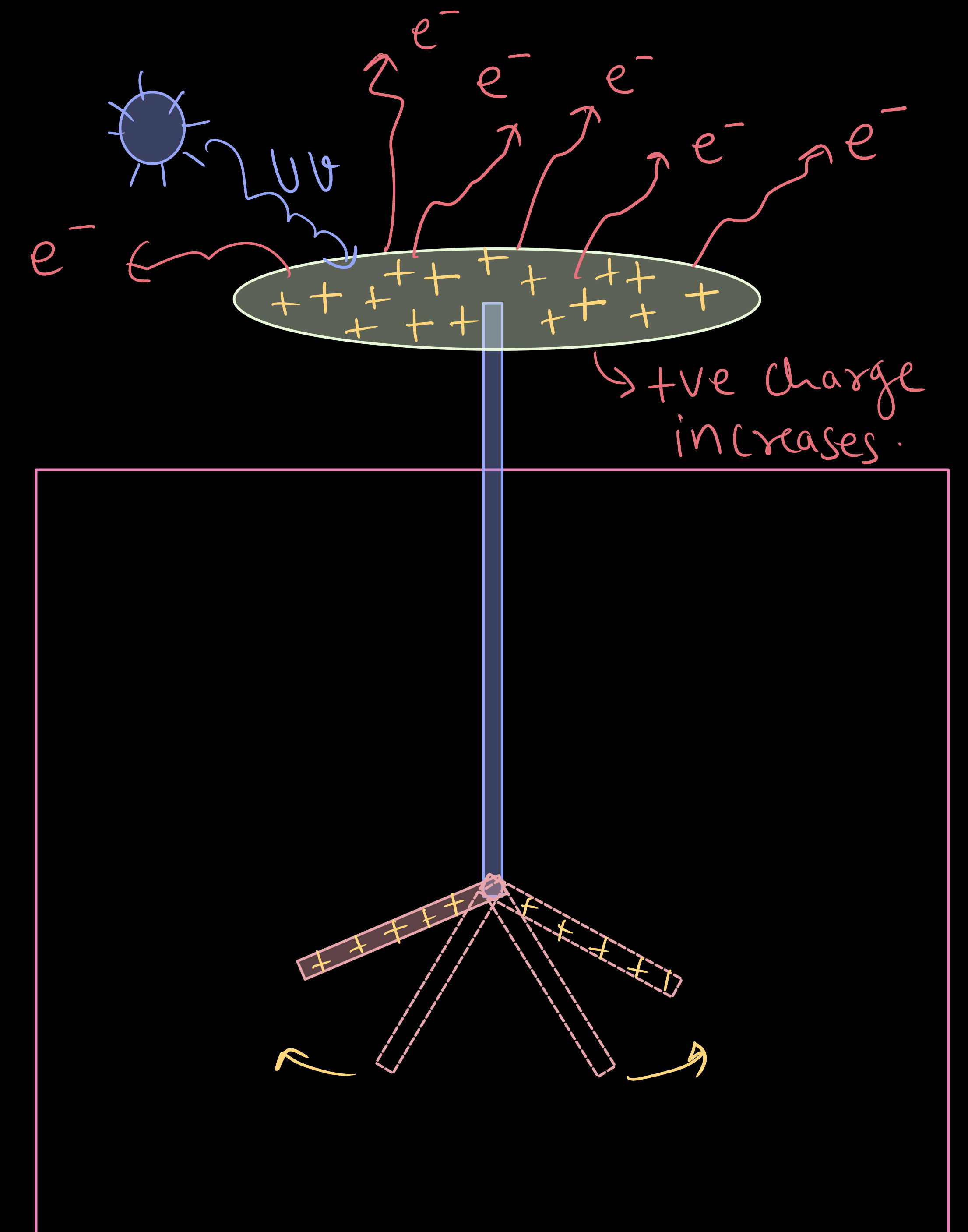
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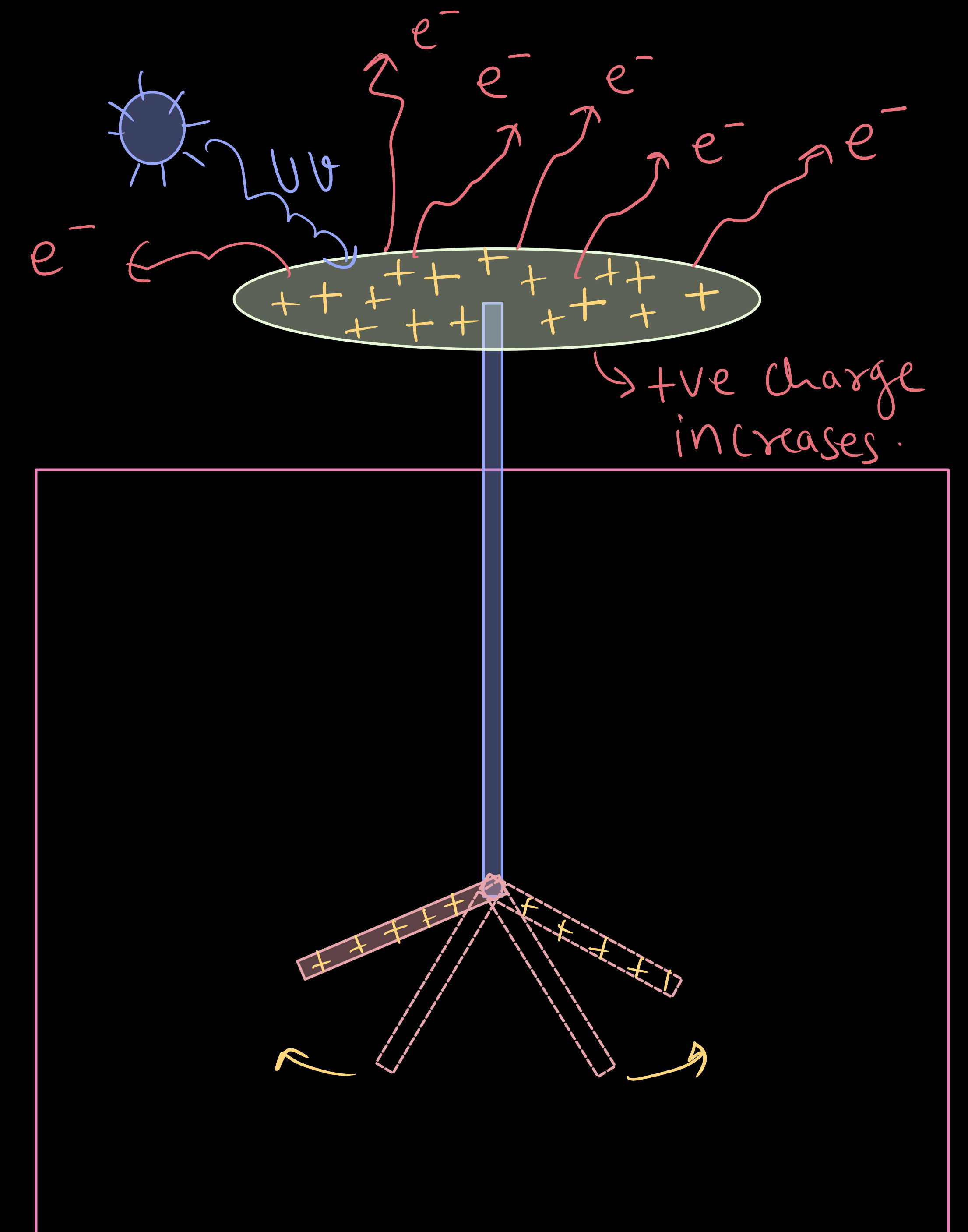
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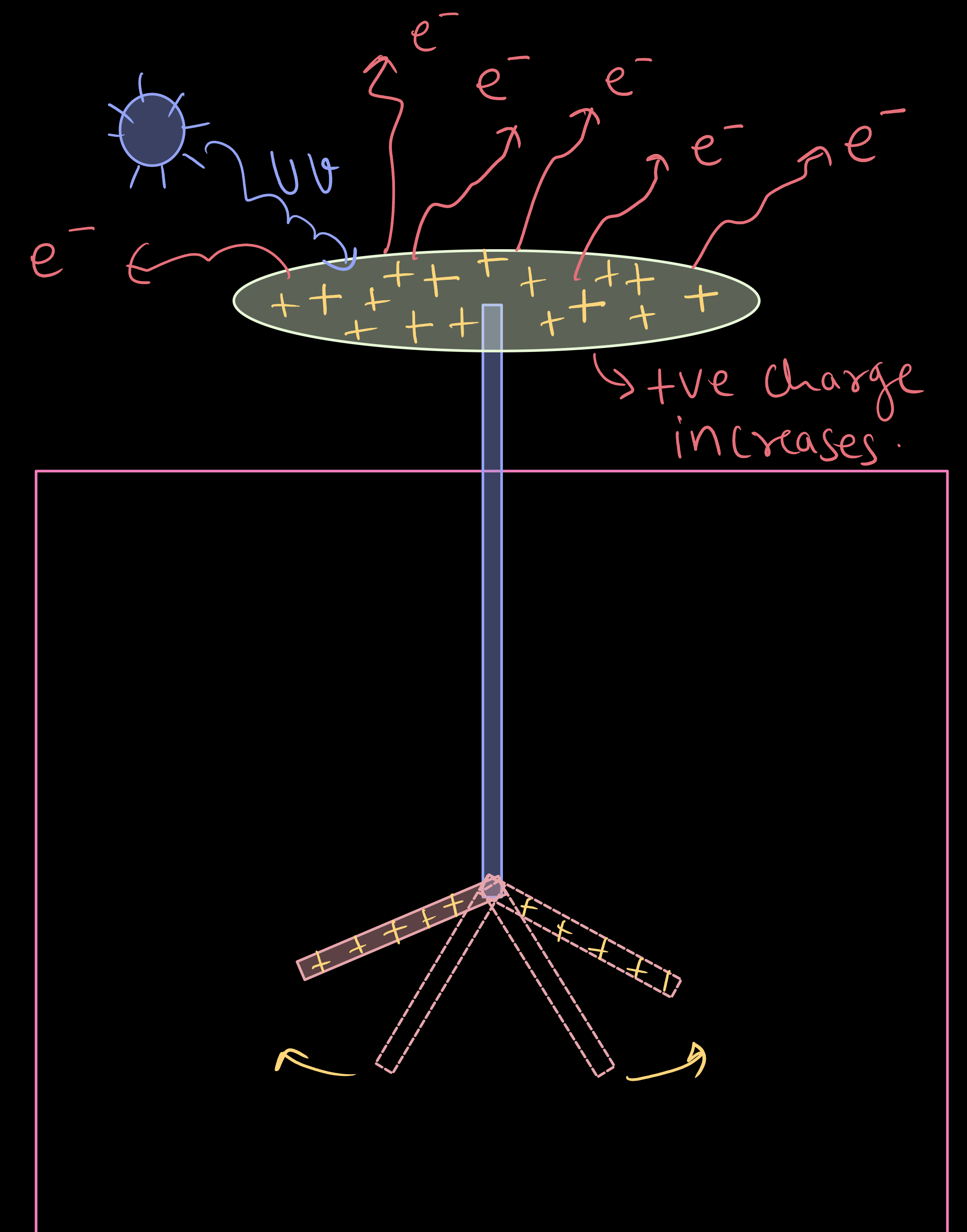
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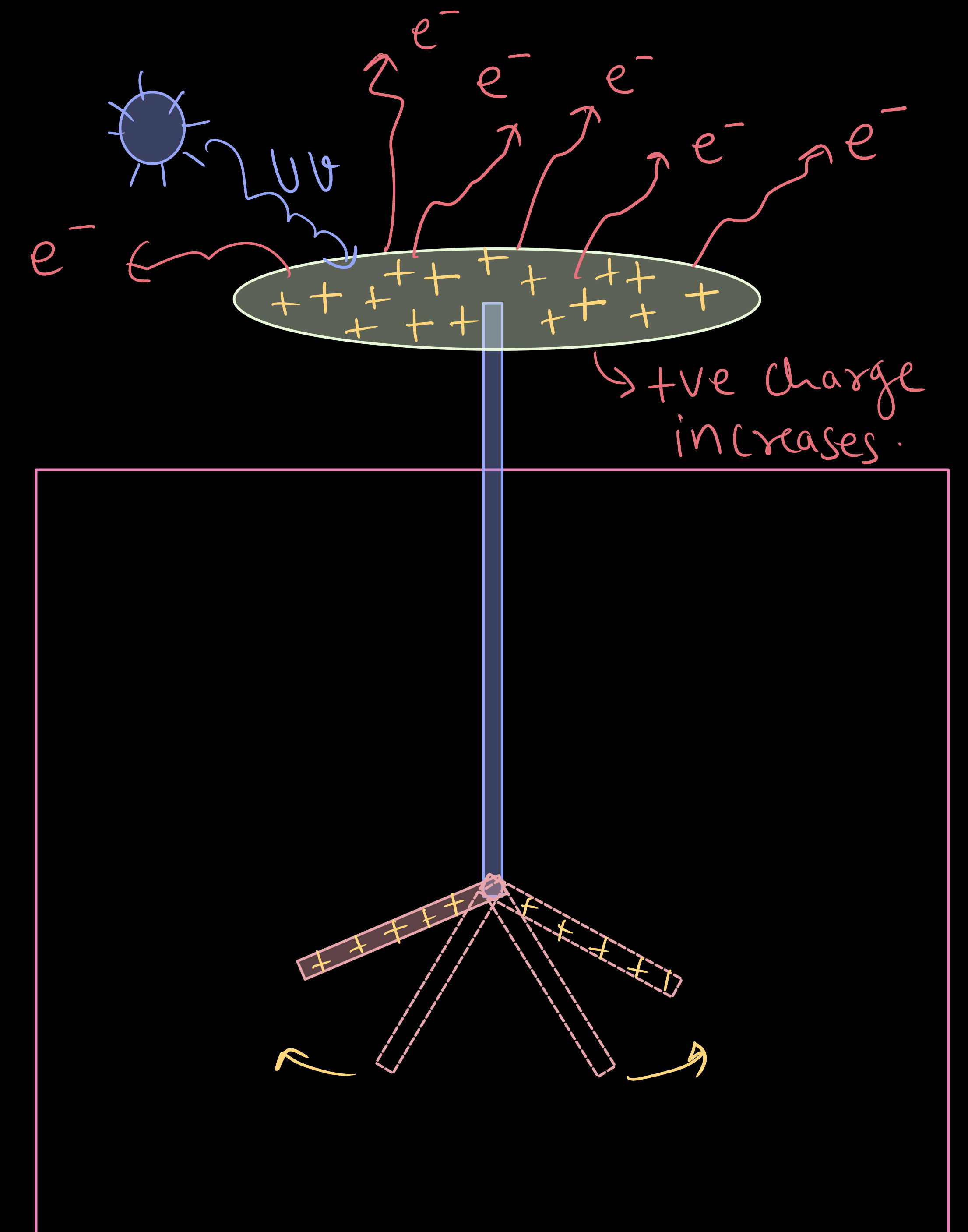
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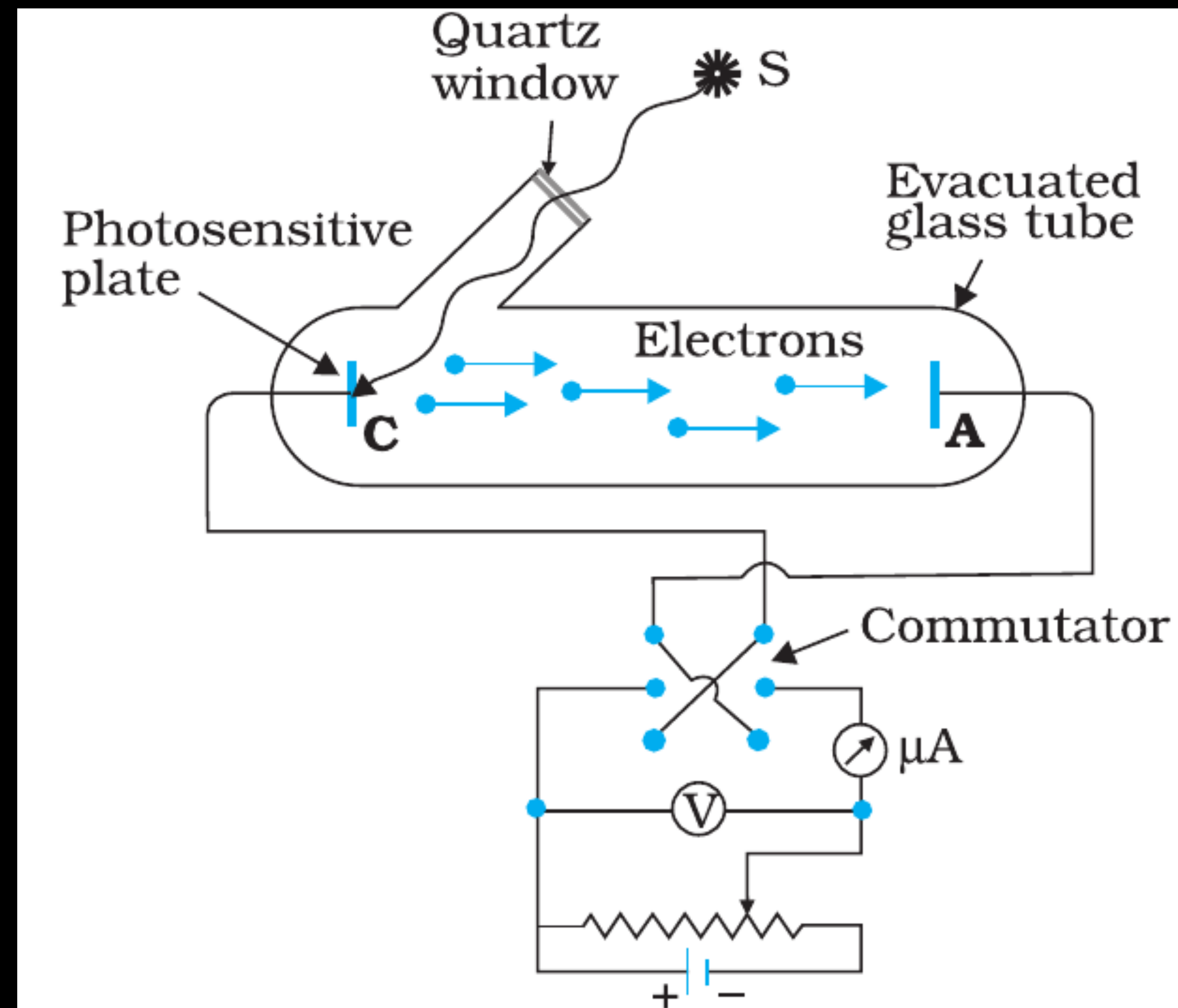


FIGURE 11.1 Experimental arrangement for study of photoelectric effect.

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- * The arrangement used for the experimental study of the photoelectric effect consists of an evacuated glass/quartz tube having a thin photosensitive plate C and another metal plate A.
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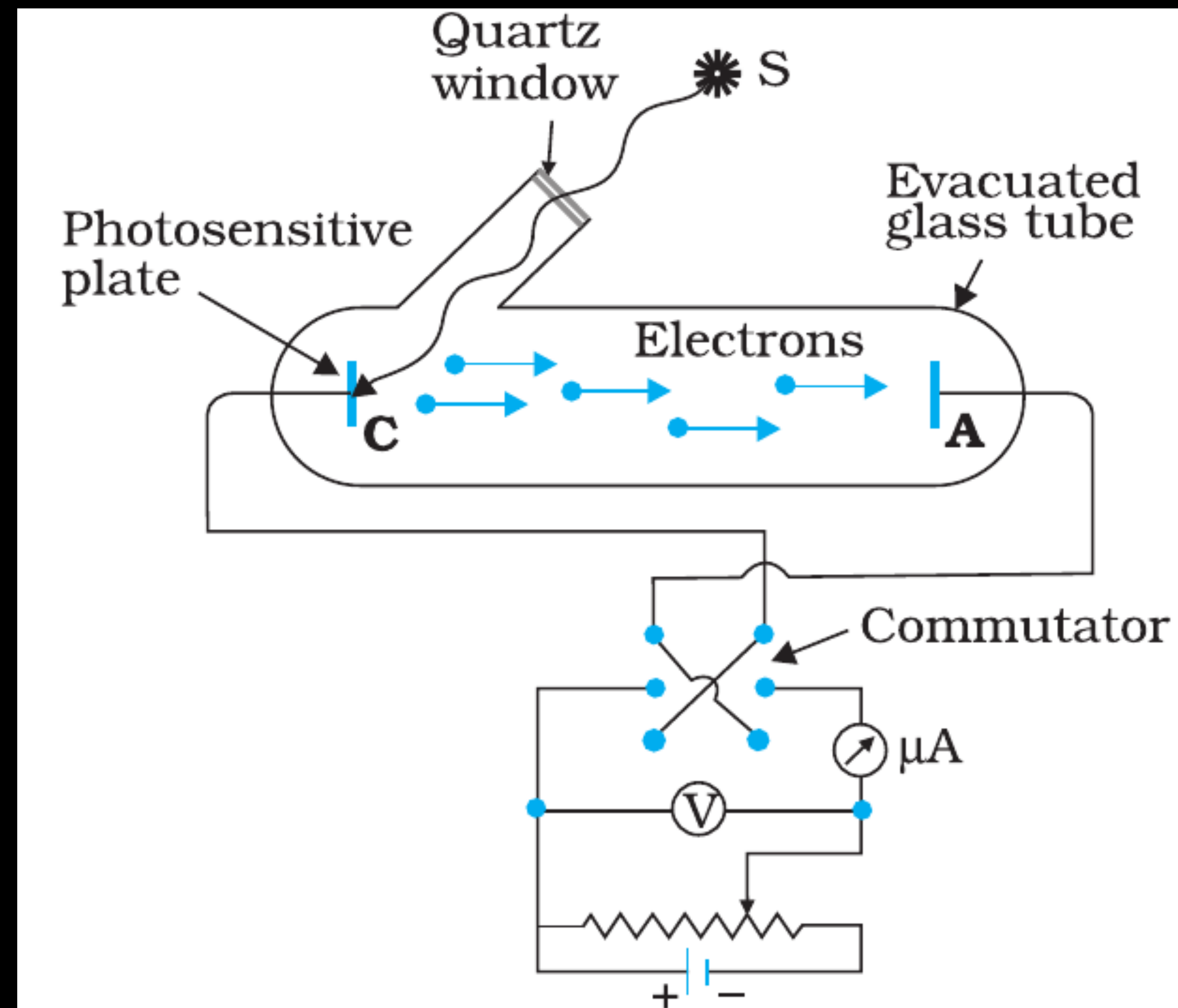


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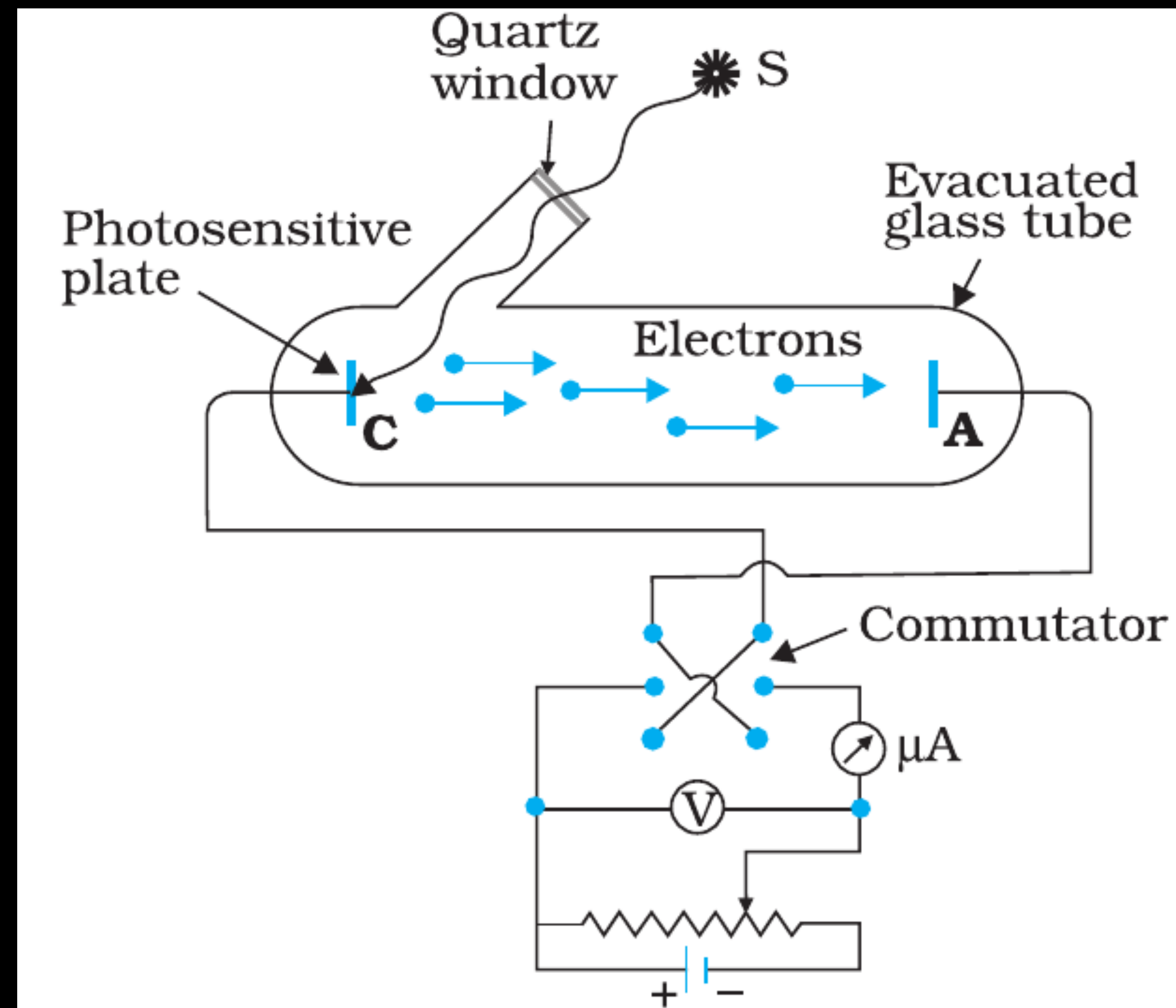


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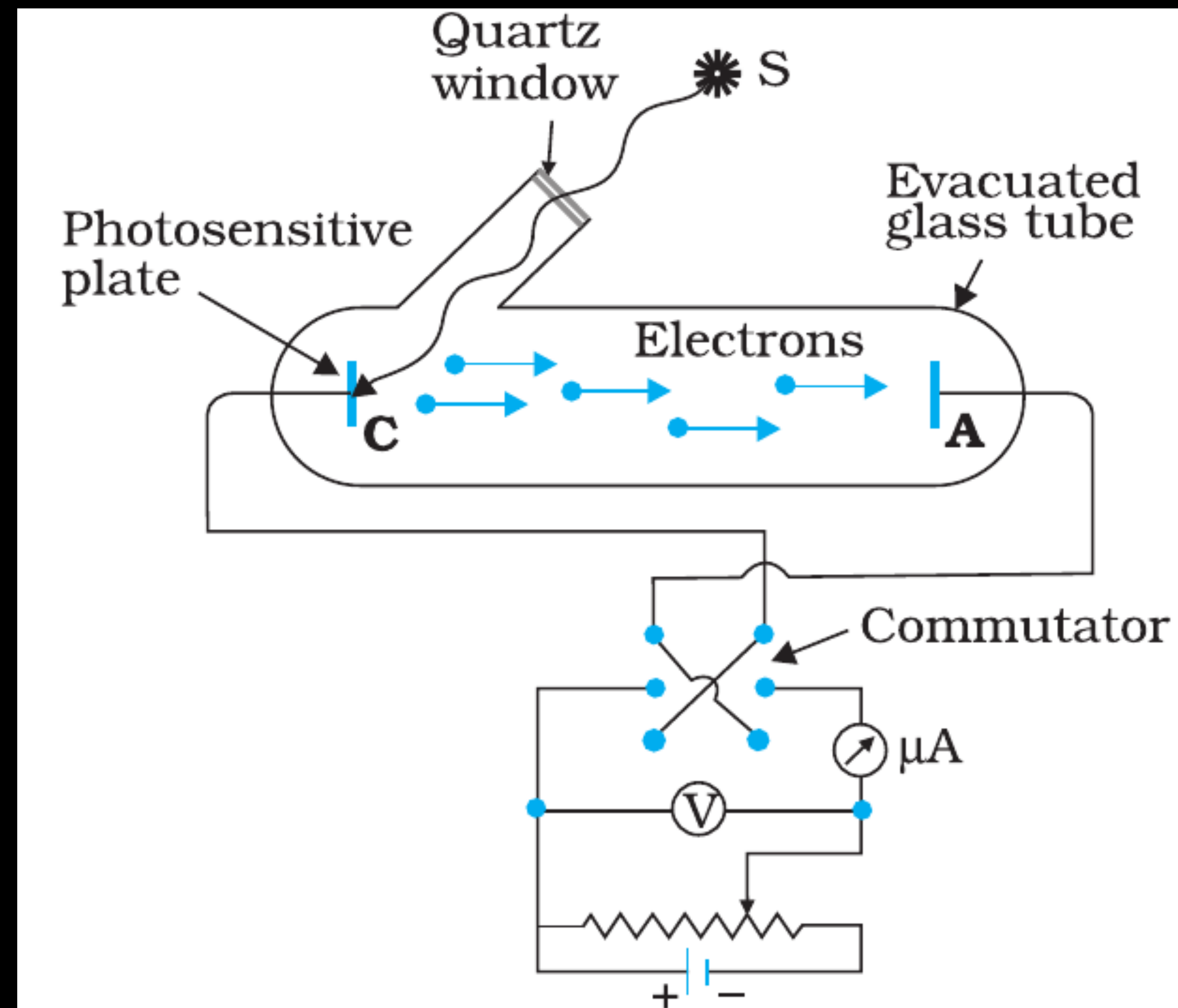


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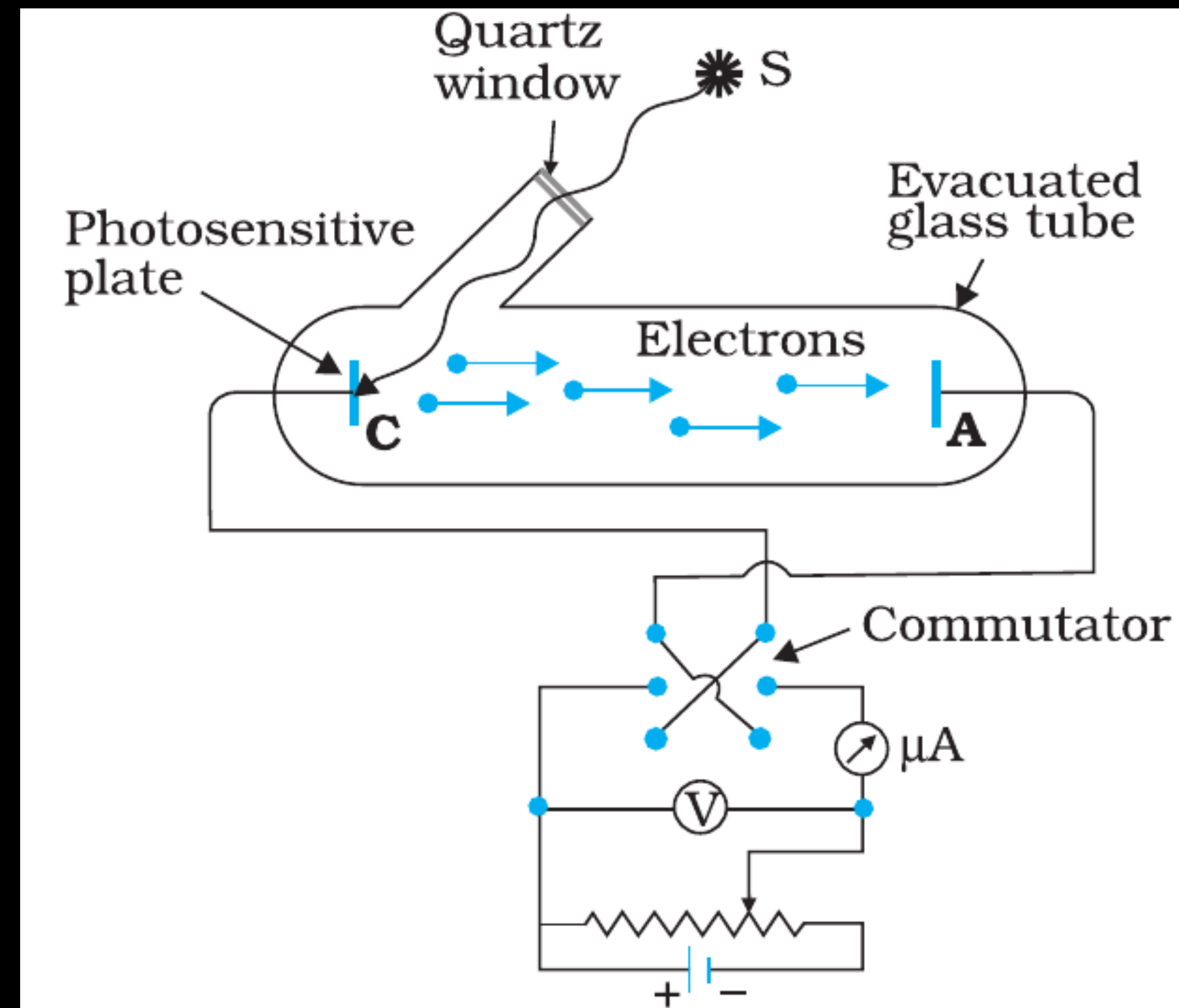


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- * The potential difference between the emitter and collector plates is measured by a voltmeter (V) whereas the resulting photo current flowing in the circuit is measured by a microammeter (μA).
- * We can use the experimental arrangement of to study the variation of photocurrent with (a) intensity of radiation, (b) frequency of incident radiation, (c) the potential difference between the plates A and C, and (d) the nature of the material of plate C

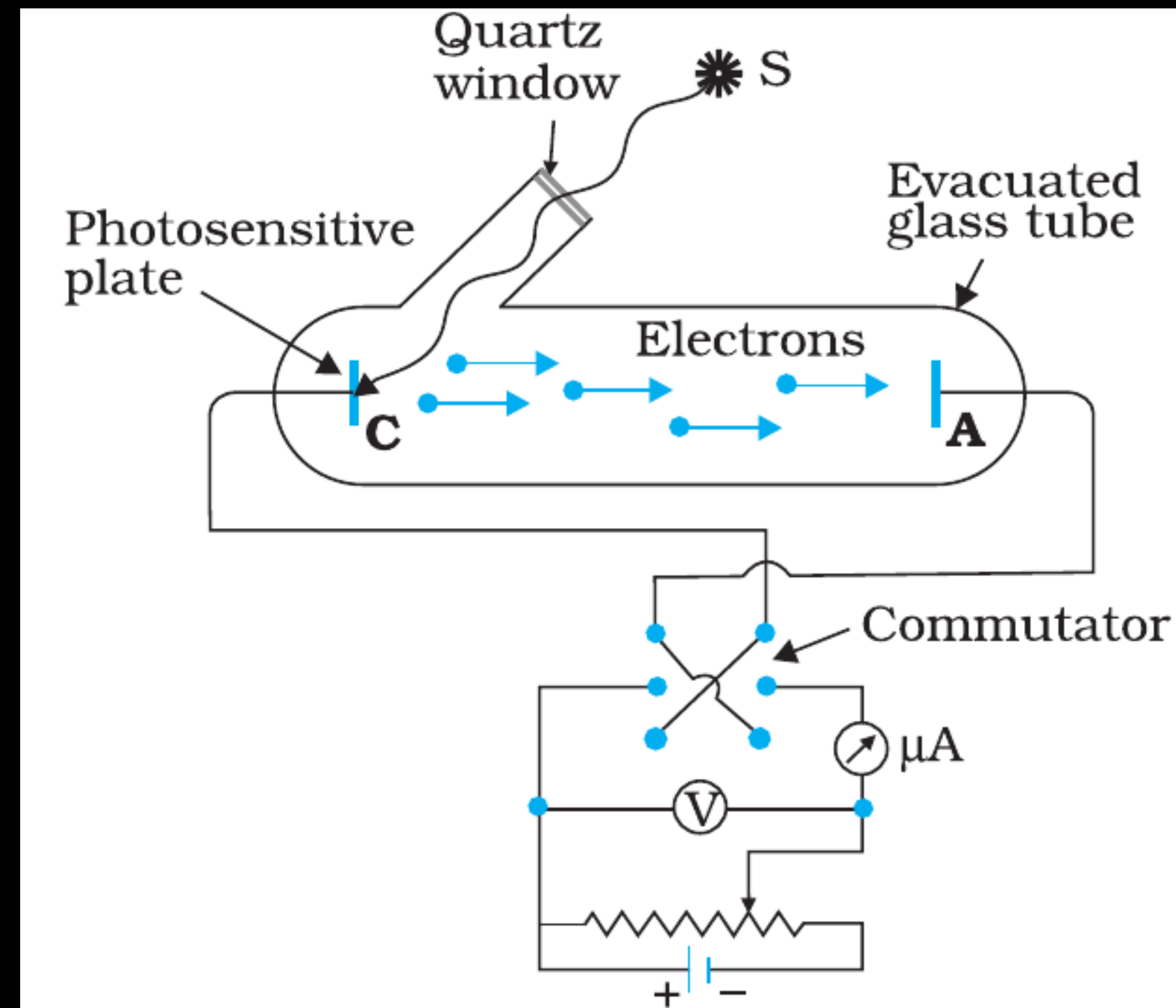


FIGURE 11.1 Experimental arrangement for study of photoelectric effect.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(1) Effect of intensity of light on photocurrent

(Keep frequency and potential constant, vary only intensity.)

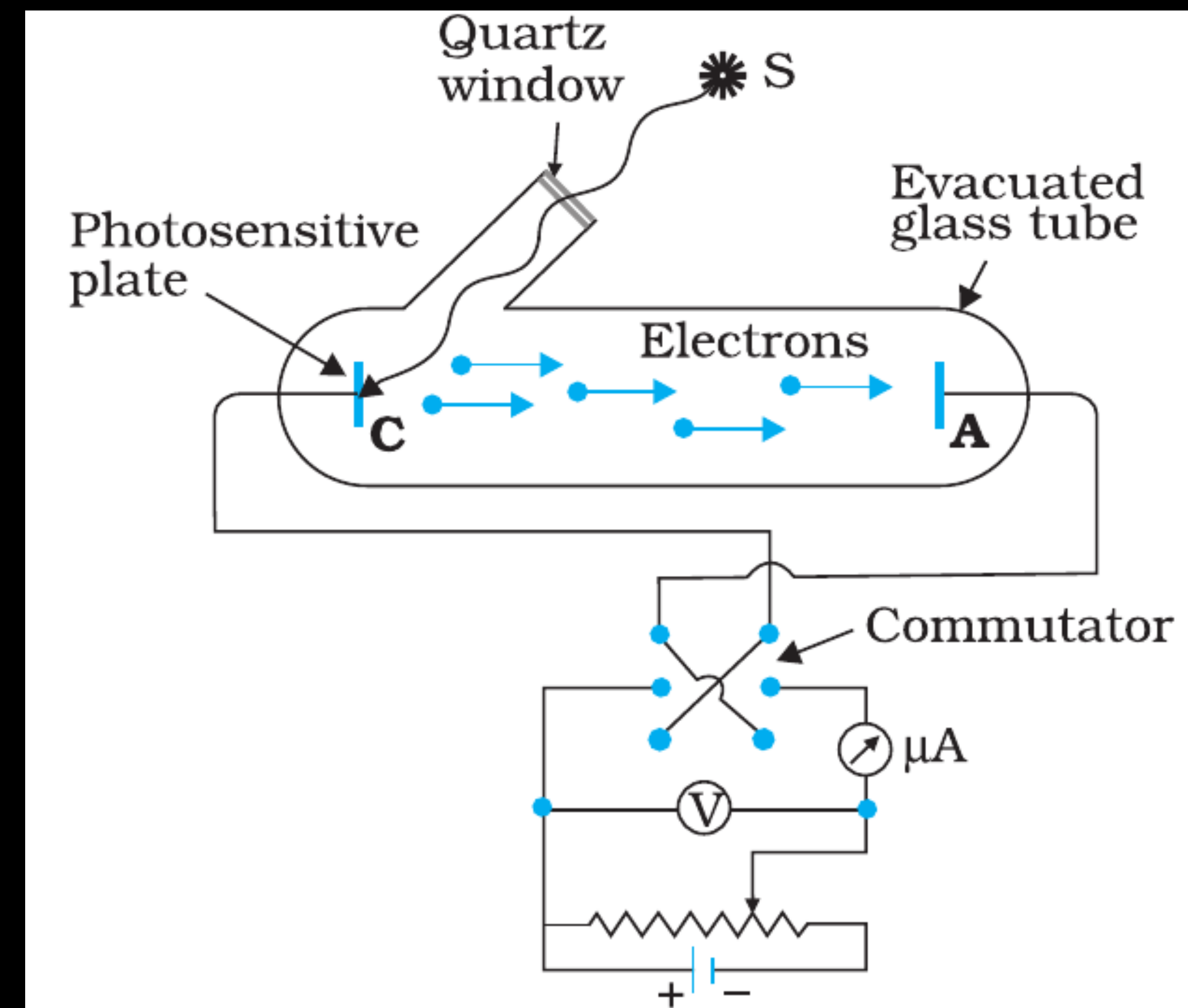


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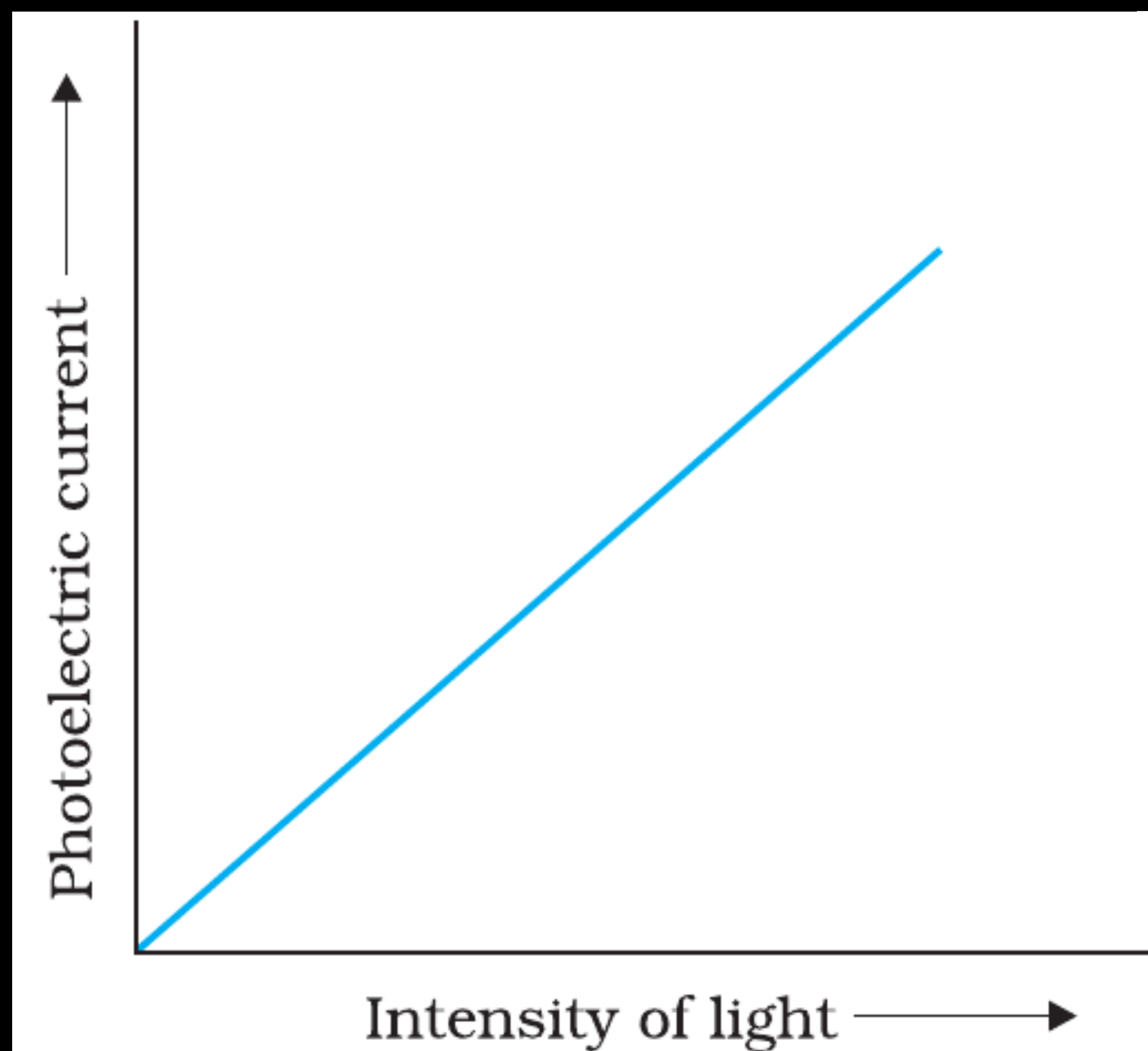


FIGURE 11.2 Variation of Photoelectric current with intensity of light.

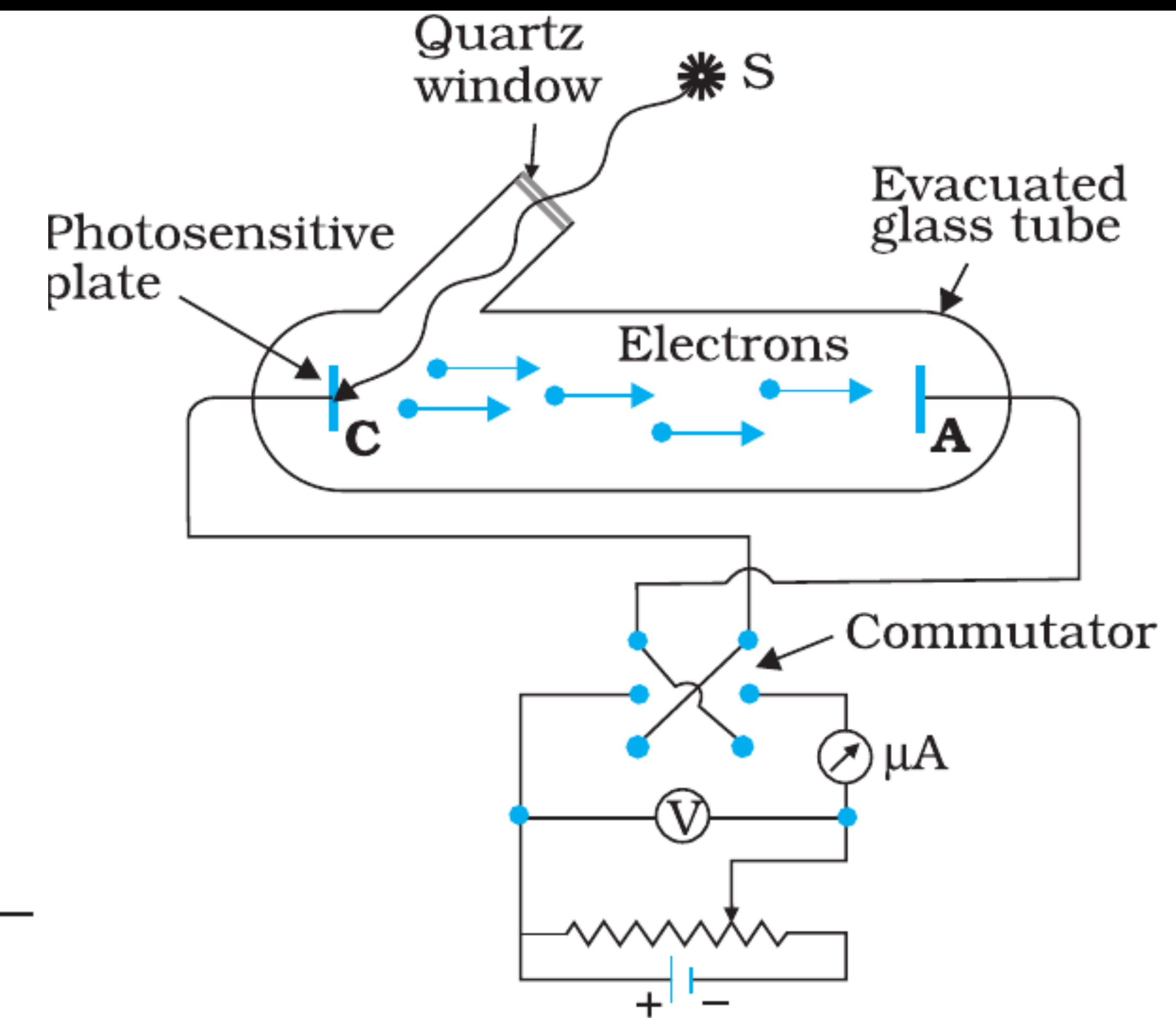


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* The number of photoelectrons emitted per second (or photocurrent) is directly proportional to the intensity of incident radiation.

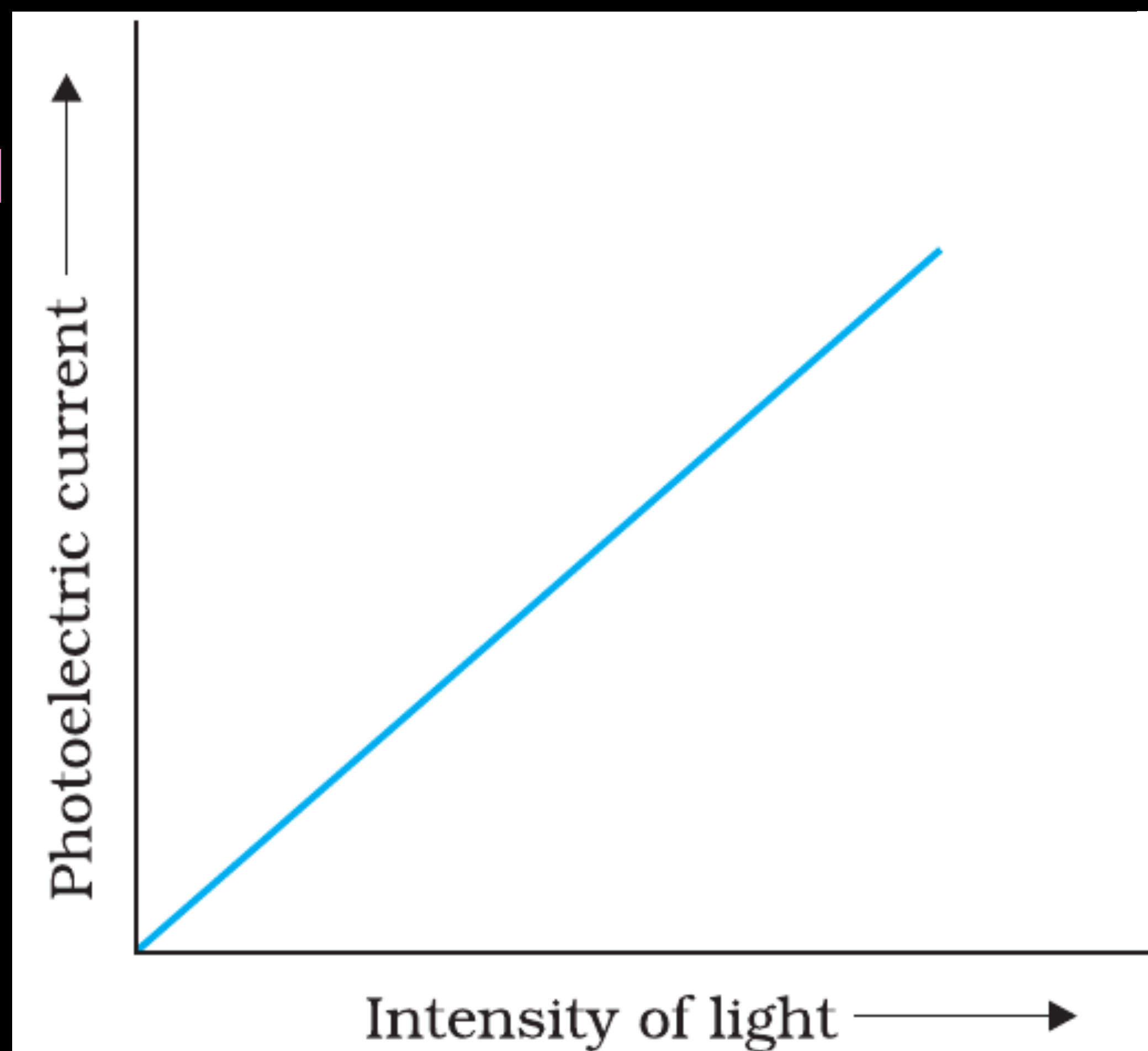


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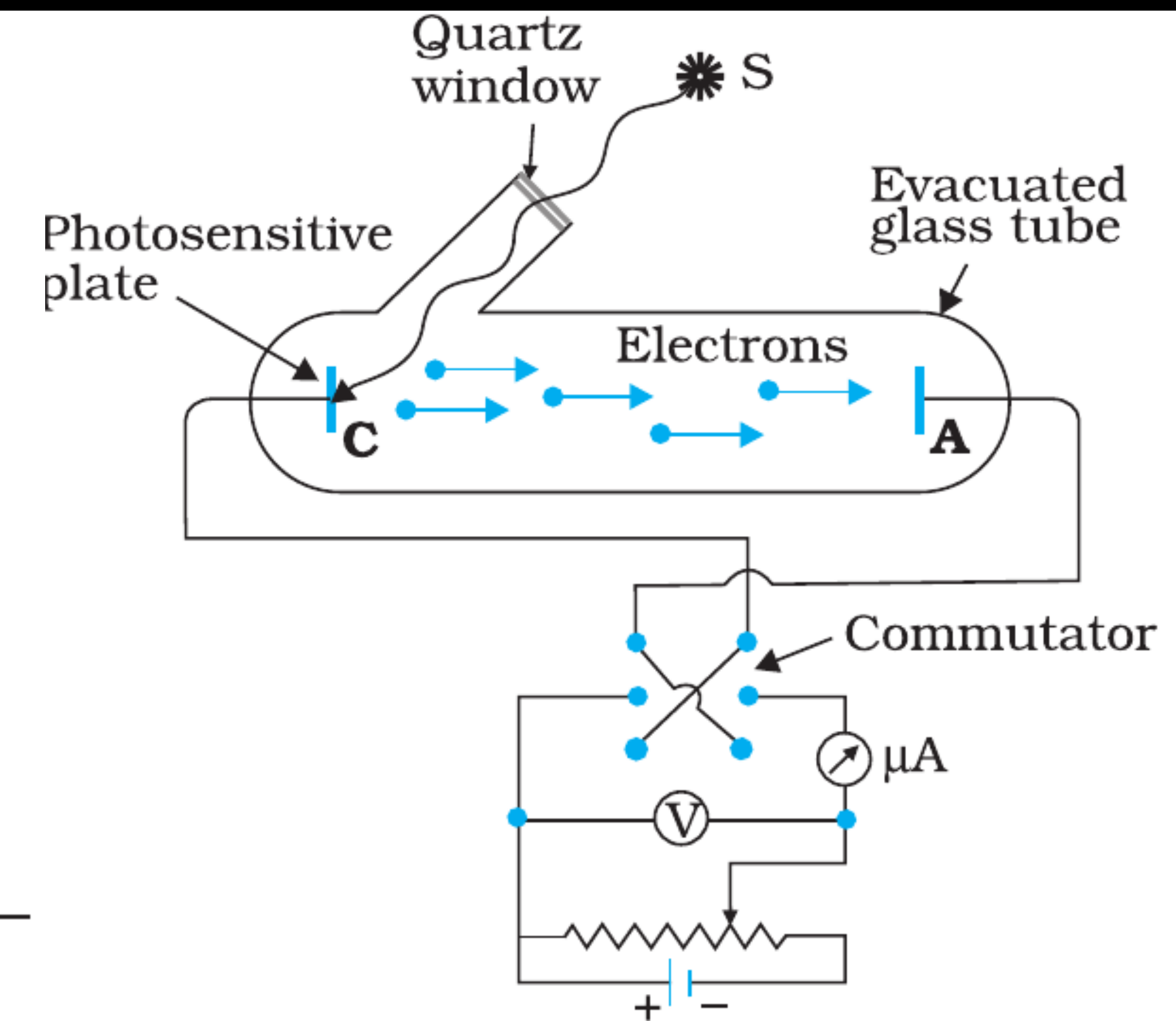


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EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(2) Effect of potential on photoelectric current

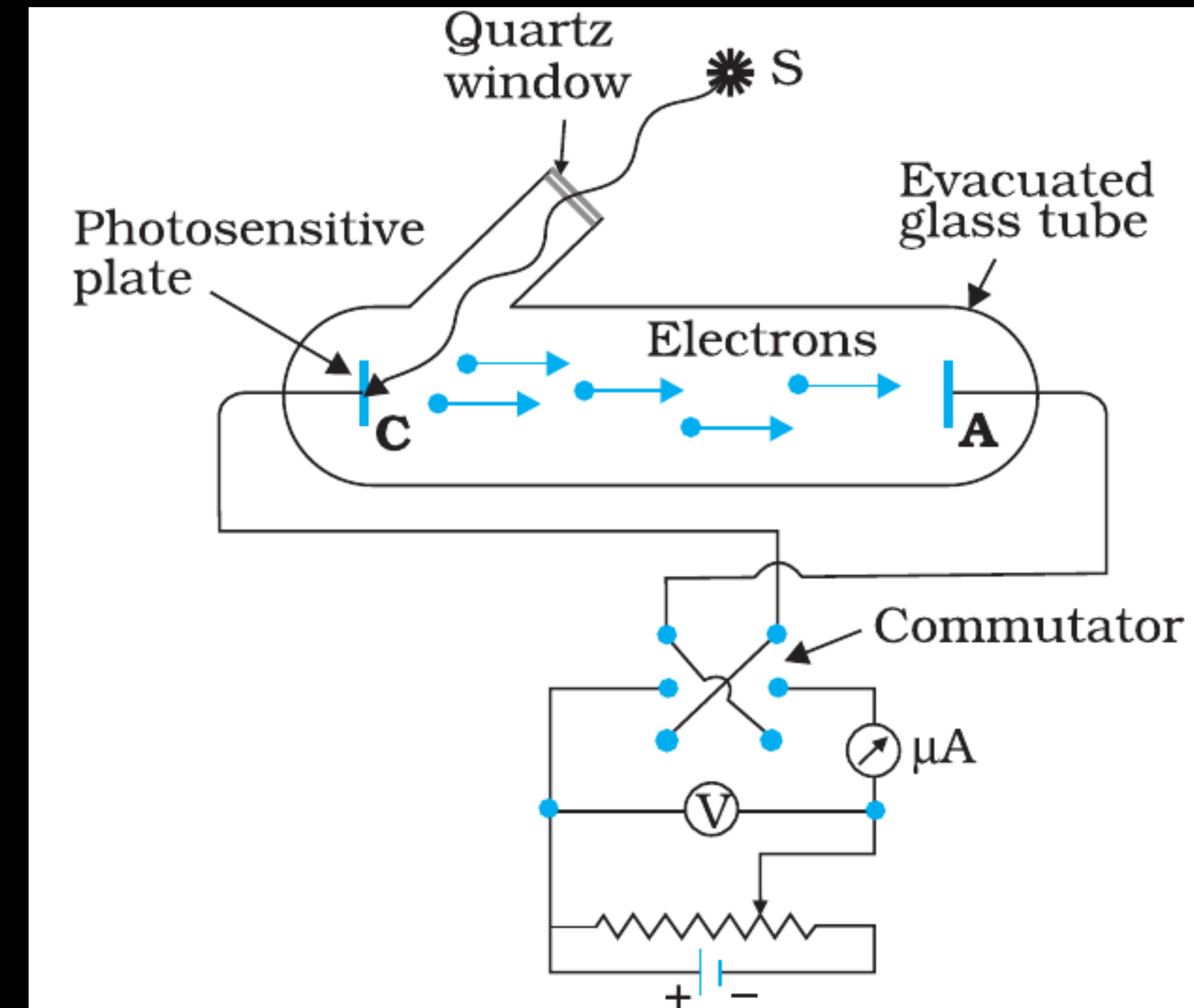


FIGURE 11.1 Experimental arrangement for study of photoelectric effect.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(2) Effect of potential on photoelectric current

(i) Positive Potential (Accelerating Potential)

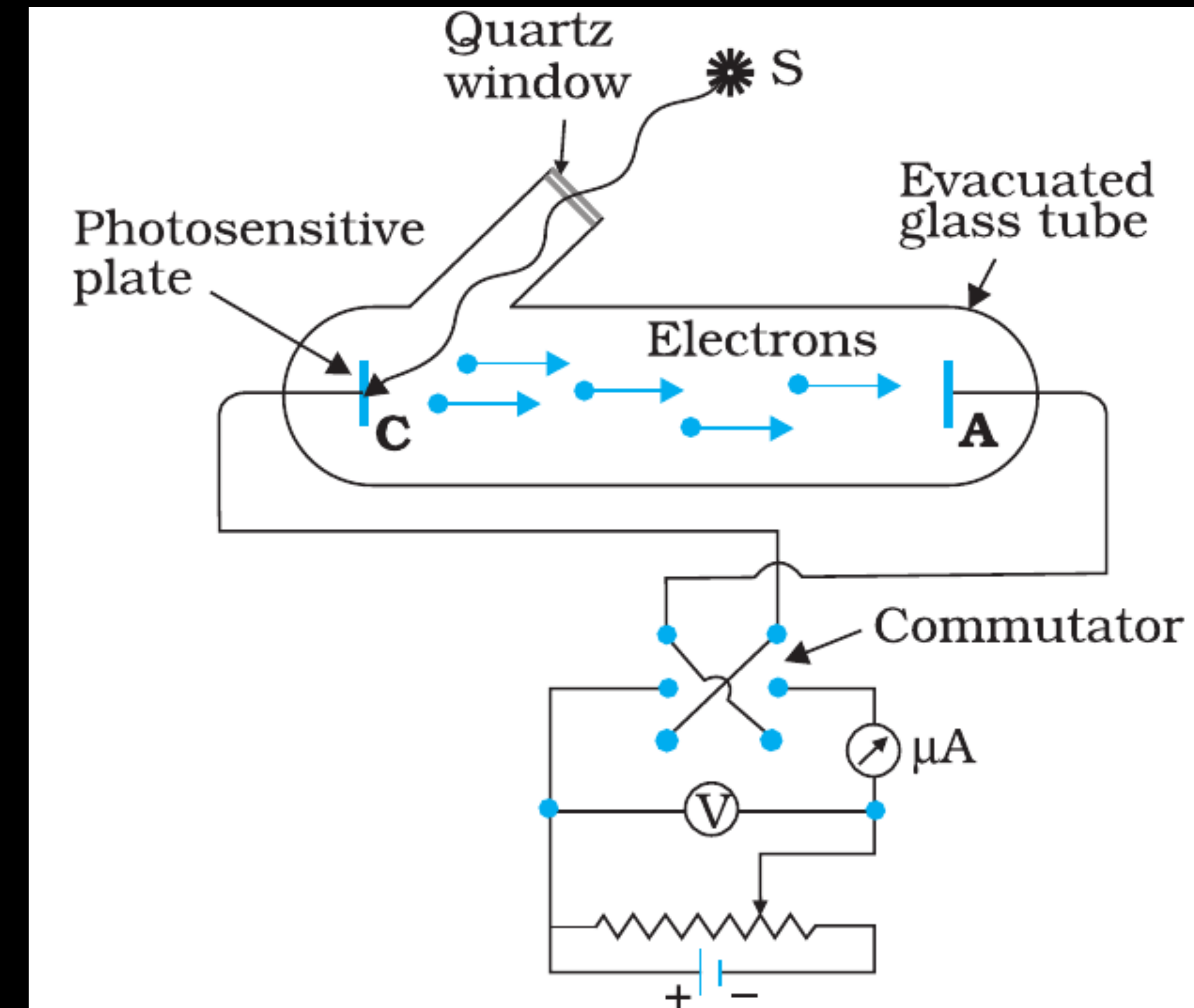


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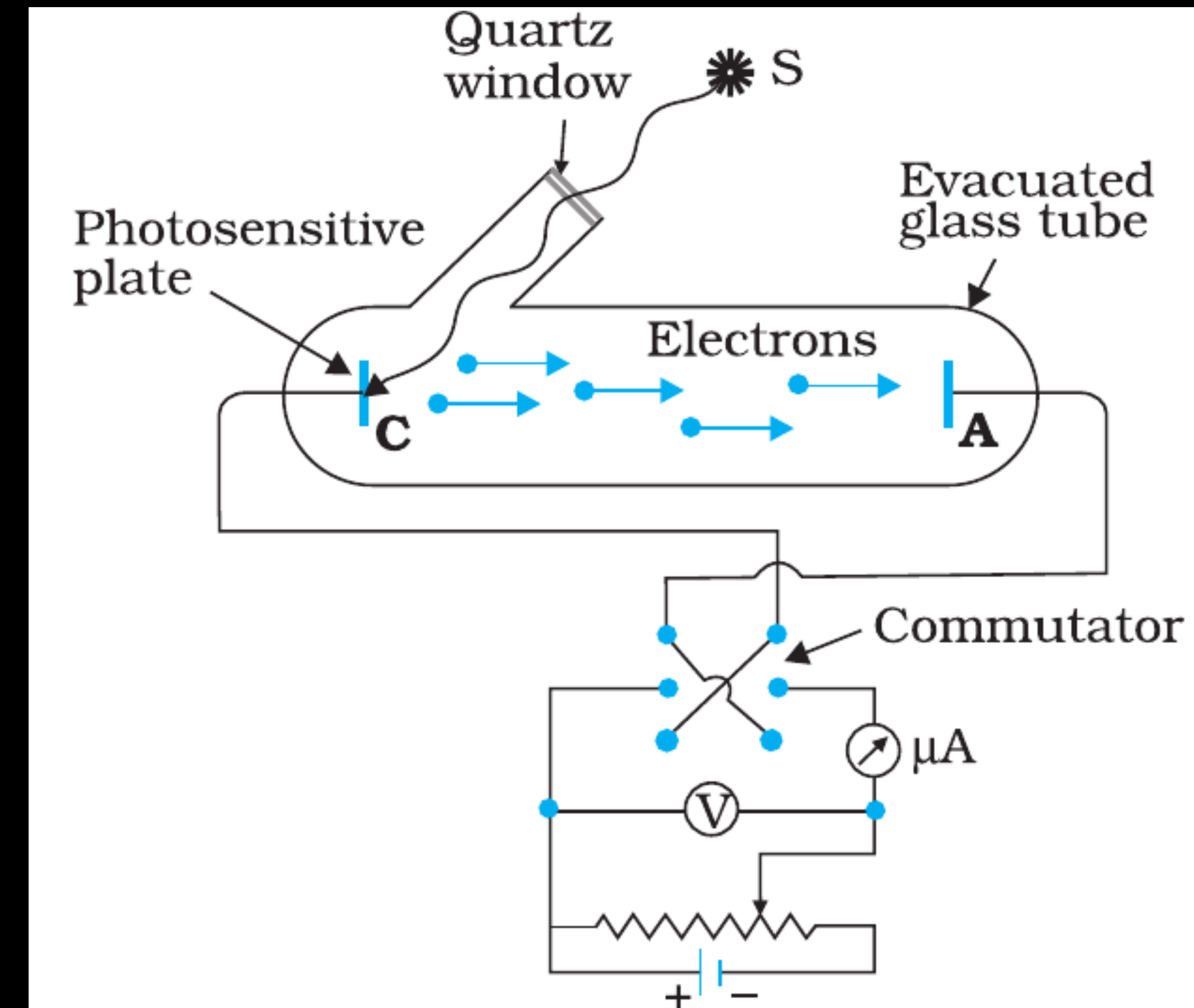


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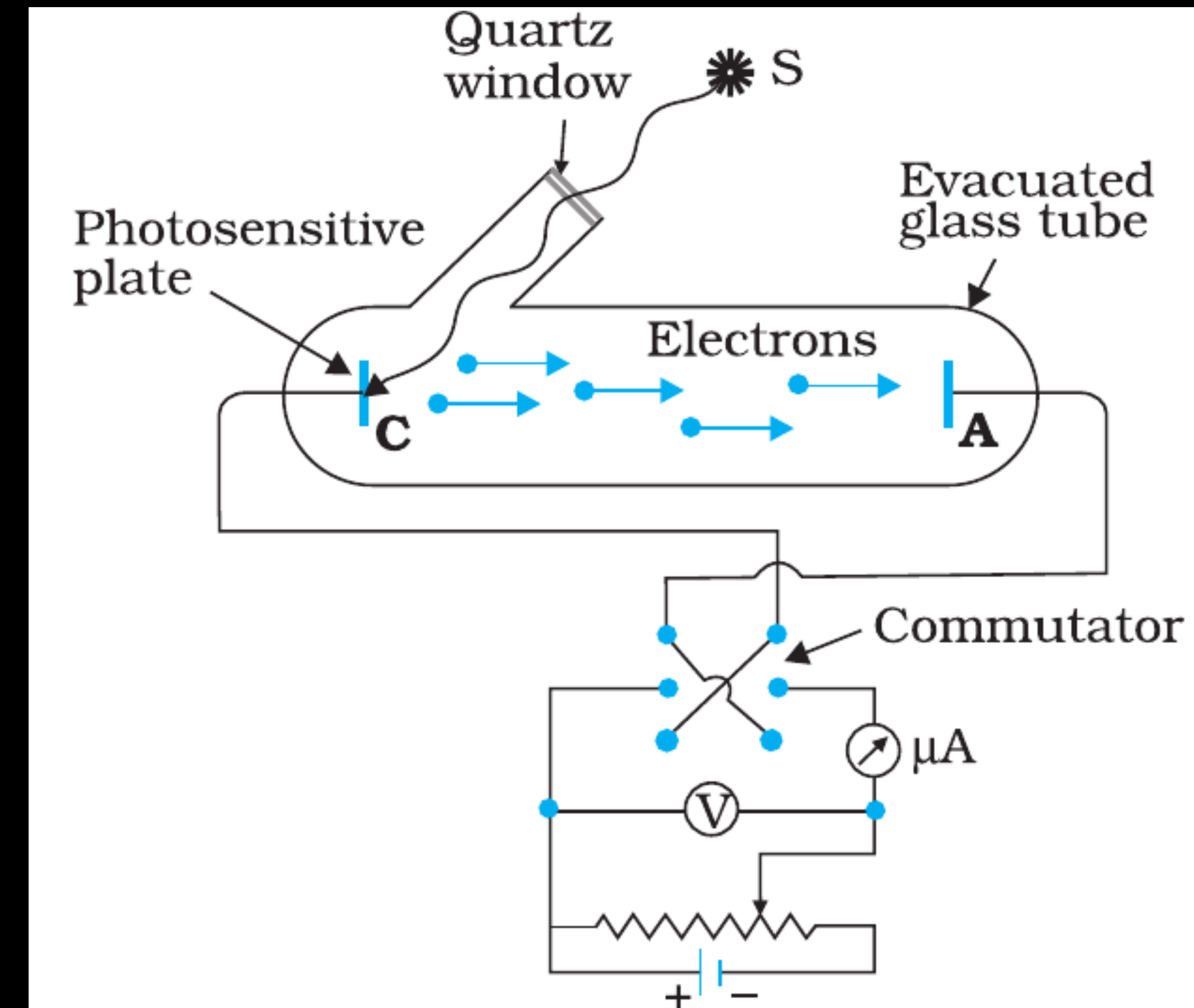


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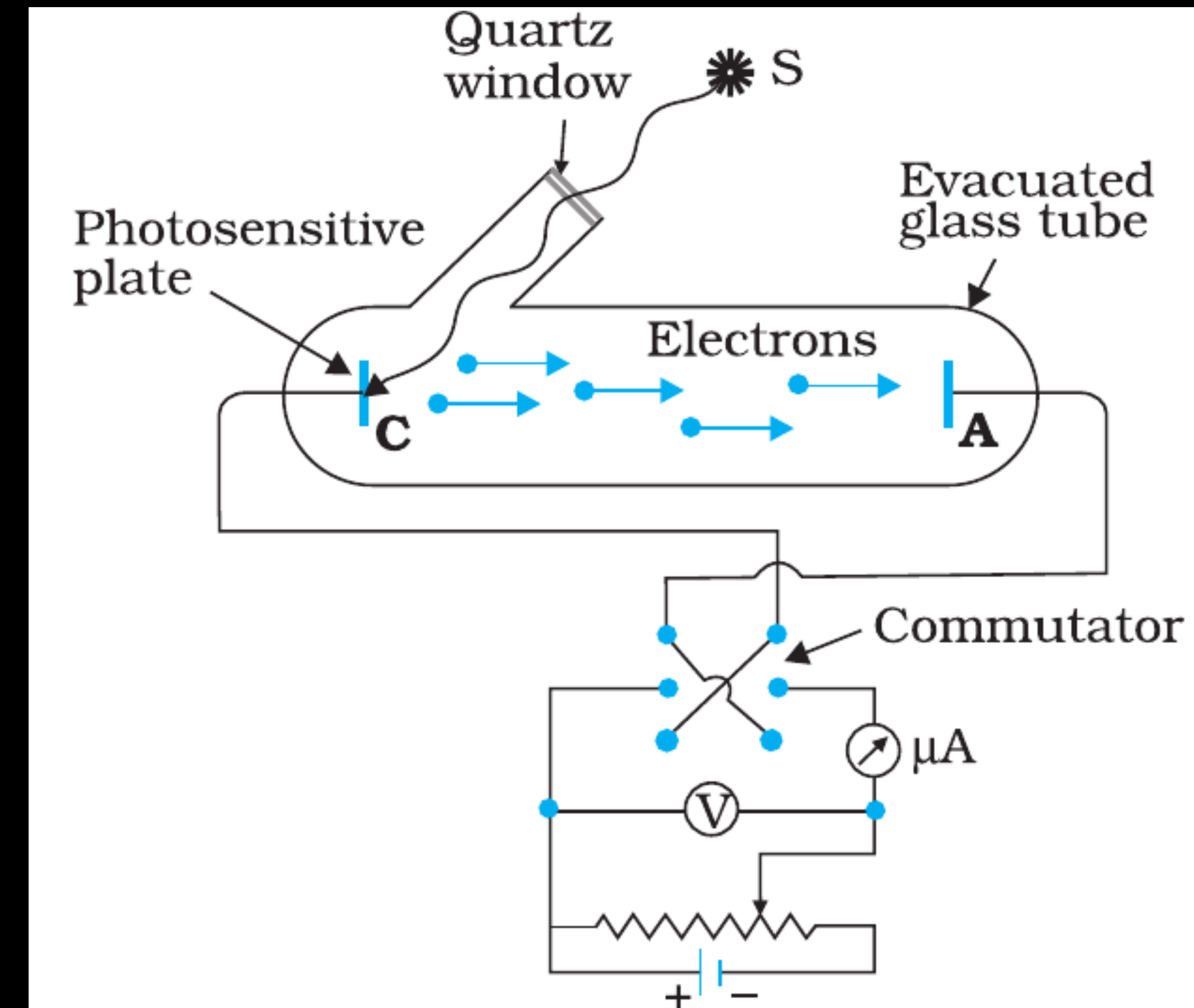


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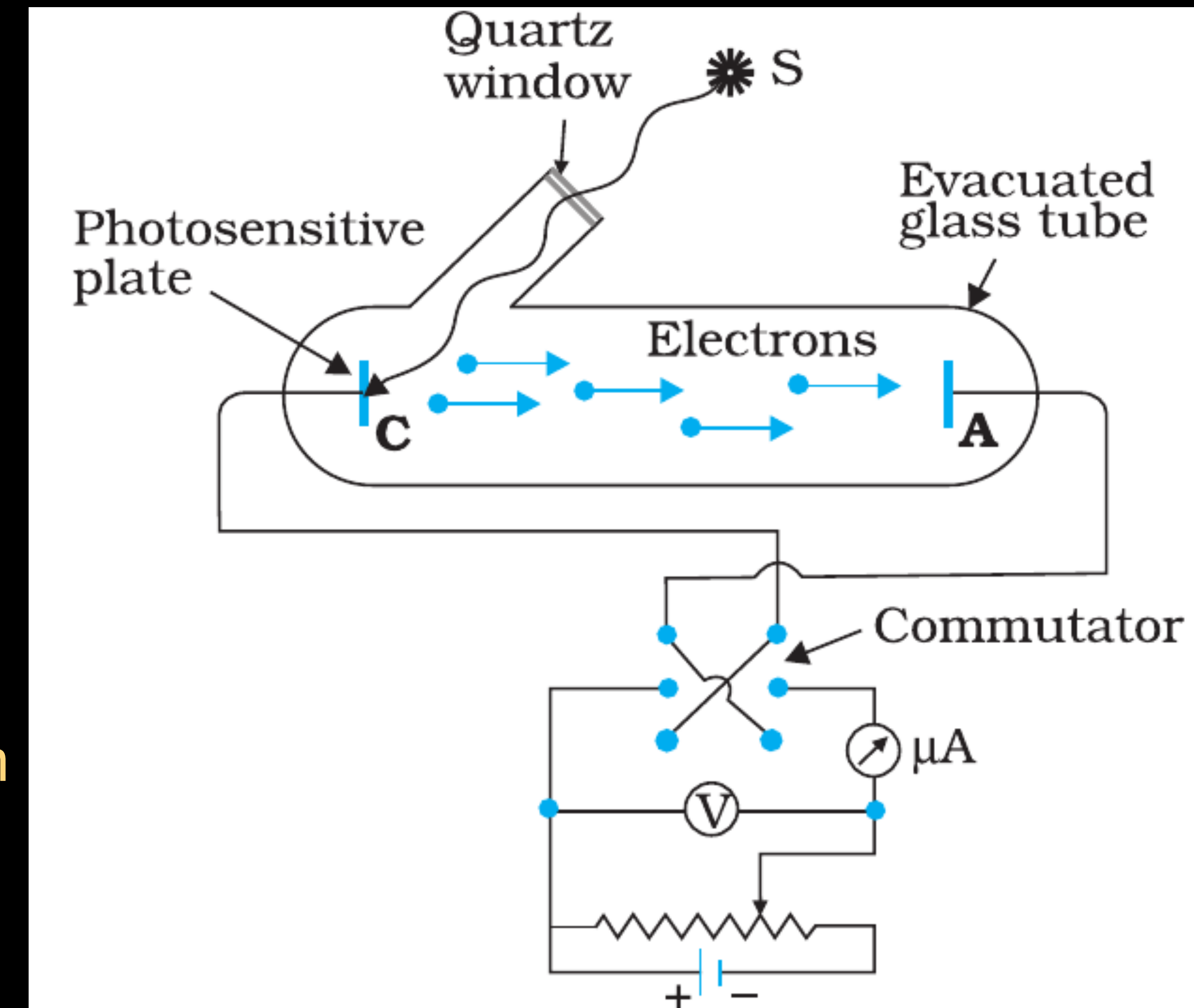


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(ii) Negative Potential (Retarding Potential) and make it increasingly negative gradually.

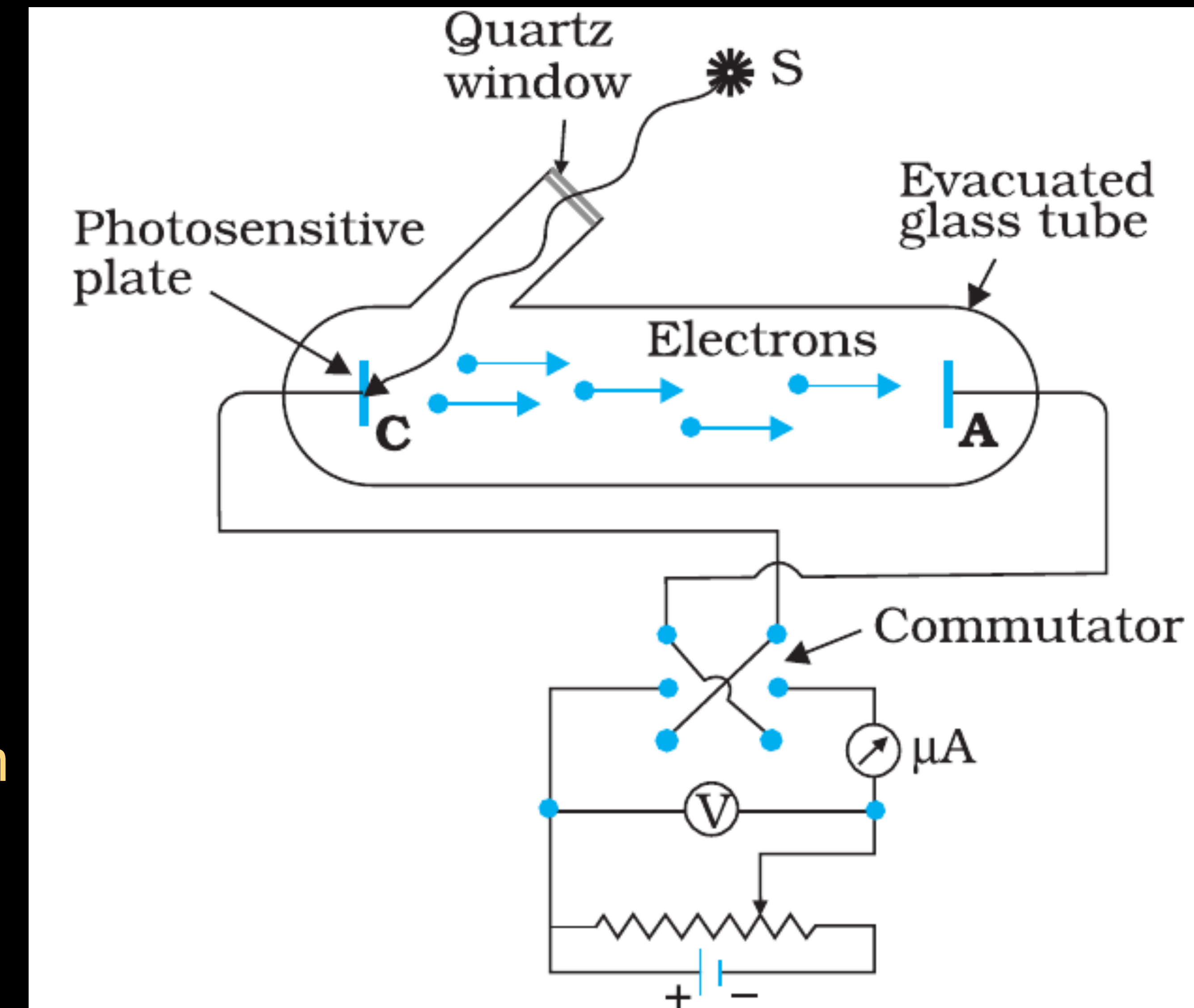


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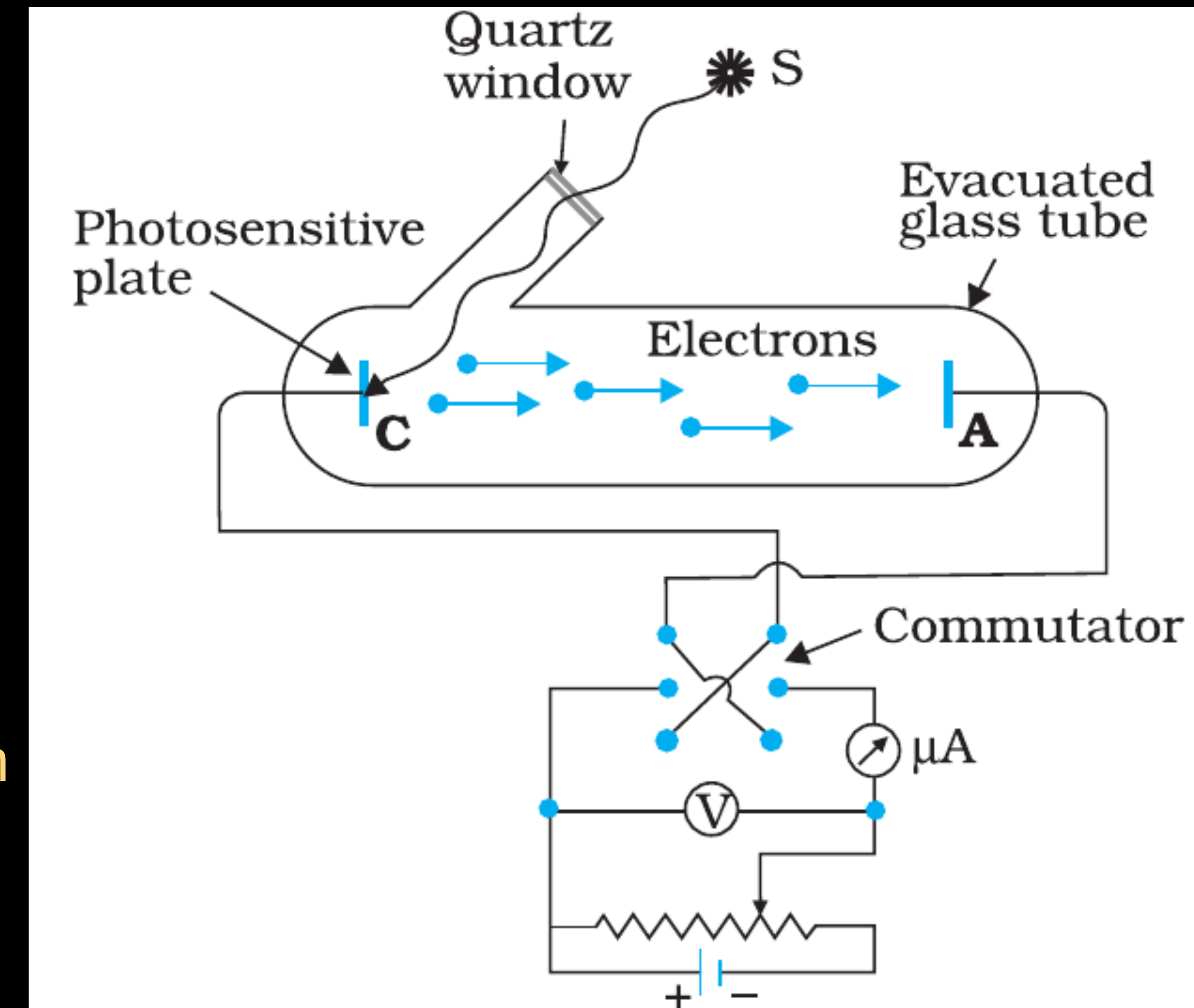


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- * When the polarity is reversed, the electrons are repelled and only the sufficiently energetic electrons are able to reach the collector A.
- * At a certain negative voltage (V_0) called stopping potential: Even the most energetic electrons cannot reach A. Photocurrent becomes zero.

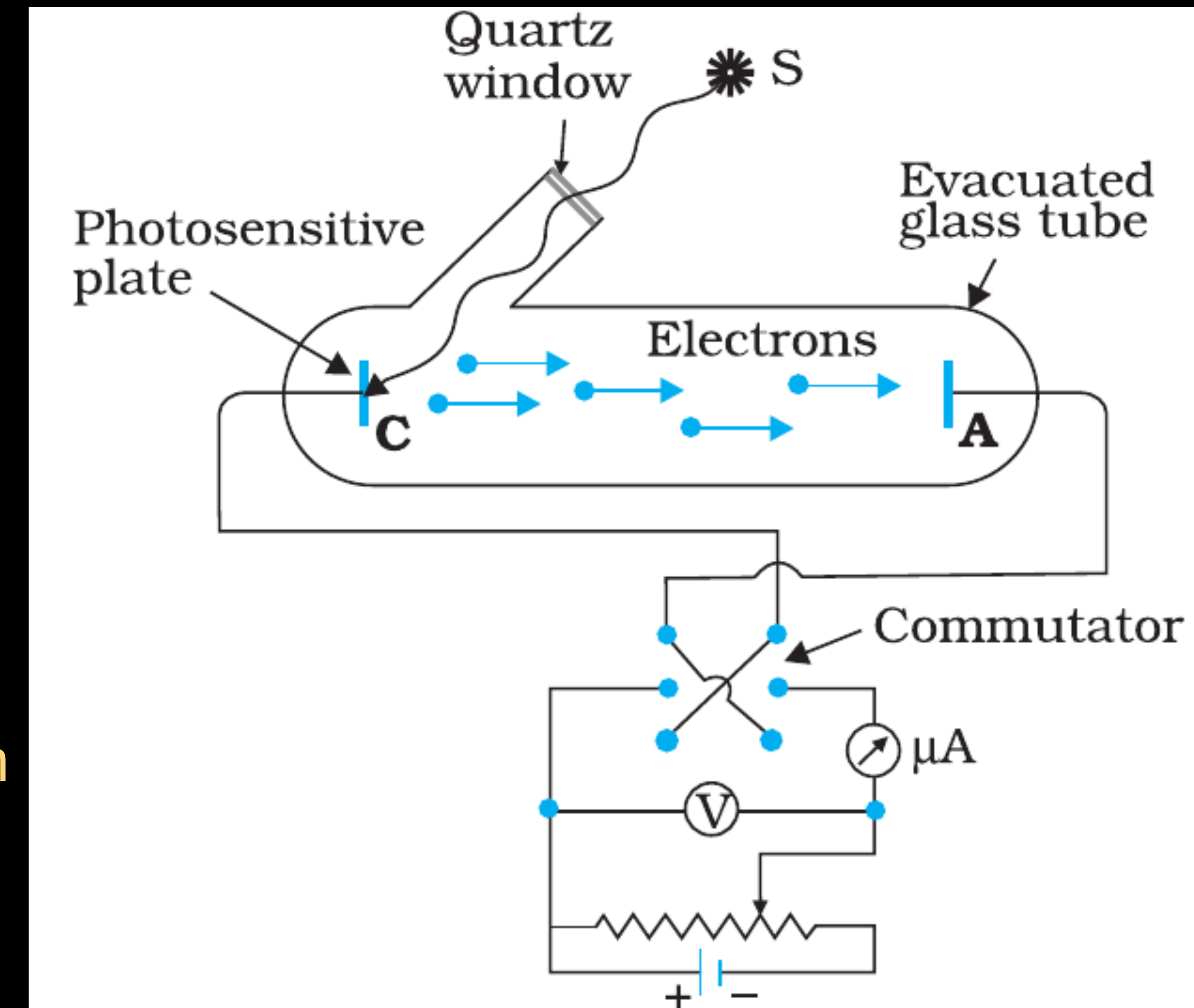


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EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(2) Effect of potential on photoelectric current

Stopping Potential (V_0) :

For a particular frequency of incident radiation, the minimum negative (retarding) potential V_0 given to the plate A for which the photocurrent stops or becomes zero is called the cut-off or stopping potential.

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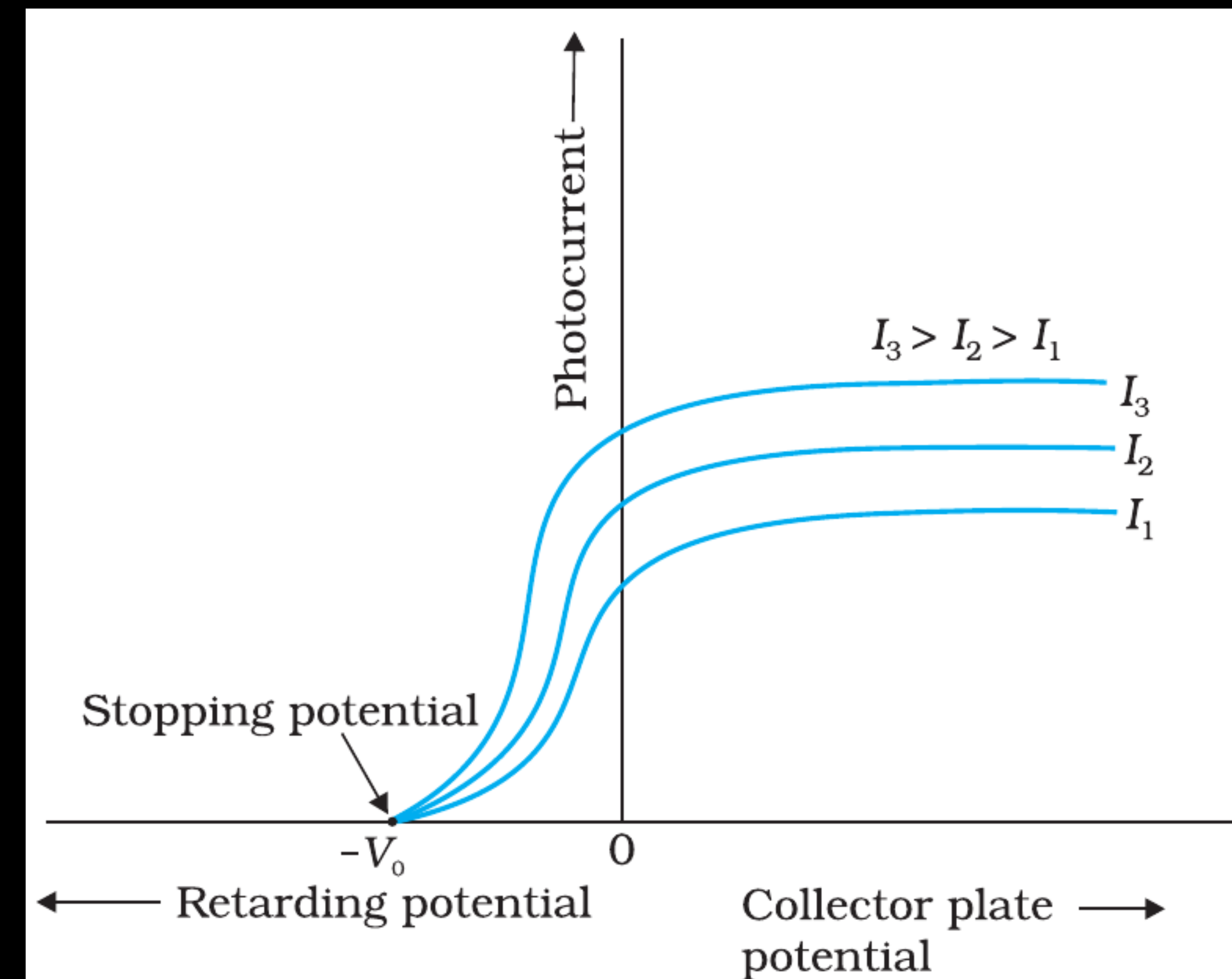


FIGURE 11.3 Variation of photocurrent with collector plate potential for different intensity of incident radiation.

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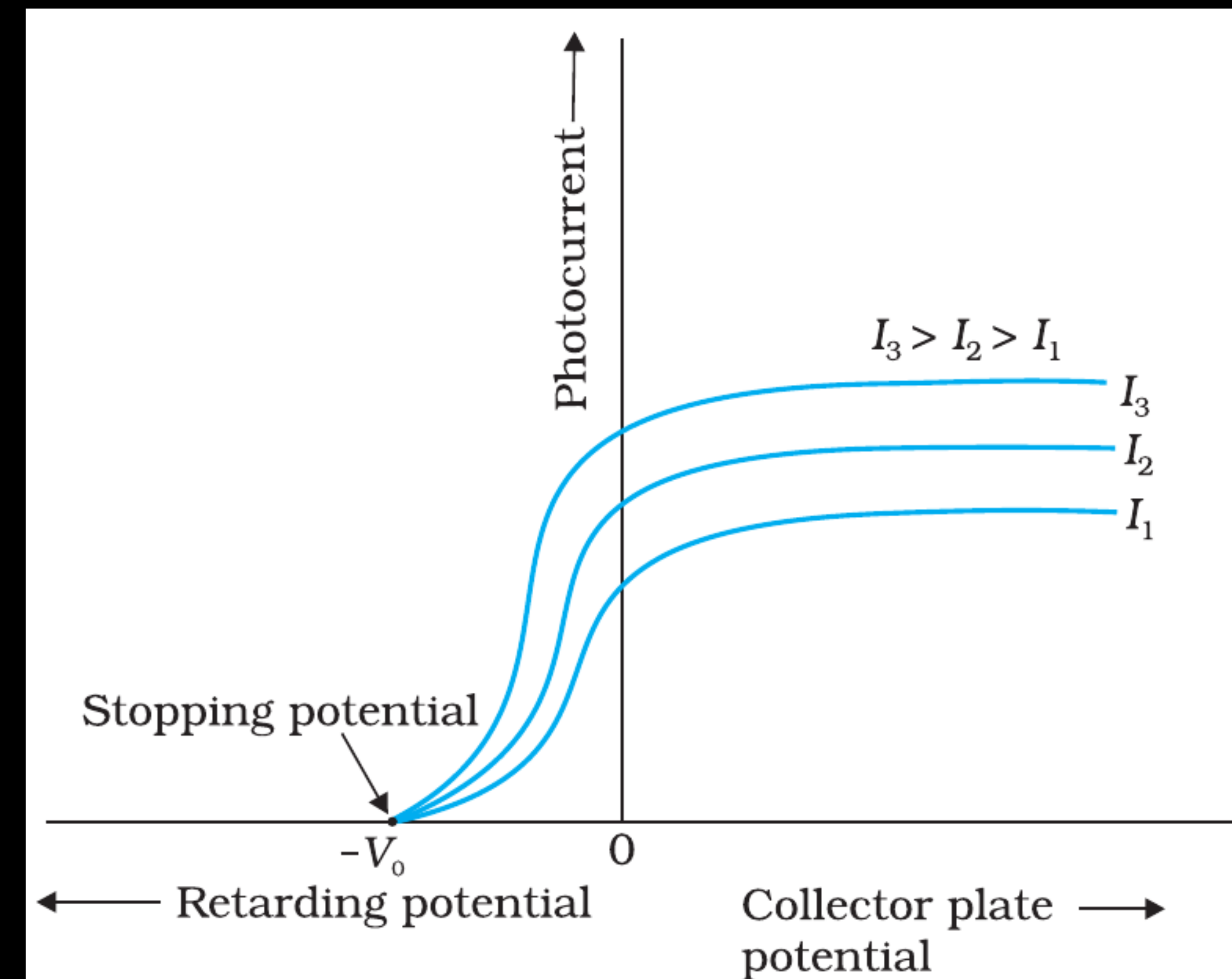


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- * Photoelectric current is zero when the stopping potential is sufficient to repel even the most energetic photoelectrons, with the maximum kinetic energy ($K_{\max}$), so that

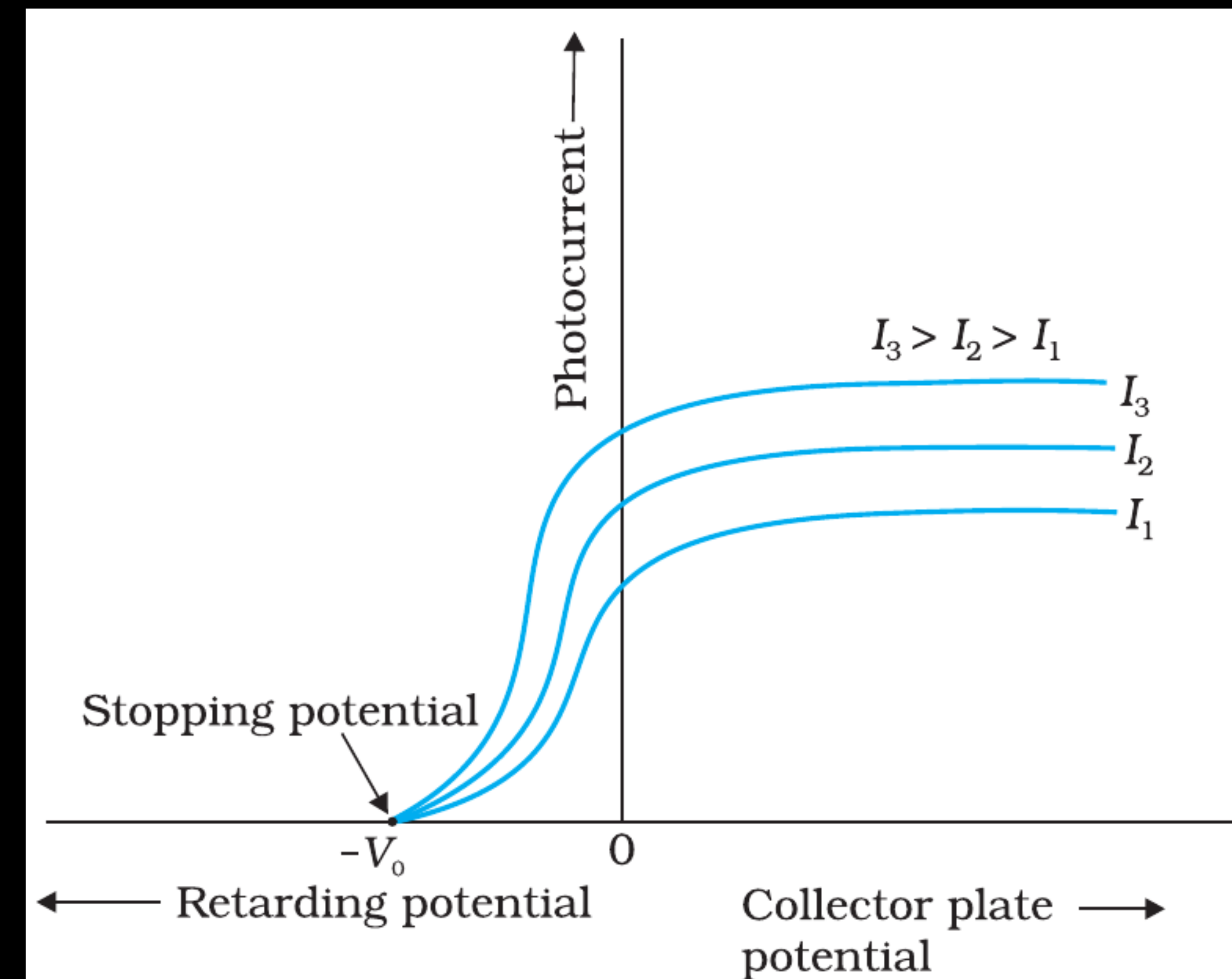


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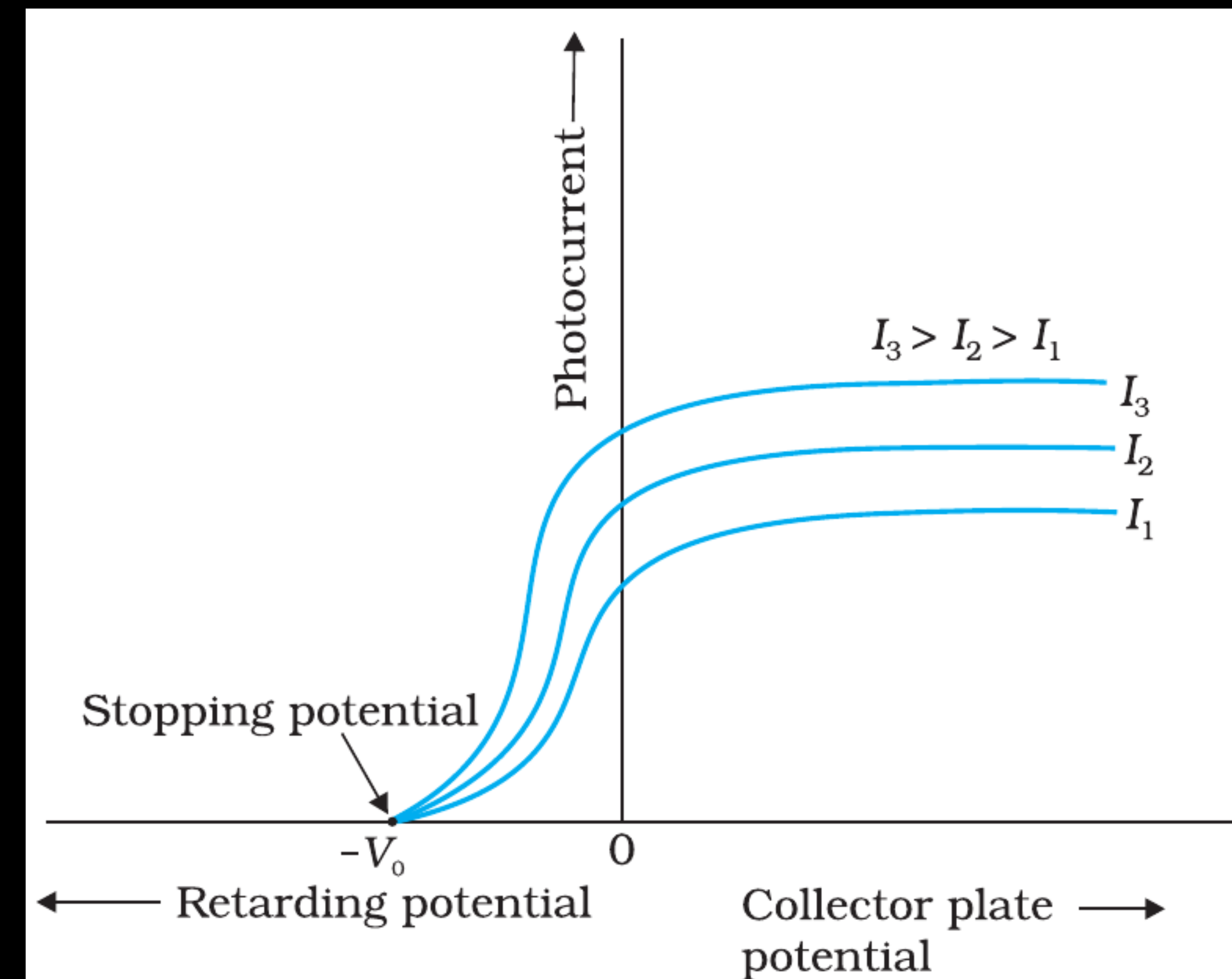


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- * For higher intensity I_2 and I_3 ($I_3 > I_2 > I_1$). We note that the saturation currents are now found to be at higher values.

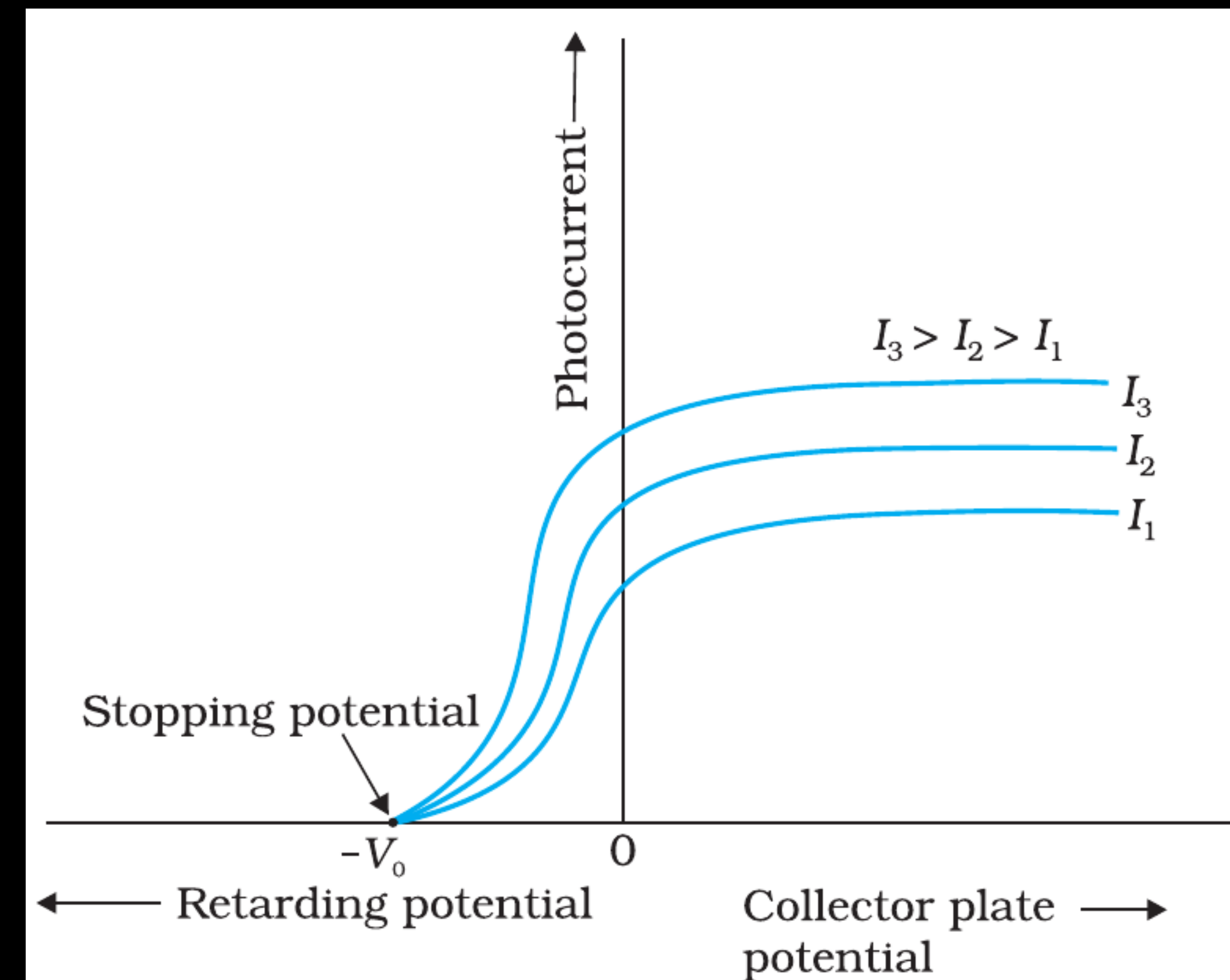


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* For a given frequency of the incident radiation, the stopping potential is independent of its intensity.

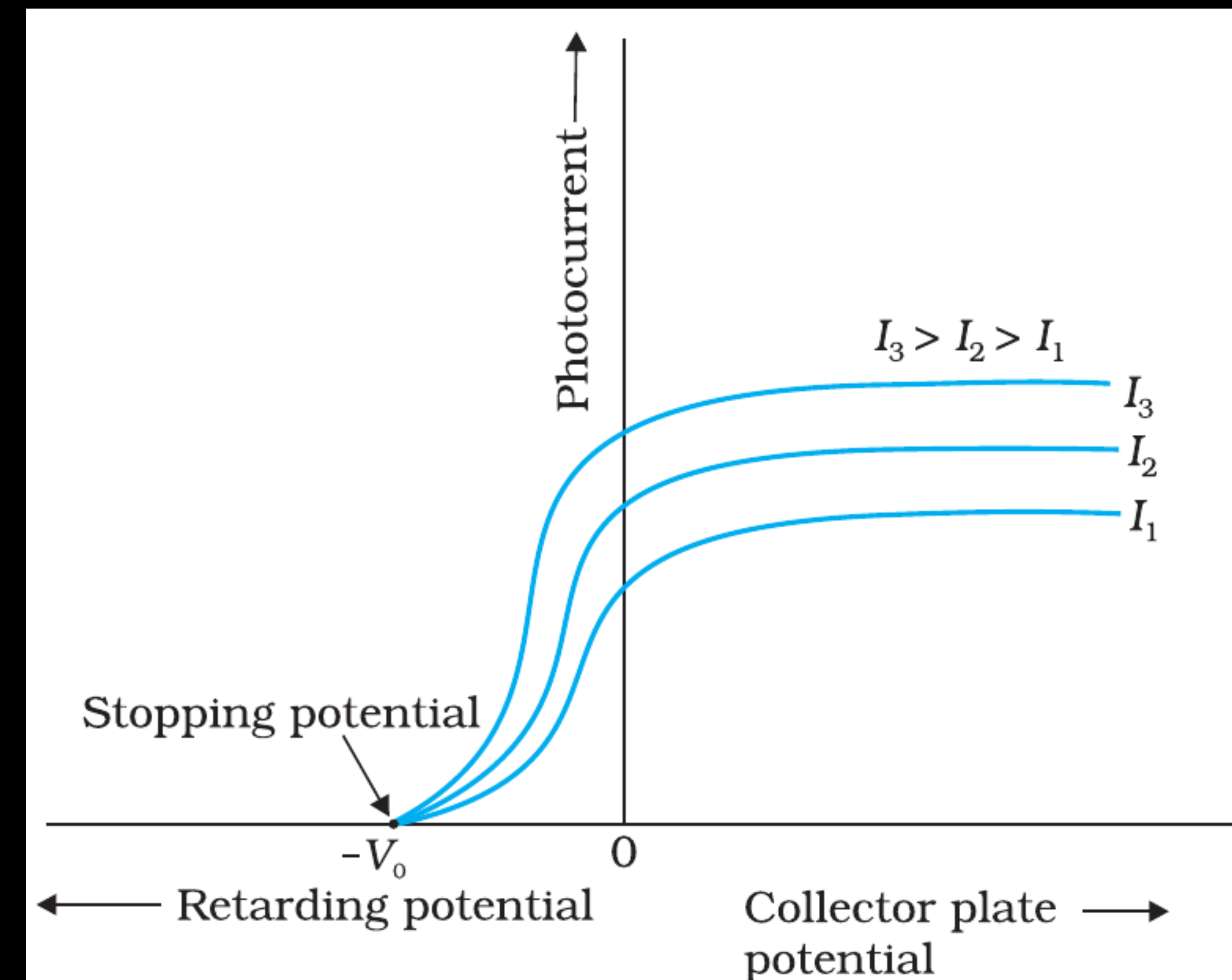


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- * For a given frequency of the incident radiation, the stopping potential is independent of its intensity.

- * In other words, the maximum kinetic energy of photoelectrons depends on the light source and the emitter plate material, but is independent of intensity of incident radiation.

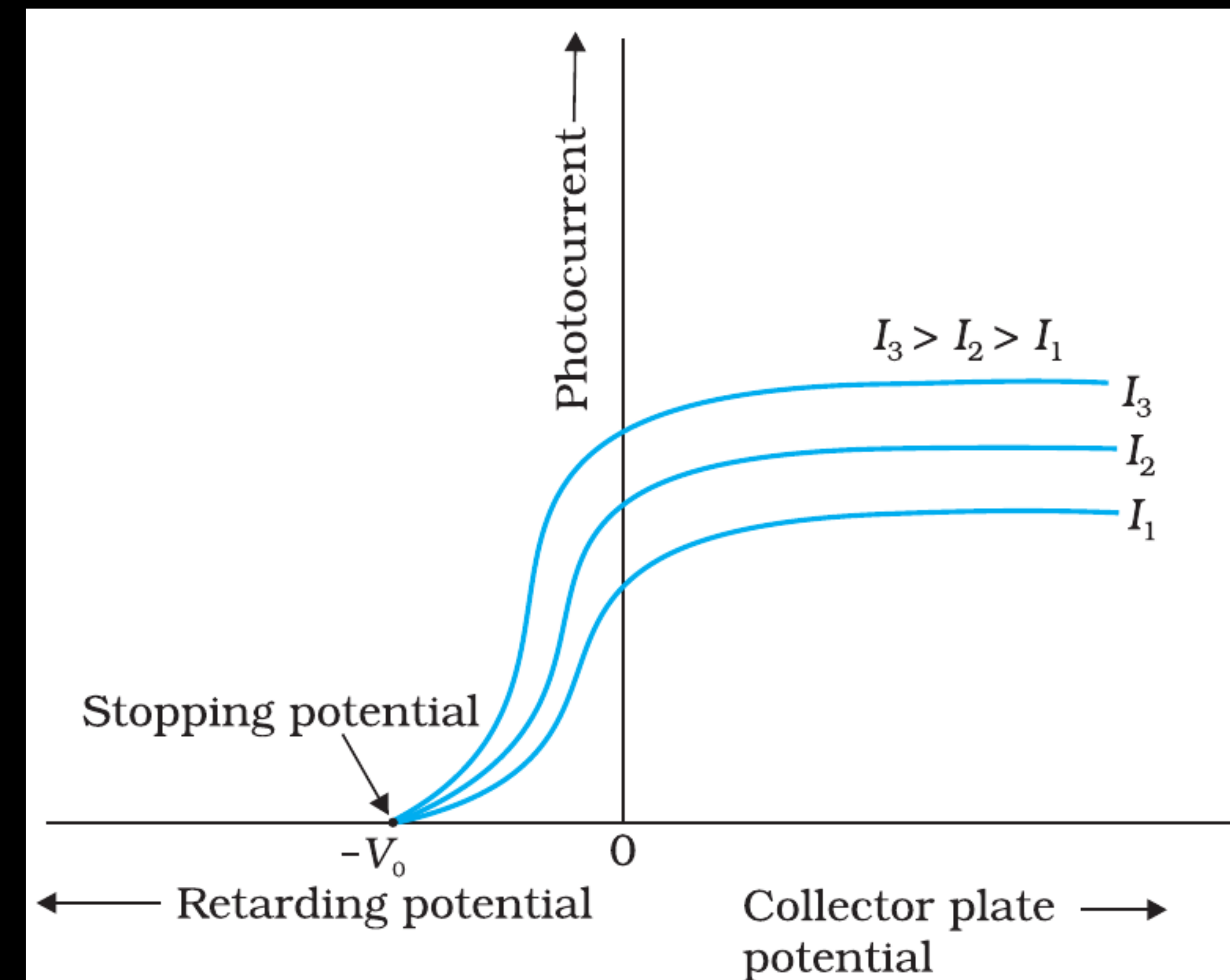


FIGURE 11.3 Variation of photocurrent with collector plate potential for different intensity of incident radiation.

EXPERIMENTAL STUDY OF PHOTOELECTRIC EFFECT

(3) Effect of frequency of incident radiation on stopping potential

(Keep intensity constant; vary frequency.)

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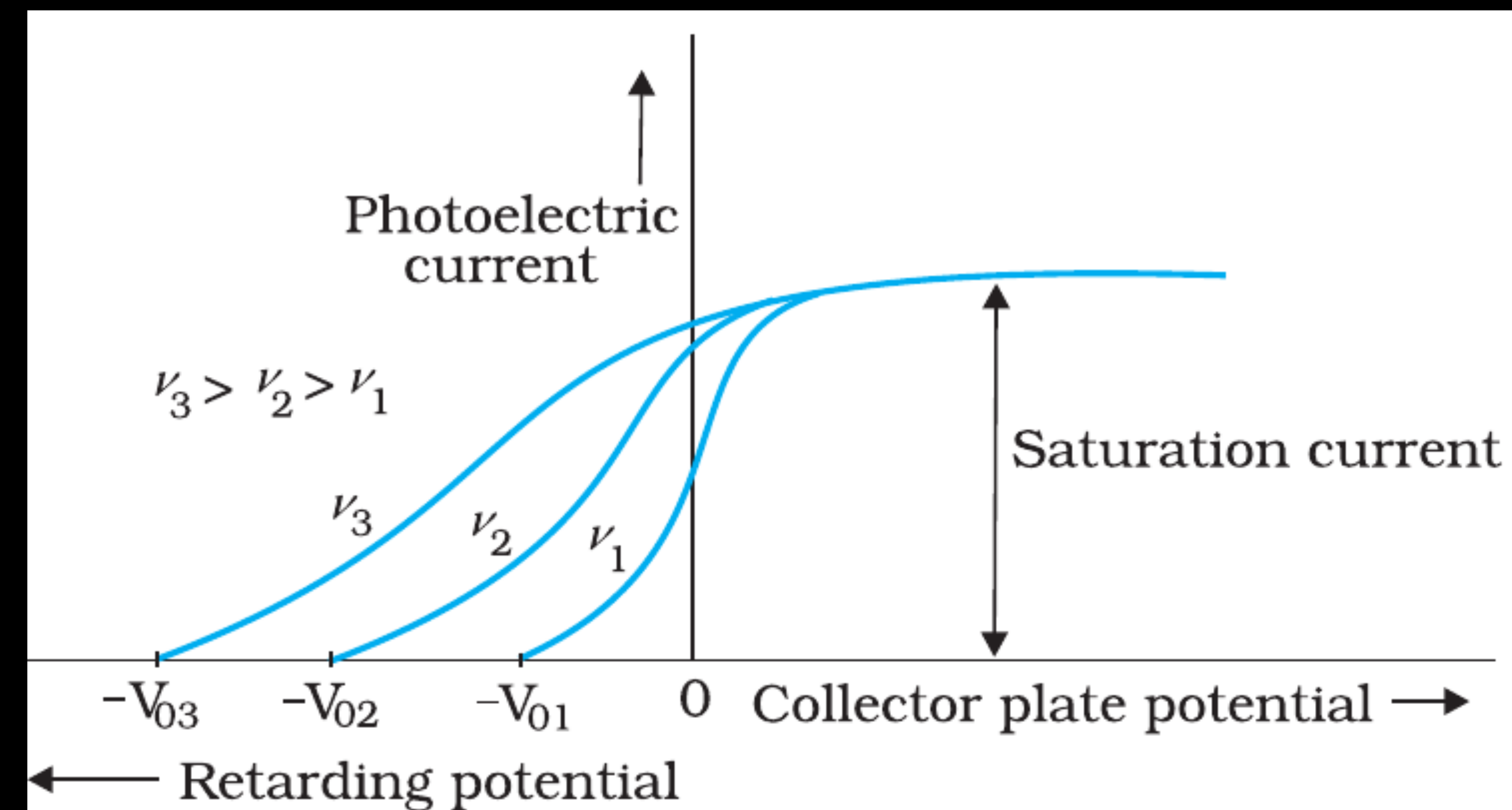


FIGURE 11.4 Variation of photoelectric current with collector plate potential for different frequencies of incident radiation.

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- * adjust the same intensity of light radiation at various frequencies and study the variation of photocurrent with collector plate potential.
- * We obtain different values of stopping potential but the same value of the saturation current for incident radiation of different frequencies.

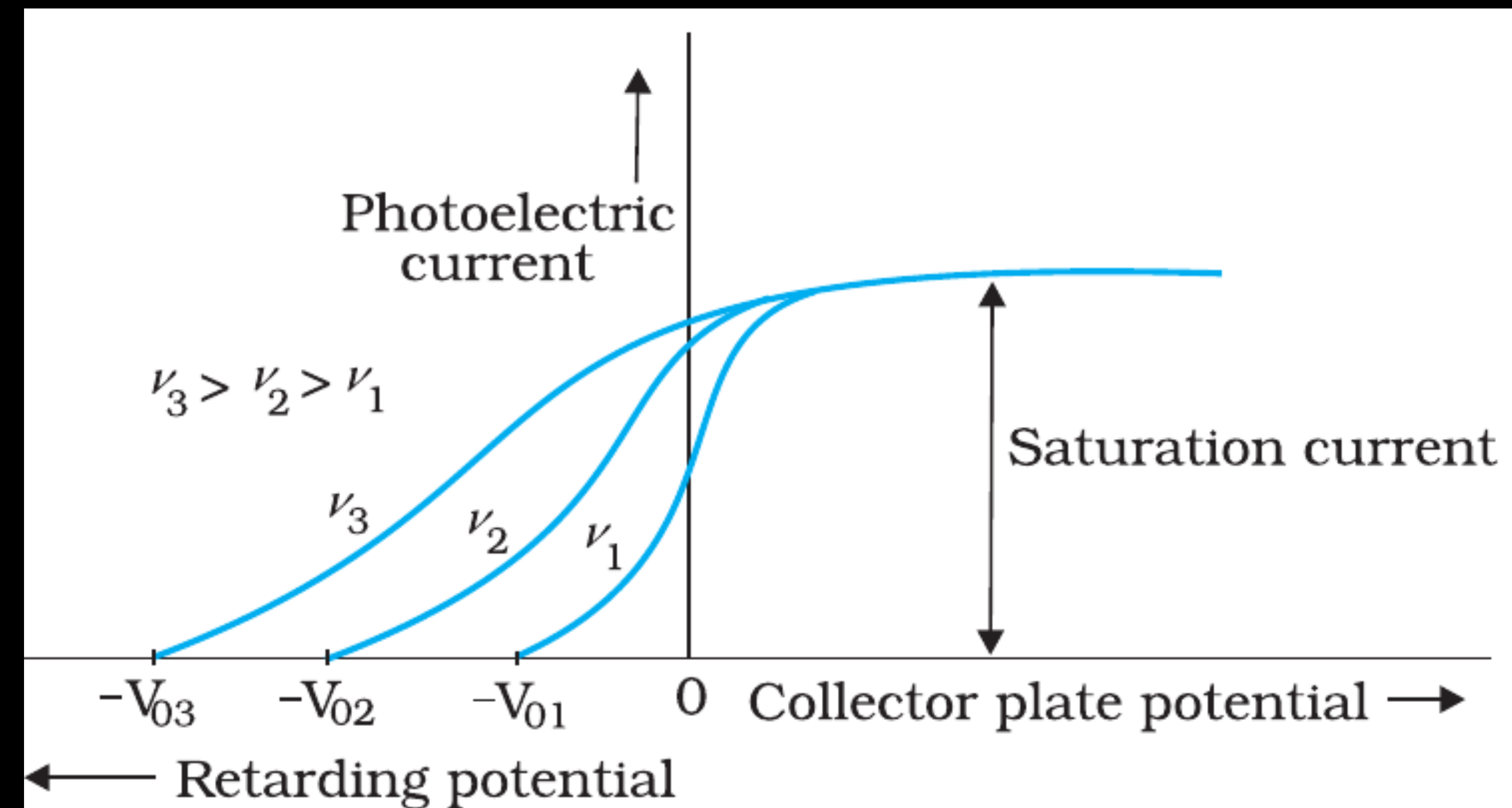


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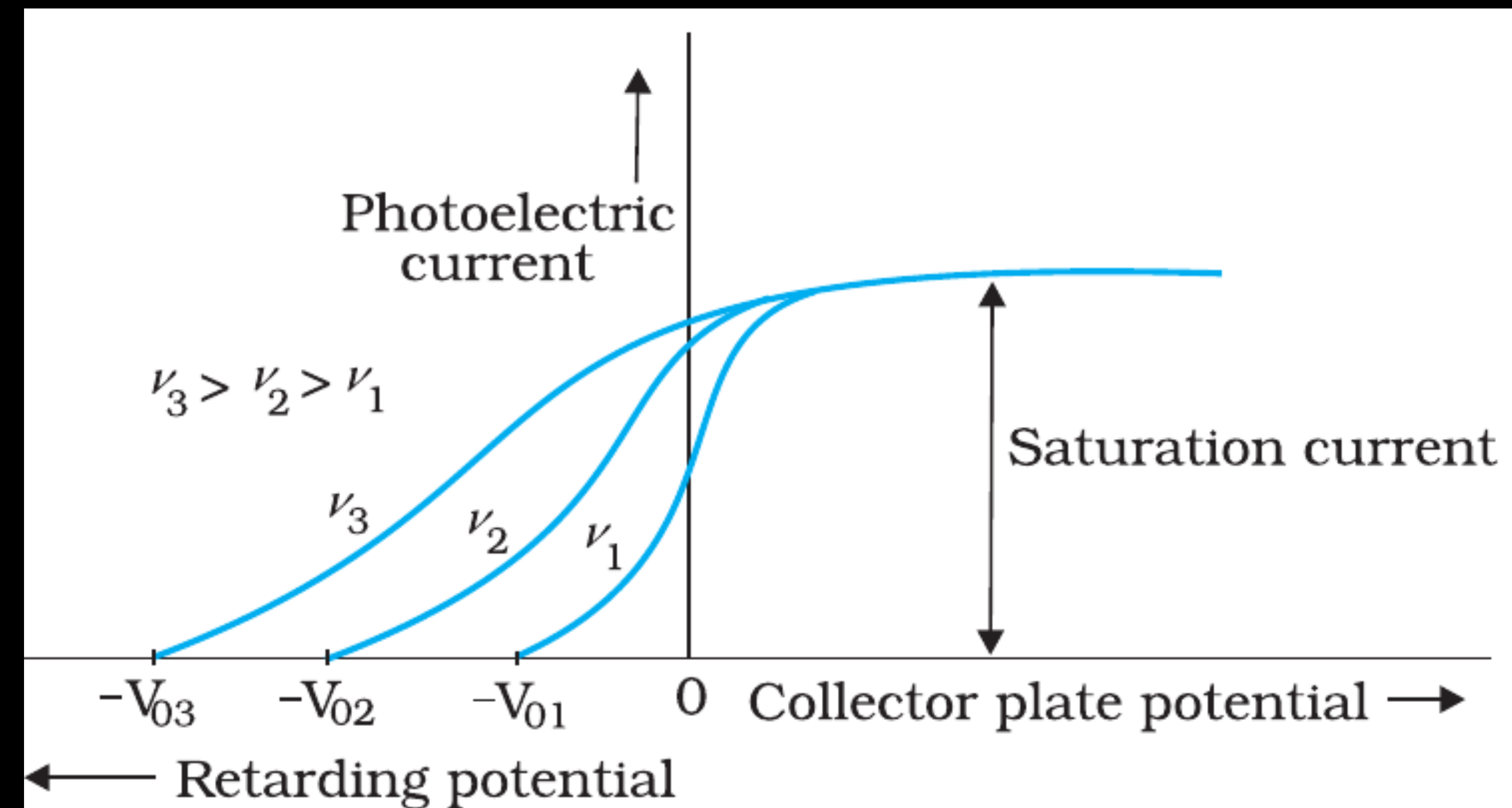


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- * The energy of the emitted electrons depends on the frequency of the incident radiations.
- * The stopping potential is more negative for higher frequencies of incident radiation.

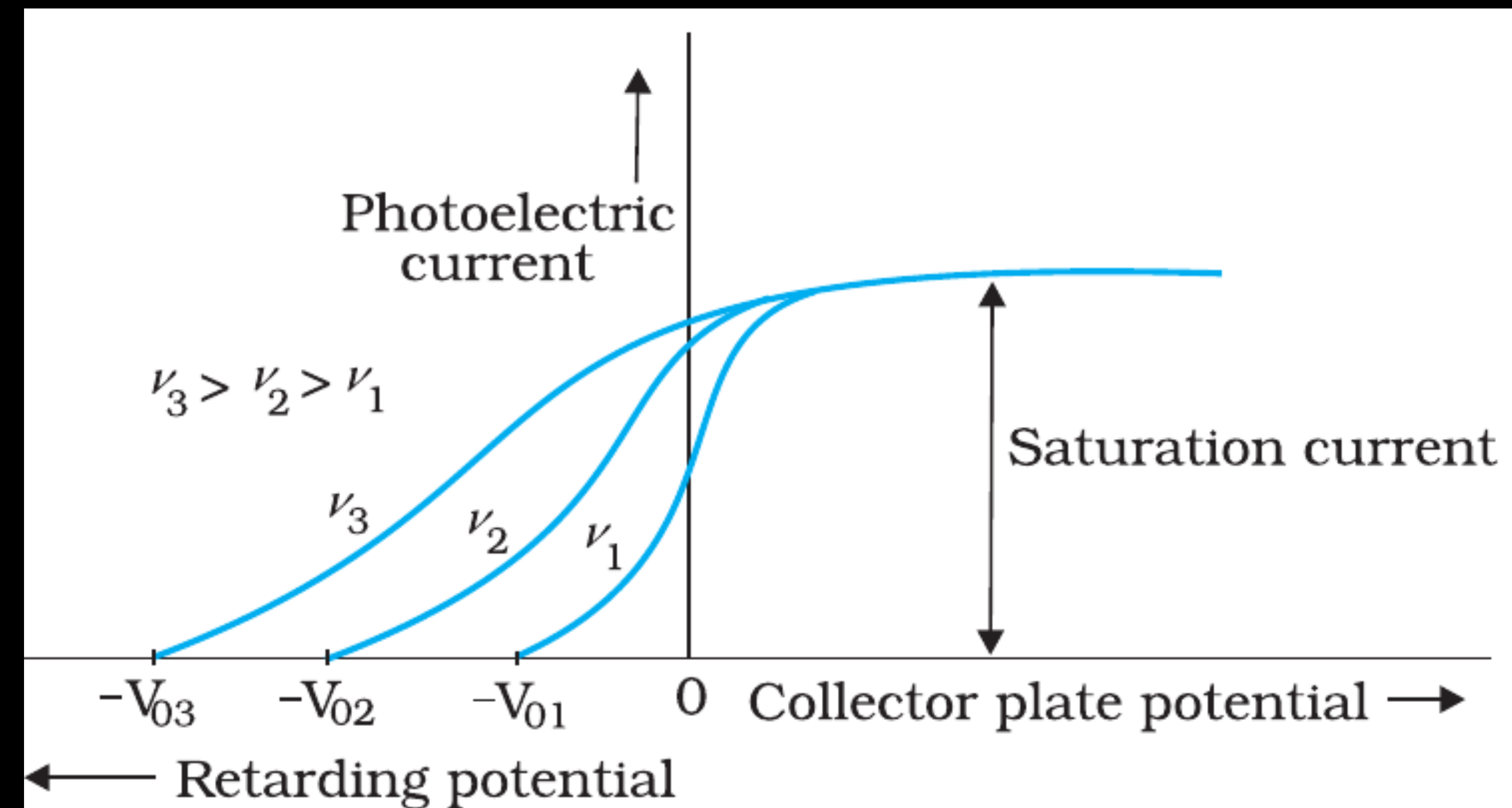


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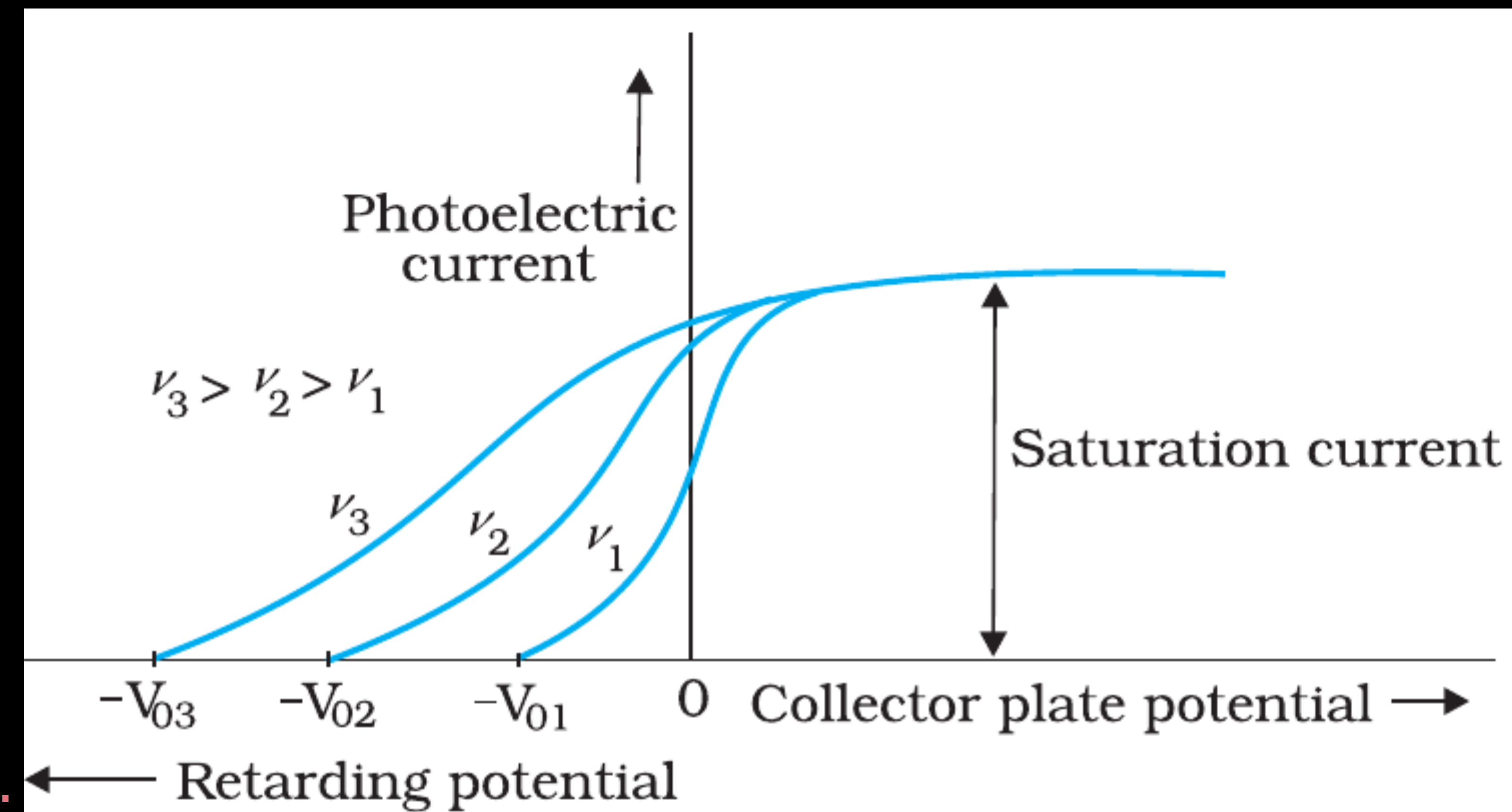


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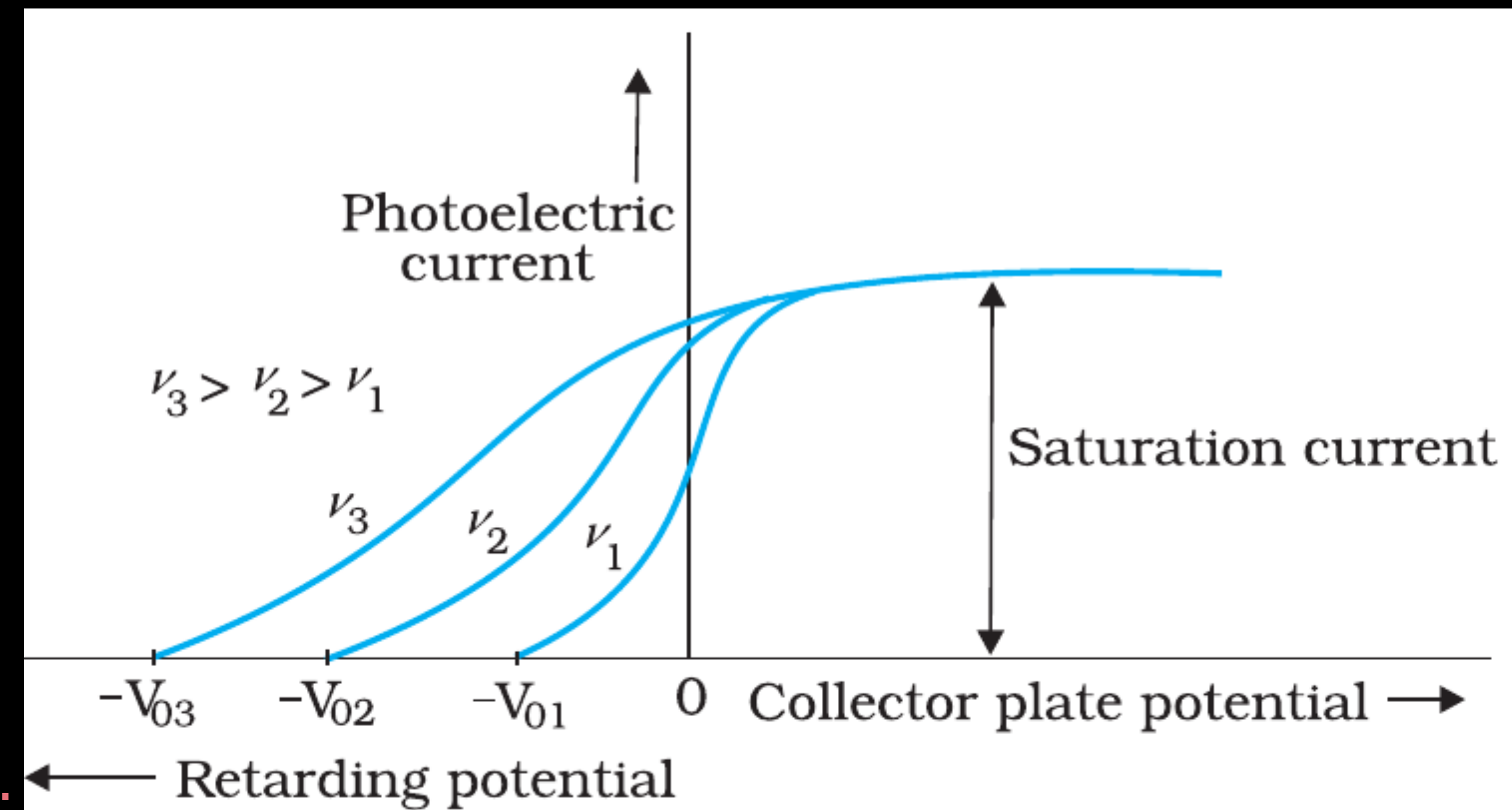


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Graph between the frequency of incident radiation and the corresponding stopping potential for different metals

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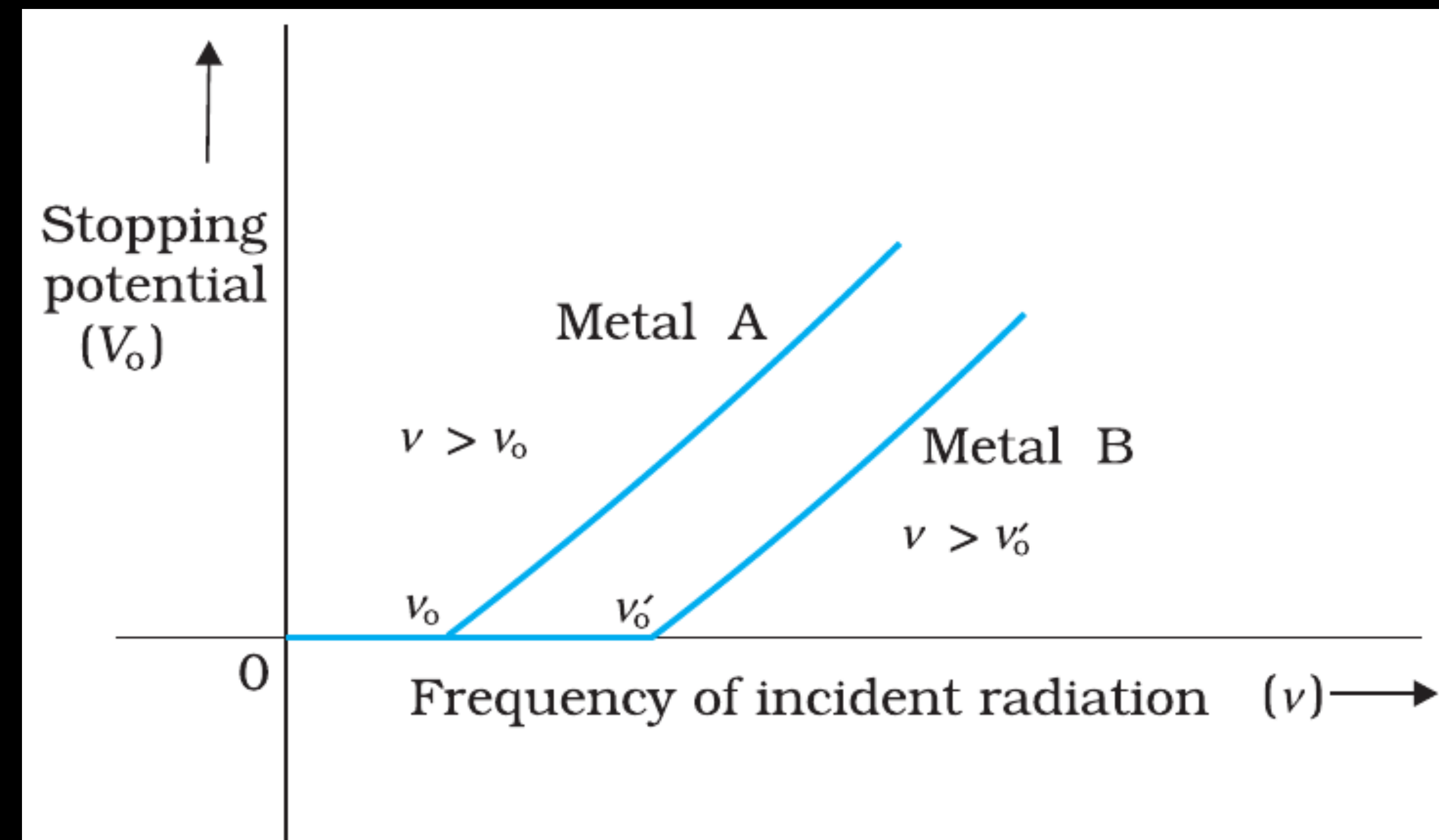


FIGURE 11.5 Variation of stopping potential V_0 with frequency ν of incident radiation for a given photosensitive material.

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(i) The stopping potential V_0 varies linearly with the frequency of incident radiation for a given photosensitive material.

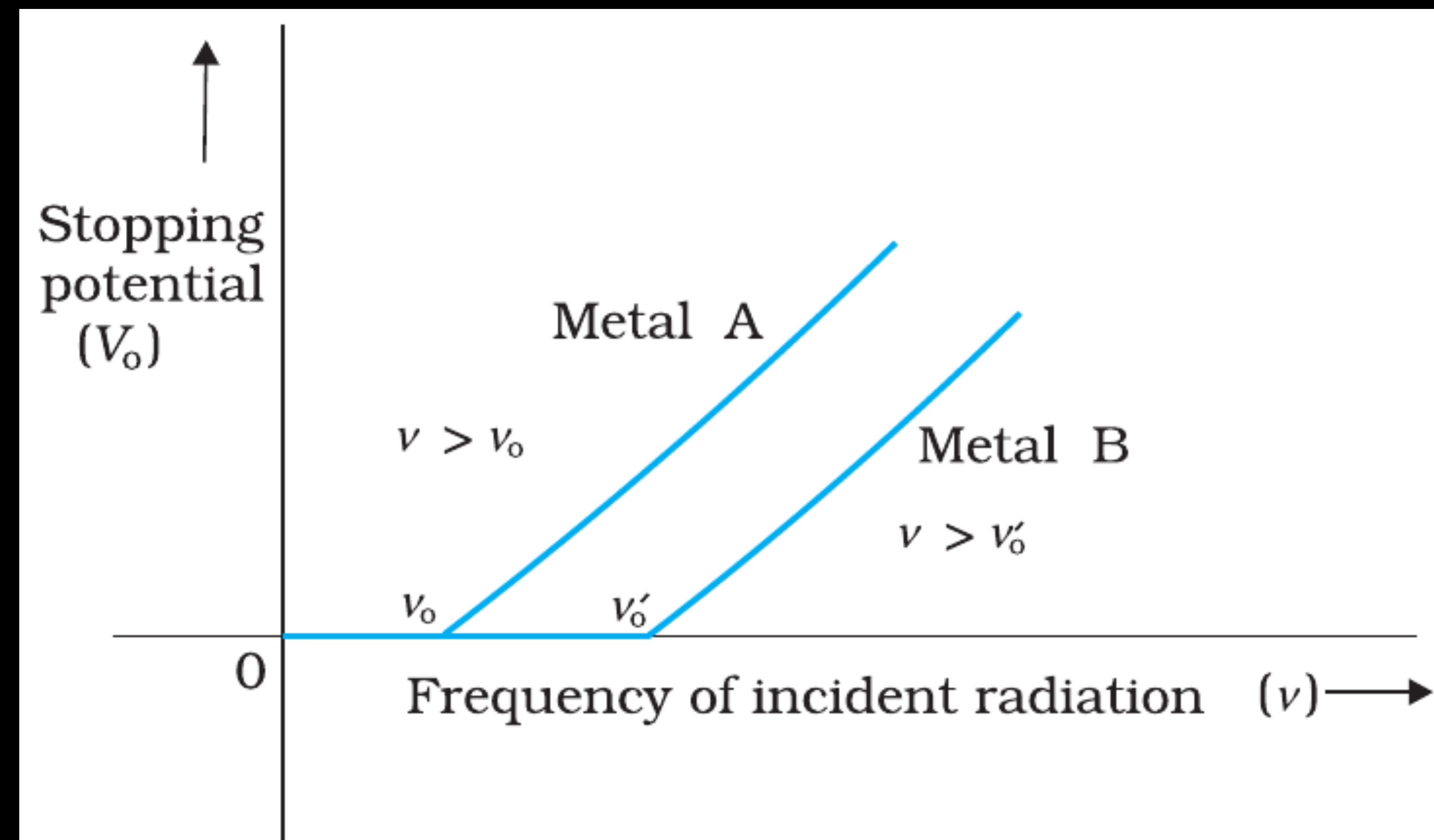


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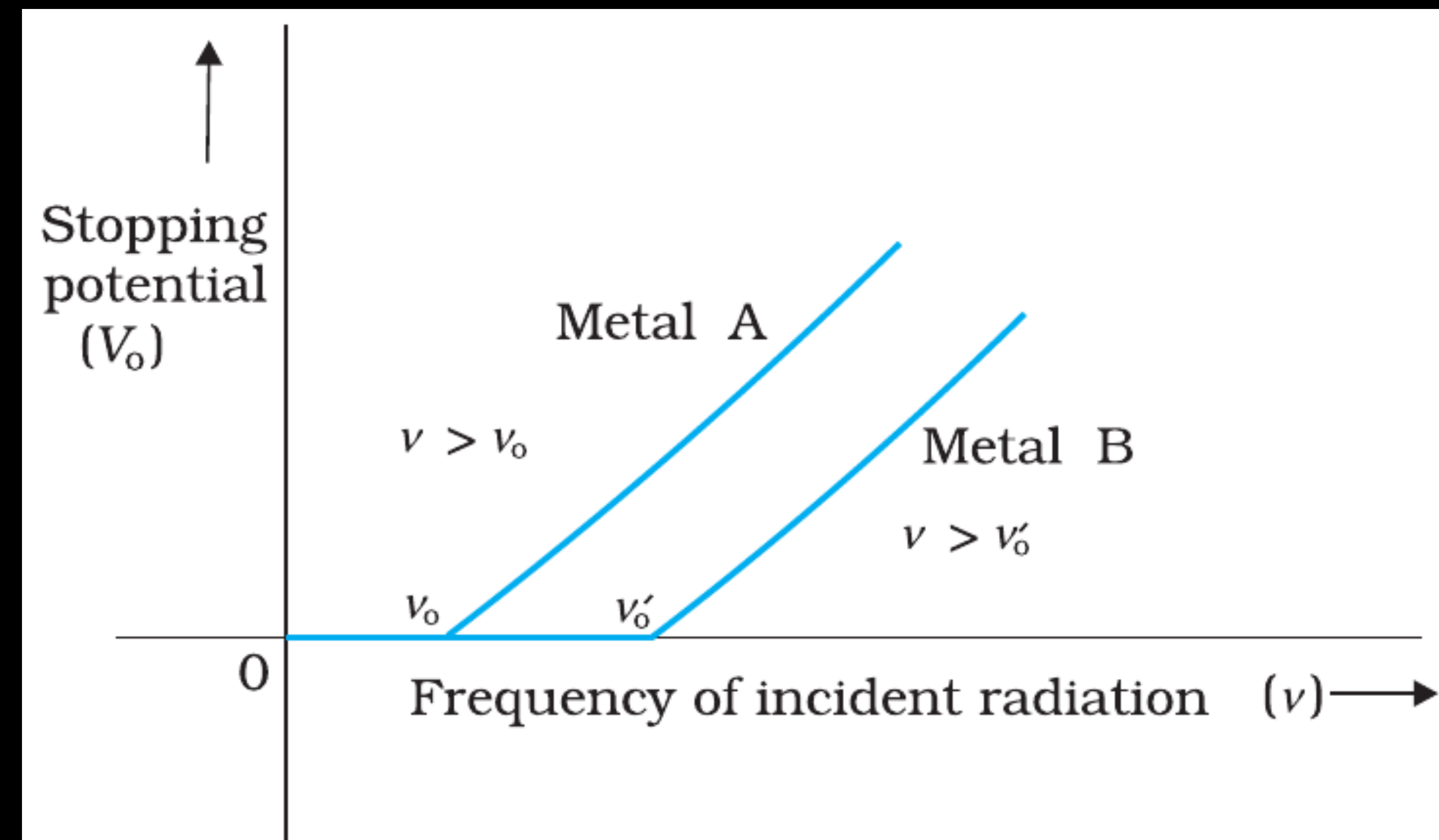


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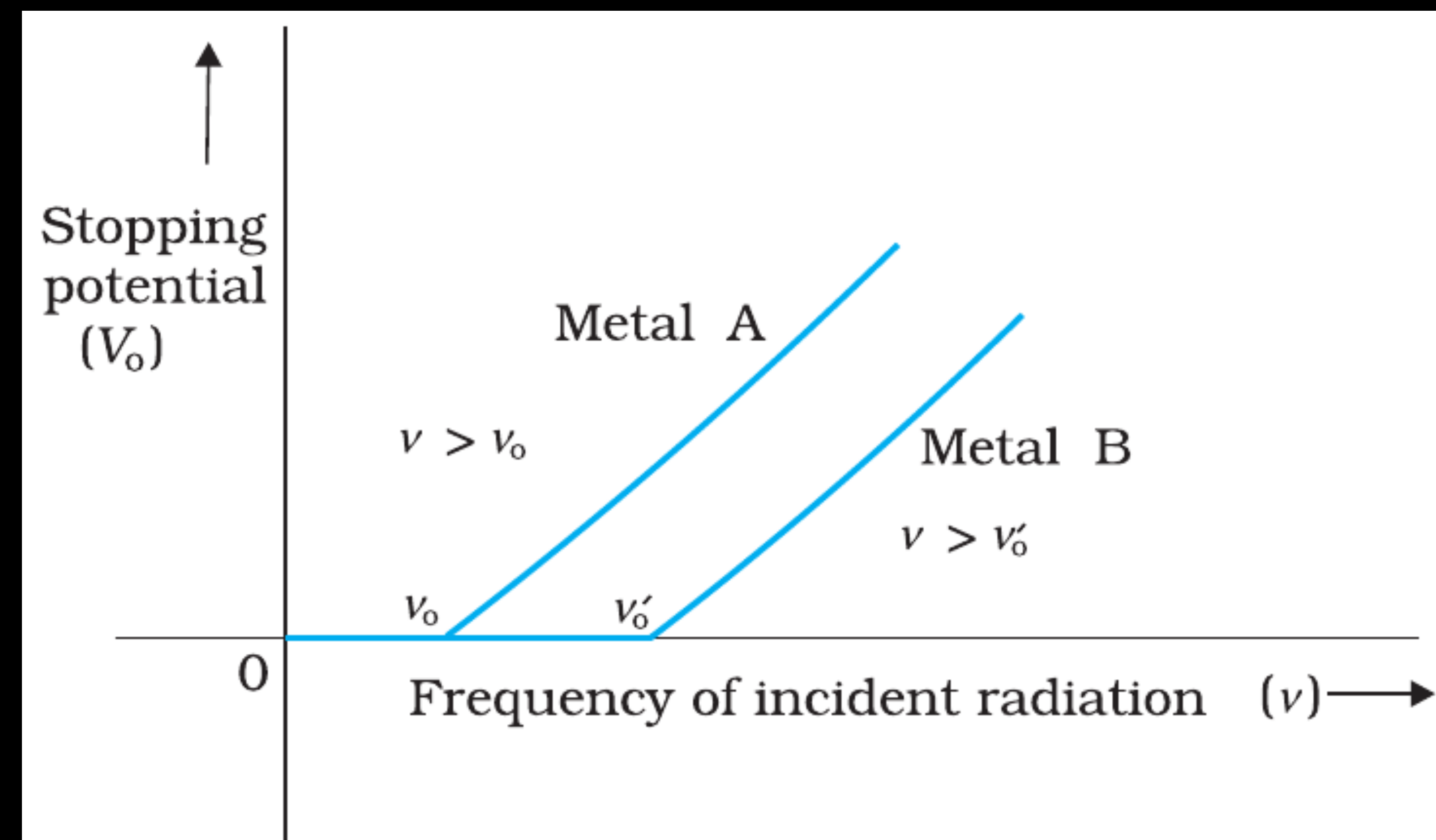


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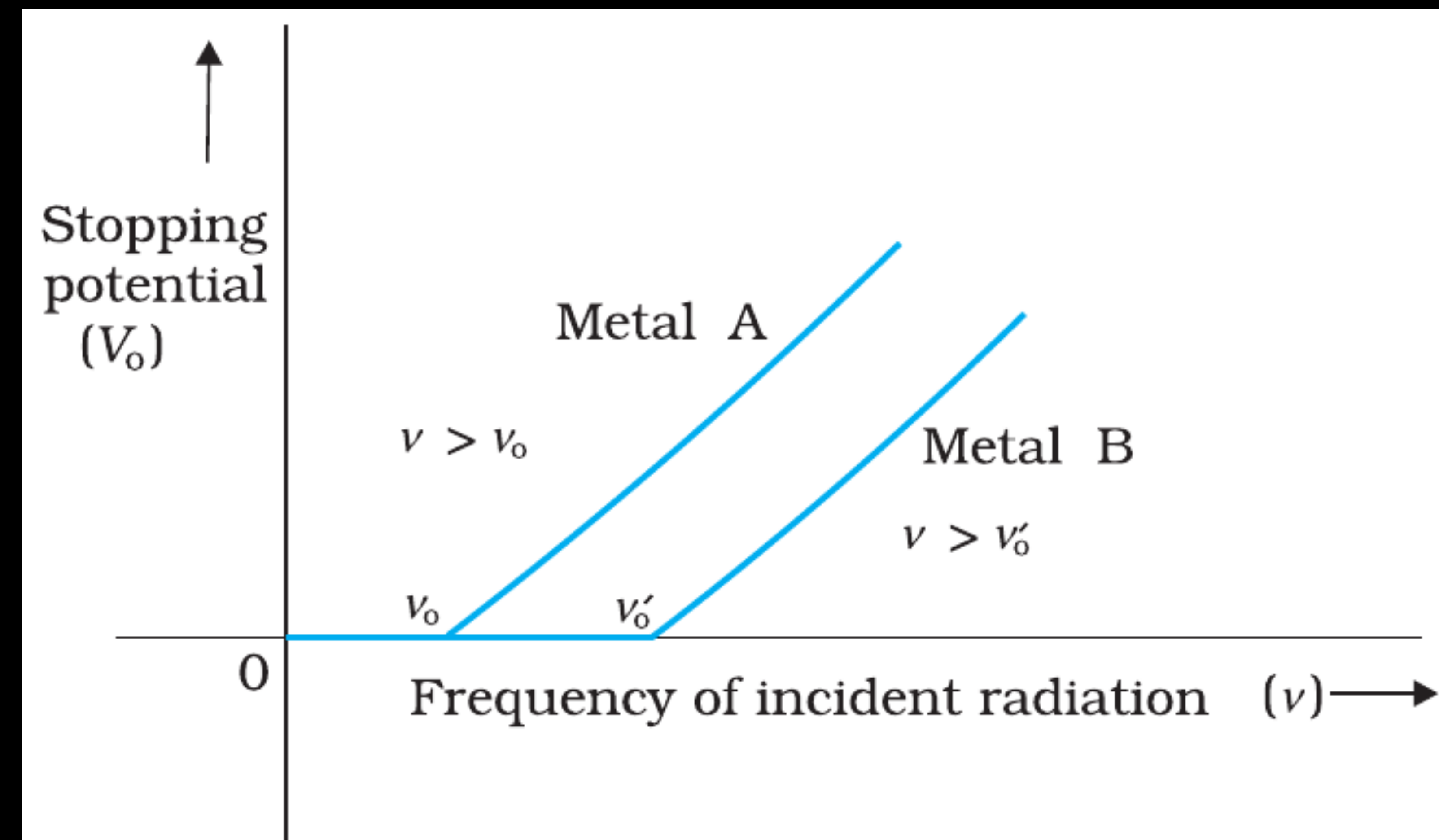


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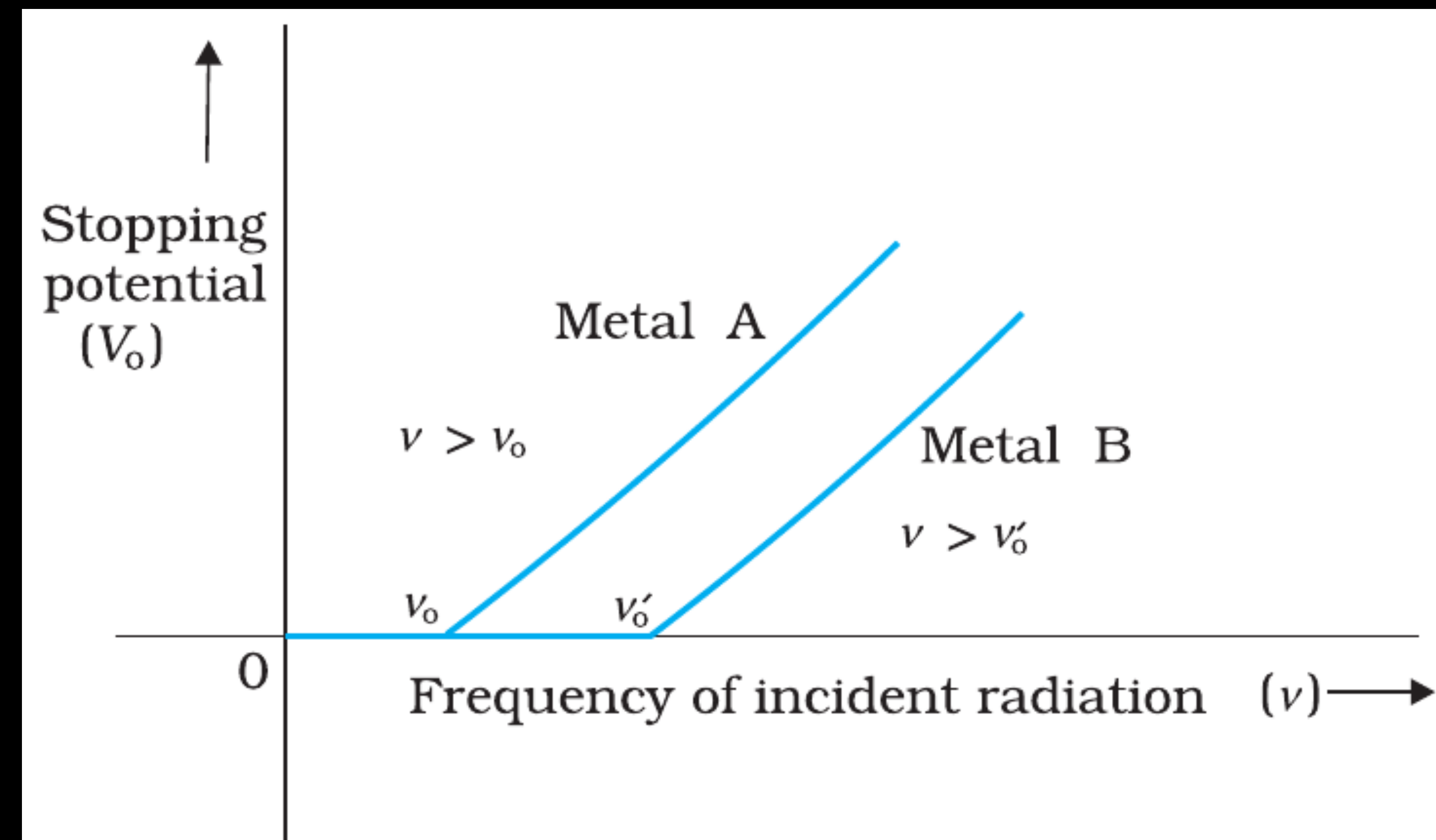


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- (iv) The photoelectric emission is an instantaneous process without any apparent time lag ($\sim 10^{-9}$ s or less), even when the incident radiation is made exceedingly dim.

PHOTOELECTRIC EFFECT AND WAVE THEORY OF LIGHT

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- * By the late 19th century, the wave nature of light was well established.
- * Wave theory successfully explained:
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 - ✓ (ii) Higher intensity → higher energy per electron
 - * Intensity corresponds to the square of amplitude.
 - * So wave theory predicts:
 - * Higher intensity should increase the maximum kinetic energy of emitted electrons.
 - * More intense light should eject electrons with larger energies.

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 - * In the wave picture, the absorption of energy by electron takes place continuously over the entire wavefront of the radiation.
 - * Since a large number of electrons absorb energy, the energy absorbed per electron per unit time turns out to be small.
 - * Explicit calculations estimate that it can take hours or more for a single electron to pick up sufficient energy to overcome the work function and come out of the metal.

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* Experimental Observations (Contradictions):

- (i) Kinetic energy depends only on frequency, not intensity.
- (ii) Threshold frequency exists ($\nu \geq \nu_0$ required).
- (iii) No time lag — emission is instantaneous ($< 10^{-9}$ s).
- (iv) Low-frequency light cannot eject electrons, no matter the intensity.
- (v) Intensity only changes photocurrent, not electron energy.

EINSTEIN'S PHOTOELECTRIC EQUATION: ENERGY QUANTUM OF RADIATION

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★ Basis of Einstein's Explanation

- * Einstein used Planck's quantum theory (1900), which states:
 - * Light consists of discrete packets of energy called photons (or quanta).
 - * Each photon has energy: $E = h\nu$
where, h = Planck's constant, ν = frequency of light
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★ When a photon strikes an electron in the metal:

- * Energy of photon = $h\nu$
- * This energy is used in two parts:
 - (i) To overcome the work function (ϕ_0). The minimum energy needed to free an electron from the metal surface.
 - (ii) The electron is emitted with maximum kinetic energy = Remaining energy

* Thus, $K_{\max} = h\nu - \phi_0$ → Einstein's photoelectric equation.

- * More tightly bound electrons will emerge with kinetic energies less than the maximum value.

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* Stopping Potential (V_0) Relation

The most energetic electrons are stopped by a retarding potential

$$K_{\max} = e V_0$$

Substitute into Einstein's equation:

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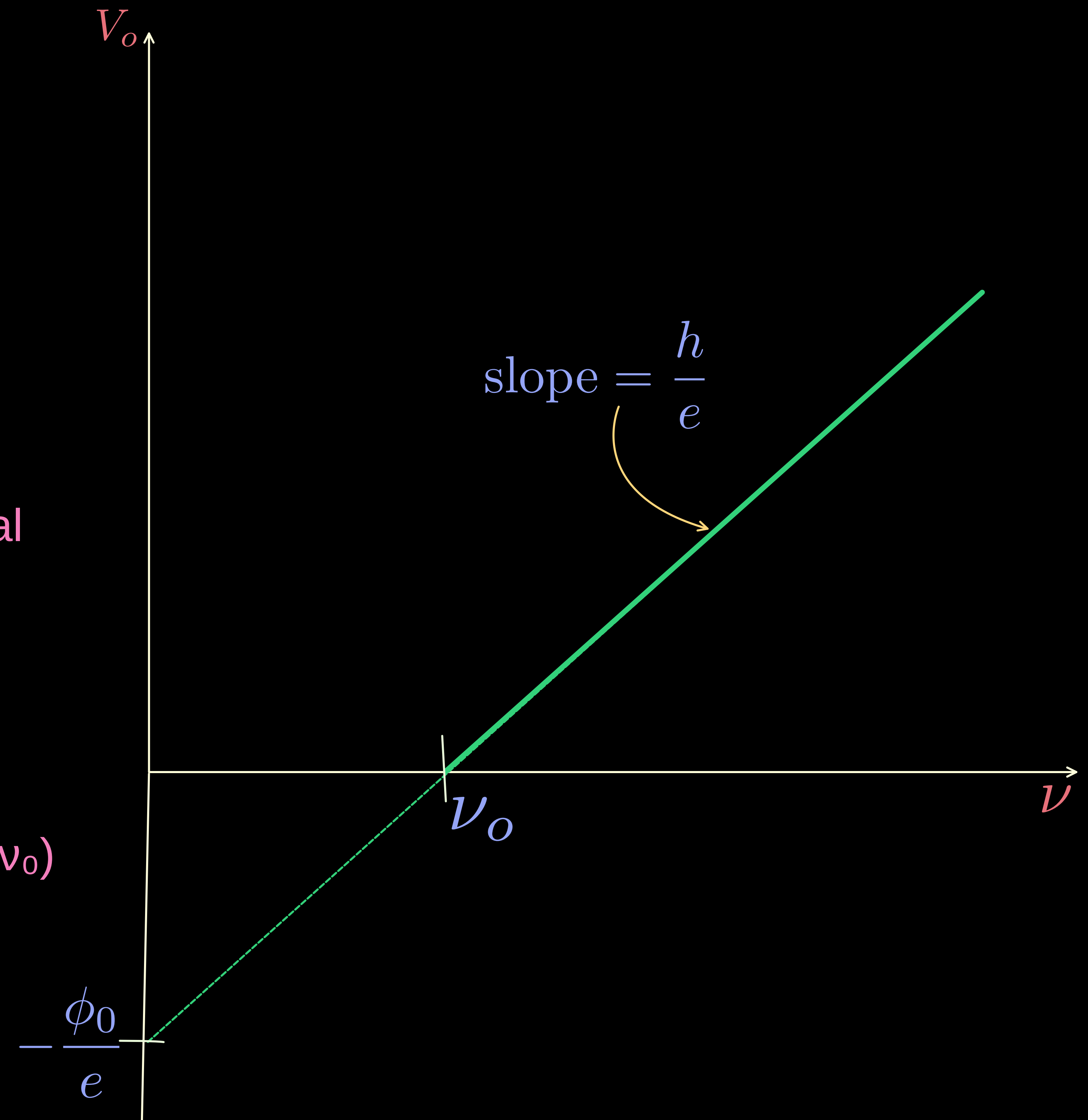
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- * It predicts that the V_0 versus ν curve is a straight line with
- * slope = (h/e) , independent of the nature of the material.
- * y-Intercept = $-h\nu_0/e = -\phi_0/e$

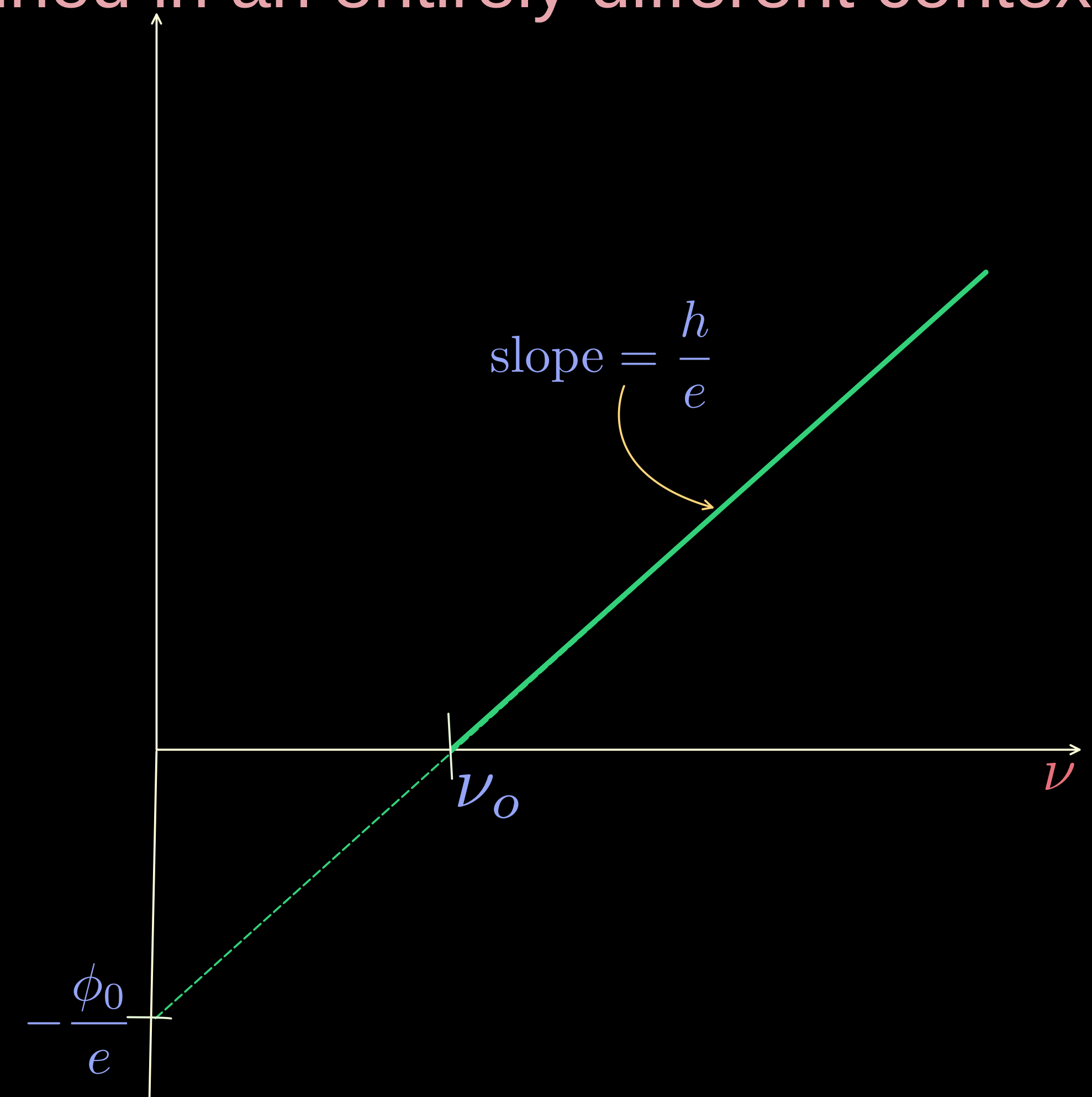


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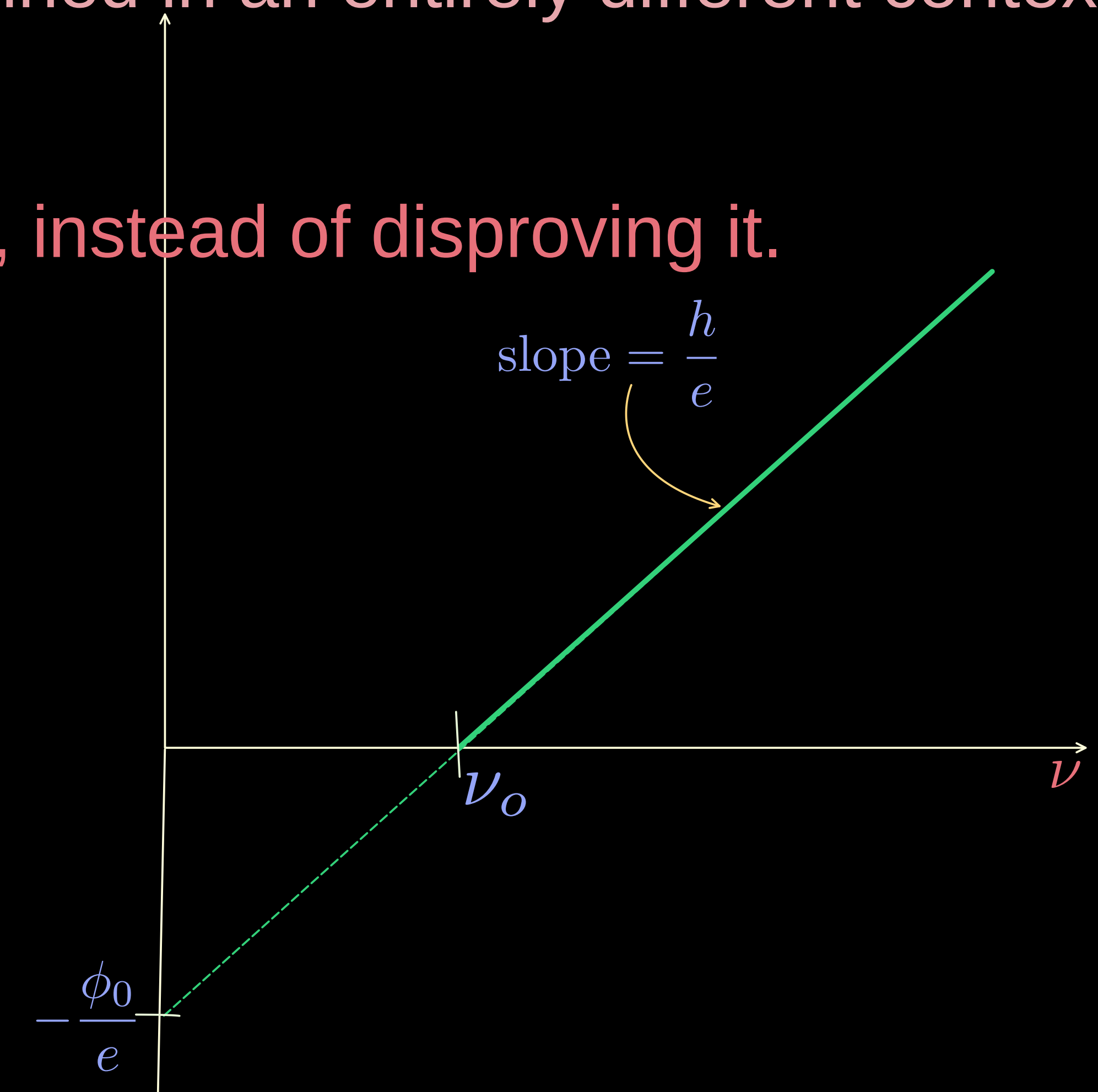
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- * In this way, in 1916, Millikan proved the validity of Einstein's photoelectric equation, instead of disproving it.



PARTICLE NATURE OF LIGHT: THE PHOTON

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- * A definite value of energy as well as momentum is a strong sign that the light quantum can be associated with a particle. This particle was later named photon.
- * The particle-like behaviour of light was further confirmed, in 1924, by the experiment of A.H. Compton (1892-1962) on scattering of X-rays from electrons.

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- * (v) In a photon-particle collision (such as photon-electron collision), the total energy and total momentum are conserved. However, the number of photons may not be conserved in a collision. The photon may be absorbed or a new photon may be created.

Example 11.1 Monochromatic light of frequency 6.0×10^{14} Hz is produced by a laser. The power emitted is 2.0×10^{-3} W. (a) What is the energy of a photon in the light beam? (b) How many photons per second, on an average, are emitted by the source?

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Solution

(a) Each photon has an energy

$$\begin{aligned} E &= h \nu = (6.63 \times 10^{-34} \text{ J s}) (6.0 \times 10^{14} \text{ Hz}) \\ &= 3.98 \times 10^{-19} \text{ J} \end{aligned}$$

(b) If N is the number of photons emitted by the source per second, the power P transmitted in the beam equals N times the energy per photon E , so that $P = N E$. Then

$$\begin{aligned} N &= \frac{P}{E} = \frac{2.0 \times 10^{-3} \text{ W}}{3.98 \times 10^{-19} \text{ J}} \\ &= 5.0 \times 10^{15} \text{ photons per second.} \end{aligned}$$

WAVE NATURE OF MATTER: DE-BROGLIE HYPOTHESIS

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* “The dual aspect of matter is evident in the de Broglie relation. On the left-hand side, λ is a wave attribute, while on the right-hand side, momentum p is a particle attribute. Planck's constant h relates the two attributes.”

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This is the reason why macroscopic objects in our daily life do not show wave-like properties.

On the other hand, in the sub-atomic domain, the wave character of particles is significant and measurable.

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- * Substituting the numerical values of h , m , e , we get,

$$\lambda = \frac{1.227}{\sqrt{V}} \text{ nm}$$

V is the magnitude of accelerating potential in volts.